

Environmental Currying and Causal Internal Homs: Dimension-Profile and Representability Obstructions in Quantum, Classical, and Higher-Order Process Theories

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Abstract

Can a process with a fixed environmental input be represented naturally as one into another physical system? Categorically, does tensoring with the environment have a right adjoint, or causal internal hom? The dimension-profile method introduced here records the number of independent parameters in each family of processes. Any adjunction preserves convex mixtures and therefore these counts. The intercept principle turns systematic violations into geometric or arithmetic no-go tests.

Applied to quantum and classical channels, instruments, comb states, and superchannels, this yields complete or conditional classifications. Quantum channels fail an all-input arithmetic test; even exceptional dimension matches have incompatible boundary geometry. Classical channel dimensions match, but products of simplices are not simplices. Nontrivial instruments fail grade by grade. Adding a fixed channel type admits a right adjoint only when the added type is trivial. Comb-state dimensions are derived separately from arbitrary higher-order process spaces.

Dimension is only the first layer. The analysis separates affine shape, positivity with ancillas, and naturality; constructs the exact linear candidate for a causal internal hom; and proves that universal evaluators assemble into an adjunction. Finite tomography, symmetry, and robustness make the converse testable. Extensions treat generalized probabilistic theories, composites without local tomography, positive extension, and finite operator-system realization.

Keywords: Environmental currying; causal internal hom; dimension-profile obstruction; quantum channels and superchannels; higher-order quantum processes; generalized probabilistic theories.

1 Introduction

Standard background on adjunctions, monoidal categories, and enriched categories may be found in [7, 10]. In a symmetric monoidal category, a right adjoint to $- \otimes E$ gives natural bijections

$$\mathcal{C}(X \otimes E, Y) \cong \mathcal{C}(X, RY). \quad (1)$$

Such an adjunction is environmental currying: it represents an E -input process as a state or process with values in an internal-hom object. In unrestricted finite-dimensional linear categories this is standard. For nontrivial environments in the quantum, classical, instrument, and deterministic-superchannel categories treated below, such right adjoints do not exist.

The notation T_E is category-relative. On quantum systems it denotes the system functor

$$T_E^{\text{sys}}(A) = A \otimes E,$$

whereas diagonal environmental extension on channel types denotes

$$T_E^{\text{pair}}(A, B) = (A \otimes E, B \otimes E).$$

More generally, a fixed channel type $K = (F, G)$ defines

$$T_K(A, B) = (A \otimes F, B \otimes G).$$

The choices $K = (E, I)$ and $K = (I, E)$ are respectively input-only and output-only type decoration. Their right-adjoint conditions have different coefficient laws and are classified jointly in Theorem 11.6.

The first obstruction records only affine dimensions. For each codomain Y , the function

$$p_Y(X) = \dim_{\text{aff}} \mathcal{C}(X, Y)$$

is its dimension profile. Pullback by T_E sends p_Y to $p_Y \circ T_E$. If (1) exists affinely, then

$$T_E^* p_Y = p_{RY}.$$

Thus the realizable profile locus must be invariant. A nonzero affine hyperplane containing all profiles gives an intercept obstruction when its normal covector is rescaled. A discrete profile locus gives an arithmetic obstruction.

The dimension-profile condition is necessary but not sufficient. Deterministic classical channels exhibit this distinction: transformed dimensions are realizable, but for a nontrivial environment and codomain the required channel polytope is a nontrivial product of simplices and cannot be the simplex state space of a classical object. Quantum channels furnish a second separation: some individual dimensions agree, but the channel body's nonclosed extreme-point set cannot match the closed pure-state boundary of a quantum state space. This yields parallel necessary tests of increasing structural resolution: dimension profile; affine shape and extreme-boundary topology; ancillary complete-order structure; and naturality and composition. The ordering does not assert that passing one test implies the next. For compact normalized bases, affine isomorphism already determines the generated ordinary cone, so affine-base and generated-cone representability are one test rather than two.

The principal results are as follows.

- (i) An ordinary adjunction is automatically affine when T_E is affine and composition is separately affine. The concrete no-go theorems therefore obstruct ordinary, not merely enriched, right adjoints.
- (ii) The dimension-profile theorem is formulated without a preferred basis, and the relation-space theorem recovers coefficient comparison while retaining all dependencies among chosen factors.
- (iii) The causal internal-hom vector space $W(E, Y)$ is constructed and shown to give exact currying of the normalization fibers at the linear level. The tensor-unit state-space test is separated from affine storage retracts and from full adjoint representability. Objectwise evaluators with unique factorization for every test object are shown to assemble automatically into the right-adjoint functor.
- (iv) For deterministic quantum channels, the object-dimension equation is completely classified and its exact arithmetic profile defect is calculated. A nonclosed extreme-boundary theorem then gives the complete affine classification of channel bodies that can be quantum state spaces.
- (v) Deterministic classical channels show that profile invariance is not sufficient: a vertex-count mismatch gives a strict polytopal obstruction.

- (vi) Quantum and classical instruments are treated as graded convex categories. Full grade rigidity controls the unit, counit, right adjoint, and every grade-wise transposition. Isolated grades and their compatibility are classified arithmetically; both the quantum and classical fixed-grade affine state-space problems are completely classified.
- (vii) A recurrence and closed sum are proved for the affine dimension of deterministic comb states. Explicit profile hypotheses give conditional right- and dimension-regular left-adjoint obstructions for any separately specified higher-map category. For deterministic superchannels, decoration by a fixed channel type has a right adjoint exactly for the tensor unit; input and output decoration are treated by distinct coefficient and arithmetic mechanisms. This is combined with complete tensor-unit classifications, a polygonal-section invariant, relative right-adjoint reflection, canonical set-valued and convex-causal completions, compilation of finite presheaves into fixed autonomous classical combs with exact minimal program registers, a stochastic naturality counterexample, environmental evaluator assembly, a tomographic codec criterion, and an exact finite tester-frame theorem. The latter reconstructs factorable program statistics from a calibrated outcome-response matrix, proves the sharp independent-probability budget and stability constant, and admits a concrete faithful-ancilla quantum realization. Symmetry-orbit theorems then replace full tester calibration by an exact isotypic multiplicity-rank condition; one nonconstant seed orbit suffices on an irreducible program direction. Compact Haar frames admit exact finite weighted quadratures and remain complete under an explicit orbit-implementation error threshold.
- (viii) Hom-presheaf nonrepresentability is isolated, and compact CP representability is shown not to imply universal broadcasting. Normalized restriction is characterized through the unit, counit, and right adjoint, and finite-dimensional examples show that these causal properties are independent of ordinary adjointness.
- (ix) Compact closure supplies the positive comparison: $- \otimes E \dashv - \otimes E^*$. In the finite-dimensional completely positive realization, the Bell unit and counit show that this adjunction restricts to trace-preserving maps exactly when $d_E = 1$. The optimal triangle defect within the normalized weighted-cup, trace-nonincreasing-cap family is $1 - \delta^{-2}$. Terminal compact closure, by contrast, collapses every hom-set to a singleton.
- (x) Profile separation and affine rank deficiency are quantified exactly. For a fixed hom-set metric, approximate inverse and naturality defects give explicit triangle bounds, while inradii convert dimension mismatch into sharp uniform reconstruction lower bounds. An explicit Hausdorff example proves that no such operational bound follows from dimension alone. Typed Petz recovery, Bayesian inversion, and opposite entropy-change examples separate those notions from right-adjoint representability.
- (xi) Restricted GPT process bodies are homogenized into canonical ordered vector spaces, removing the maximal-transformation hypothesis from the tensor-unit order criterion and proving that affine-base and generated-order representability coincide. Matrix transposition gives an explicit strict boundary at ancillary complete order. A finite-amplification theorem reduces all ancillary levels to the single level $\max\{m, n\}$. Fixed-embedding existence is reduced to semialgebraic Choi feasibility. The C^* -envelope determines intrinsic finite realizability and minimal size; finite covers give an explicit Shilov-ideal algorithm. A finite completely order-separating matrix-state certificate is exactly the additional data needed to construct such a cover, while no size bound depends only on operator-system dimension. For composites without local tomography, an exact linear theorem separates product-state relations from global quotient degrees. Positive extension is characterized by quotient-cone duality, transverse closedness, facial-direction and dual complementary-face criteria for nice cones, and a general semialgebraic decision with power-law conditioning, together with LP or Choi-SDP regimes and sharp failure examples.

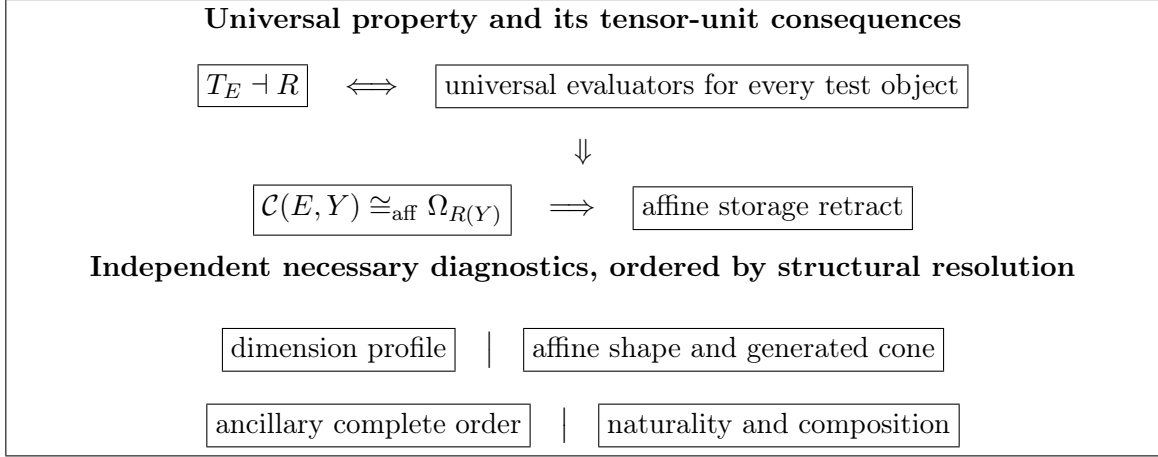


Figure 1: Logical status of the representability criteria. The upper line contains one equivalence followed by genuine implications. The lower line lists distinct necessary diagnostics; adjacency does not assert implication or sufficiency.

Figure 1 records the logical status of the principal representability data used throughout the article.

Related companion work. Two-sided adjointability, finite-memory process storage and evaluation, quantitative compression bounds, and exact process recovery are studied in [1]. Higher-order superchannels and deterministic combs of arbitrary finite order are treated in [2]. The facial geometry of channel Choi sets, environment-independent factorization, and associated information-theoretic invariants are developed in [3]. Additive valuations on decorated channel categories, quadratic channel invariants, and continuous compression bounds are studied in [4]. The left-adjoint boundary, normalization-free completely positive currying, exact sequential-comb ranks, and their relation to additive dynamics and approximation are synthesized in [6].

2 Affine Adjunctions and Automatic Affinity

Throughout, convex sets are nonempty and have finite-dimensional affine hulls unless stated otherwise; categorical uses of convex structure in probability are reviewed in [11, 12]. The term *hom-set* denotes the underlying categorical set, whereas *hom-space* emphasizes its convex or affine structure. The noun *internal hom* is unhyphenated; *internal-hom* is used only as a compound modifier. For such a set C and any $x_0 \in \text{aff}(C)$, define

$$\dim_{\text{aff}} C = \dim(\text{aff}(C) - x_0).$$

Lemma 2.1 (Affine extension). An affine map $\Phi : C_1 \rightarrow C_2$ between nonempty convex sets has a unique affine extension

$$\bar{\Phi} : \text{aff}(C_1) \rightarrow \text{aff}(C_2).$$

Proof. First, Φ preserves arbitrary affine relations. Suppose $x_i \in C_1$ and

$$\sum_i t_i x_i = 0, \quad \sum_i t_i = 0.$$

Let $P = \{i : t_i > 0\}$, $N = \{i : t_i < 0\}$, and $S = \sum_{i \in P} t_i = -\sum_{i \in N} t_i$. If $S = 0$, all coefficients vanish. Otherwise,

$$\sum_{i \in P} \frac{t_i}{S} x_i = \sum_{i \in N} \frac{-t_i}{S} x_i$$

is an equality of convex combinations. Affinity gives the corresponding equality after applying Φ , and multiplication by S yields $\sum_i t_i \Phi(x_i) = 0$.

Every $z \in \text{aff}(C_1)$ has an affine representation $z = \sum_i s_i x_i$ with $\sum_i s_i = 1$. Define

$$\bar{\Phi}(z) = \sum_i s_i \Phi(x_i).$$

Two representations differ by an affine relation, so the preceding paragraph makes this definition independent of the representation. It is affine and extends Φ . Uniqueness follows because C_1 affinely spans $\text{aff}(C_1)$. $\square$

Lemma 2.2 (Inverse of an affine bijection). A bijective affine map between convex sets has affine inverse and preserves affine dimension.

Proof. Let $\phi : C \rightarrow D$ be affine and bijective. For $y_1, y_2 \in D$, put $x_i = \phi^{-1}(y_i)$. For $0 \leq t \leq 1$, affinity gives

$$\phi(tx_1 + (1-t)x_2) = ty_1 + (1-t)y_2.$$

Applying ϕ^{-1} to this equality yields

$$\phi^{-1}(ty_1 + (1-t)y_2) = t\phi^{-1}(y_1) + (1-t)\phi^{-1}(y_2),$$

so ϕ^{-1} is affine.

By Lemma 2.1, ϕ and ϕ^{-1} extend uniquely to affine maps

$$\bar{\phi} : \text{aff}(C) \rightarrow \text{aff}(D), \quad \bar{\phi}^{-1} : \text{aff}(D) \rightarrow \text{aff}(C).$$

The composite $\bar{\phi}^{-1} \circ \bar{\phi}$ agrees with the identity on C . Both maps are affine, and C affinely spans $\text{aff}(C)$; uniqueness of affine extension therefore gives

$$\bar{\phi}^{-1} \circ \bar{\phi} = \text{id}_{\text{aff}(C)}.$$

The other composite is the identity on $\text{aff}(D)$ by the same argument. Hence the affine hulls are affinely isomorphic, and their direction spaces have equal dimension. Thus $\dim_{\text{aff}} C = \dim_{\text{aff}} D$. $\square$

Proposition 2.3 (Automatic affinity of ordinary transposition). Let $\mathcal{C}$ have convex hom-sets and separately affine composition. Let $T : \mathcal{C} \rightarrow \mathcal{C}$ act affinely on every hom-set. If T has an ordinary right adjoint R , then every adjunction bijection

$$\Phi_{X,Y} : \mathcal{C}(TX, Y) \longrightarrow \mathcal{C}(X, RY)$$

is an affine isomorphism.

Proof. The inverse transpose is

$$\Psi_{X,Y}(g) = \epsilon_Y \circ T(g),$$

where ϵ is the counit. The functorial action of T is affine, and postcomposition by the fixed map ϵ_Y is affine. Hence $\Psi_{X,Y}$ is affine. It is bijective by the adjunction, so Lemma 2.2 shows that its inverse $\Phi_{X,Y}$ is affine. $\square$

Remark 2.4. For quantum channels, classical stochastic maps, and each fixed instrument grade, ordinary adjunctions therefore induce affine transpositions.

3 The Dimension-Profile Principle

Let $\mathcal{C}$ be a category with nonempty finite-dimensional convex hom-sets. Let $\mathcal{X}$ be a class of test objects closed under an endofunction T , and let $\mathcal{Y}$ be a class of codomain objects. Define the affine hom-dimension function

$$D : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}, \quad D(X, Y) = \dim_{\text{aff}} \mathcal{C}(X, Y),$$

and define

$$p_Y = D(-, Y) \in \mathbb{R}^{\mathcal{X}}.$$

Assume

$$U = \text{span}_{\mathbb{R}} \{p_Y : Y \in \mathcal{Y}\}$$

is finite-dimensional and $T^*U \subseteq U$, where

$$(T^*u)(X) = u(TX).$$

The realizable profile locus is

$$\mathfrak{P} = \{p_Y : Y \in \mathcal{Y}\} \subseteq U.$$

Theorem 3.1 (Dimension-profile obstruction). Suppose that for each $Y \in \mathcal{Y}$ there is an object $RY \in \mathcal{Y}$ and, for every $X \in \mathcal{X}$, an affine bijection

$$\mathcal{C}(TX, Y) \cong \mathcal{C}(X, RY).$$

Then

$$p_{RY} = T^*p_Y \quad \text{for every } Y,$$

and consequently

$$T^*\mathfrak{P} \subseteq \mathfrak{P}.$$

In particular, any affine, arithmetic, or other property shared by all realizable profiles must be shared by every transformed profile.

Proof. Affine bijections preserve affine dimension, so

$$D(TX, Y) = D(X, RY)$$

for every X, Y . Therefore

$$(T^*p_Y)(X) = p_Y(TX) = D(TX, Y) = D(X, RY) = p_{RY}(X).$$

The equality holds pointwise on $\mathcal{X}$, proving $T^*p_Y = p_{RY} \in \mathfrak{P}$. □

Corollary 3.2 (Affine-hyperplane obstruction). Suppose $\ell \in U^*$ is nonzero, $c \neq 0$, and

$$\ell(p_Y) = c \quad \text{for every } Y, \quad \ell \circ T^* = \lambda \ell.$$

If the representability condition of Theorem 3.1 holds, then $\lambda = 1$.

Proof. For every Y ,

$$c = \ell(p_{RY}) = \ell(T^*p_Y) = \lambda \ell(p_Y) = \lambda c.$$

Since $c \neq 0$, $\lambda = 1$. □

Remark 3.3 (Intercept terminology). Suppose a two-dimensional profile space has linearly independent coordinates f_{bulk} and f_{norm} , and every realizable profile has the form

$$p_Y = \alpha(Y)f_{\text{bulk}} + c f_{\text{norm}}, \quad c \neq 0.$$

The fixed coefficient c , arising from the normalization constraint, is the *intercept*. If ℓ extracts the f_{norm} coefficient, then the realizable locus is contained in the affine hyperplane

$$\ell(p) = c.$$

The intercept obstruction is therefore the two-coordinate instance of Corollary 3.2. The dimension-profile theorem does not depend on the existence or choice of these coordinates.

Remark 3.4 (Necessity, not sufficiency). Profile-locus invariance is a necessary condition. It does not preserve the full affine shape, positive cone, naturality, or composition. Section 7 gives an explicit category in which dimensions pass this test but affine representability still fails.

4 Relation Spaces and Coefficient Decompositions

The profile theorem is basis free. When a coefficient decomposition is available, the same obstruction can be expressed in a quotient that records all linear dependencies among the chosen left factors. Suppose

$$D(\xi, Y) = \sum_{i=1}^n f_i(\xi)g_i(Y), \quad f_i(T\xi) = \lambda_i f_i(\xi).$$

Define

$$q : \mathbb{R}^n \rightarrow \mathbb{R}^{\mathcal{X}}, \quad q(a) = \sum_i a_i f_i, \quad K = \ker q,$$

and put

$$g(Y) = (g_1(Y), \dots, g_n(Y))^T, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Proposition 4.1 (Relation-space theorem). The relation space is forward invariant,

$$\Lambda K \subseteq K.$$

Moreover,

$$D(T\xi, Y) = D(\xi, RY) \quad \text{for every } \xi, Y$$

if and only if

$$g(RY) - \Lambda g(Y) \in K \quad \text{for every } Y.$$

Equivalently,

$$[g(RY)] = [\Lambda g(Y)] \quad \text{in } \mathbb{R}^n / K.$$

Proof. If $a \in K$, then for every ξ ,

$$q(\Lambda a)(\xi) = \sum_i a_i \lambda_i f_i(\xi) = \sum_i a_i f_i(T\xi) = 0.$$

Thus $\Lambda a \in K$.

Furthermore,

$$D(T\xi, Y) = q(\Lambda g(Y))(\xi), \quad D(\xi, RY) = q(g(RY))(\xi).$$

These functions are equal for every ξ exactly when $q(g(RY) - \Lambda g(Y)) = 0$, which is the stated condition. $\square$

Corollary 4.2 (Independent-factor theorem). If $f_1, \dots, f_n$ are linearly independent, then

$$g_i(RY) = \lambda_i g_i(Y)$$

for every i, Y . If also $g_i(RY) = g_i(Y)$ for all Y and g_i is nonzero somewhere, then $\lambda_i = 1$.

Proof. Linear independence of $f_1, \dots, f_n$ means that

$$q(a) = \sum_i a_i f_i = 0$$

only when $a = 0$. Hence $K = \ker q = \{0\}$. Proposition 4.1 then gives

$$g(RY) - \Lambda g(Y) = 0.$$

Taking the i th component yields

$$g_i(RY) = \lambda_i g_i(Y).$$

If also $g_i(RY) = g_i(Y)$ for every Y , choose Y_0 with $g_i(Y_0) \neq 0$. Then

$$g_i(Y_0) = \lambda_i g_i(Y_0),$$

and division by $g_i(Y_0)$ gives $\lambda_i = 1$. □

5 Causal Internal Homs and the State-Space Test

At the linear level, environmental currying is governed by a causal internal-hom space. Its construction also specifies the structural role of discarding.

5.1 Linear causal internal hom

Let V_X, V_E, V_Y be finite-dimensional real vector spaces with nonzero linear functionals

$$u_X \in V_X^*, \quad u_E \in V_E^*, \quad u_Y \in V_Y^*.$$

They are the deterministic effects. Equip the algebraic tensor product $V_X \otimes V_E$ with the functional $u_X \otimes u_E$. Identifying this algebraic tensor product with the physical composite $V_{X \otimes E}$ is the local tomography hypothesis; the linear statement below does not require that physical identification.

Definition 5.1 (Causal internal-hom vector space). Define

$$W(E, Y) = \{G \in \text{Hom}(V_E, V_Y) : u_Y \circ G \in \text{span}(u_E)\}.$$

For $G \in W(E, Y)$, let $u_W(G)$ be the unique scalar satisfying

$$u_Y \circ G = u_W(G) u_E.$$

Lemma 5.1 (Linearity of the internal-hom effect). The map $u_W : W(E, Y) \rightarrow \mathbb{R}$ is linear.

Proof. If $u_Y G_i = \alpha_i u_E$, then for real scalars s, t ,

$$u_Y(sG_1 + tG_2) = (s\alpha_1 + t\alpha_2)u_E.$$

Uniqueness of the coefficient of the nonzero functional u_E gives

$$u_W(sG_1 + tG_2) = su_W(G_1) + tu_W(G_2).$$

□

Theorem 5.2 (Dimension and exact normalization-fiber currying). Let $e = \dim V_E$ and $y = \dim V_Y$. Then

$$\dim W(E, Y) = e(y - 1) + 1.$$

Moreover, ordinary vector-space currying restricts to a linear bijection between the affine normalization fibers

$$\begin{aligned} & \{F \in \text{Hom}(V_X \otimes V_E, V_Y) : u_Y \circ F = u_X \otimes u_E\} \\ & \cong \{\tilde{F} \in \text{Hom}(V_X, W(E, Y)) : u_W \circ \tilde{F} = u_X\}. \end{aligned}$$

Proof. Consider the linear map

$$Q : \text{Hom}(V_E, V_Y) \rightarrow V_E^* / \text{span}(u_E), \quad Q(G) = [u_Y \circ G].$$

It is surjective. Indeed, choose $y_0 \in V_Y$ with $u_Y(y_0) = 1$. For any $\varphi \in V_E^*$, define $G_\varphi(e) = \varphi(e)y_0$. Then $u_Y \circ G_\varphi = \varphi$. Hence

$$\text{rank } Q = \dim(V_E^* / \text{span}(u_E)) = e - 1.$$

Since $W(E, Y) = \ker Q$, rank-nullity gives

$$\dim W(E, Y) = ey - (e - 1) = e(y - 1) + 1.$$

Let $\tilde{F}$ be the ordinary curry of F . If $u_Y F = u_X \otimes u_E$, then for all $x \in V_X$ and $e' \in V_E$,

$$(u_Y \circ \tilde{F}(x))(e') = u_Y(F(x \otimes e')) = u_X(x)u_E(e').$$

Thus $\tilde{F}(x) \in W(E, Y)$ and $u_W(\tilde{F}(x)) = u_X(x)$, so $u_W \circ \tilde{F} = u_X$.

Conversely, suppose $\tilde{F} : V_X \rightarrow W(E, Y)$ satisfies $u_W \circ \tilde{F} = u_X$, and let F be its ordinary uncurrying. For every simple tensor $x \otimes e'$, the definition of u_W gives

$$\begin{aligned} u_Y(F(x \otimes e')) &= u_Y(\tilde{F}(x)(e')) \\ &= u_W(\tilde{F}(x))u_E(e') \\ &= u_X(x)u_E(e'). \end{aligned}$$

Simple tensors span $V_X \otimes V_E$, so this equality proves $u_Y \circ F = u_X \otimes u_E$ on the whole tensor product. The restricted maps are inverse because ordinary vector-space currying and uncurrying are inverse. $\square$

5.2 State-space and storage boundary

Three levels are distinguished: an affine storage retract, an affine identification with an available state space, and unique evaluation for every test object. The first is weaker than the second, while the all-test-object evaluation property is equivalent to a right adjoint.

Corollary 5.3 (State-space test). If $T_E = - \otimes E$ has a right adjoint in a monoidal category with tensor unit I , then

$$\mathcal{C}(E, Y) \cong \mathcal{C}(I, RY).$$

Whenever Proposition 2.3 applies, this is an affine isomorphism. Thus the convex set of E -input processes to Y must be the state space of an available object.

Proof. The adjunction supplies natural bijections

$$\mathcal{C}(X \otimes E, Y) \cong \mathcal{C}(X, RY)$$

for every object X . Taking $X = I$ gives

$$\mathcal{C}(I \otimes E, Y) \cong \mathcal{C}(I, RY).$$

Precomposition with the left unitor $I \otimes E \cong E$ identifies the left-hand hom-set with $\mathcal{C}(E, Y)$, proving the stated bijection. If the hypotheses of Proposition 2.3 hold, that proposition makes the adjunction bijection affine; precomposition with an isomorphism is affine as well. Hence the resulting state-space bijection is affine. $\square$

Remark 5.4 (Representability boundary). Theorem 5.2 is a statement about the algebraic tensor product. Its direct application to a physical composite uses local tomography and the monoidal factorization $u_{X \otimes E} = u_X \otimes u_E$. Theorem 19.1 below gives the corresponding relation-and-excess decomposition without local tomography. Environmental currying additionally requires the resulting normalized base, positive cone, and relevant complete-positivity structure to be represented by an object of the physical category.

Remark 5.5 (Process-storage reading of the state-space test). A right adjoint to $T_E = - \otimes E$ identifies every process space $\mathcal{C}(E, Y)$ affinely with the state space of a single object $R(Y)$, naturally in Y . Corollary 5.3 supplies this necessary tensor-unit condition; the quantum and classical examples below obstruct it by dimension and convex shape, respectively. The condition does not address arbitrary non-natural encodings, approximate storage, or state-dependent recovery.

Definition 5.2 (Universal evaluator). Let $T : \mathcal{C} \rightarrow \mathcal{D}$ be a functor and $Y \in \mathcal{D}$. A T -universal evaluator for Y is an object $R_Y \in \mathcal{C}$ and a morphism

$$\epsilon_Y : T(R_Y) \rightarrow Y$$

such that, for every $X \in \mathcal{C}$, the map

$$\Psi_{X,Y} : \mathcal{C}(X, R_Y) \longrightarrow \mathcal{D}(TX, Y), \quad \Psi_{X,Y}(g) = \epsilon_Y \circ T(g),$$

is bijective.

Theorem 5.6 (Objectwise universal-evaluator criterion). A functor $T : \mathcal{C} \rightarrow \mathcal{D}$ has a right adjoint if and only if every object $Y \in \mathcal{D}$ admits a T -universal evaluator. Given a choice (R_Y, ϵ_Y) for every Y , the object assignment $Y \mapsto R_Y$ extends uniquely to a functor $R : \mathcal{D} \rightarrow \mathcal{C}$ for which $T \dashv R$ has counit ϵ . Explicitly, for $h : Y \rightarrow Z$, the morphism $R(h) : R_Y \rightarrow R_Z$ is uniquely determined by

$$\epsilon_Z \circ T(R(h)) = h \circ \epsilon_Y. \tag{2}$$

For $T_E = - \otimes E$, this says that evaluators $\epsilon_Y : R_Y \otimes E \rightarrow Y$ with unique encodings for every ancillary test object are equivalent to an environmental right adjoint.

Proof. If $T \dashv R$, take $R_Y = R(Y)$ and let $\epsilon_Y : T(R_Y) \rightarrow Y$ be the counit. Inverse transposition is

$$g \longmapsto \epsilon_Y \circ T(g),$$

so the adjunction bijection makes every (R_Y, ϵ_Y) universal.

Conversely, choose a universal evaluator for each Y . For a morphism $h : Y \rightarrow Z$, universality of (R_Z, ϵ_Z) at $X = R_Y$ gives a unique morphism $R(h) : R_Y \rightarrow R_Z$ satisfying (2). For $h = \text{id}_Y$, the identity on R_Y satisfies this equation, so uniqueness gives

$$R(\text{id}_Y) = \text{id}_{R_Y}.$$

If $h : Y \rightarrow Z$ and $k : Z \rightarrow W$, functoriality of T and the defining equations give

$$\begin{aligned}\epsilon_W \circ T(R(k) \circ R(h)) &= \epsilon_W \circ T(R(k)) \circ T(R(h)) \\ &= k \circ \epsilon_Z \circ T(R(h)) \\ &= k \circ h \circ \epsilon_Y.\end{aligned}$$

The morphism $R(k \circ h)$ is characterized by the same equation; hence

$$R(k \circ h) = R(k) \circ R(h).$$

Thus R is a functor.

The universal bijections

$$\Psi_{X,Y} : \mathcal{C}(X, RY) \xrightarrow{\cong} \mathcal{D}(TX, Y)$$

are natural in X : if $a : X' \rightarrow X$, then

$$\Psi_{X',Y}(g \circ a) = \epsilon_Y \circ T(g) \circ T(a) = \Psi_{X,Y}(g) \circ T(a).$$

They are natural in Y by (2): for $h : Y \rightarrow Z$,

$$\begin{aligned}\Psi_{X,Z}(R(h) \circ g) &= \epsilon_Z \circ T(R(h)) \circ T(g) \\ &= h \circ \epsilon_Y \circ T(g) \\ &= h \circ \Psi_{X,Y}(g).\end{aligned}$$

Their inverses therefore give natural bijections

$$\mathcal{D}(TX, Y) \cong \mathcal{C}(X, RY),$$

which exhibit $T \dashv R$. Equation (2) also shows that any functor action with counit ϵ is forced by universality, proving uniqueness. $\square$

Proposition 5.7 (Affine storage retracts and adjunctions). Let $\mathcal{C}$ be a category with convex hom-sets, and assume that $T_E = - \otimes E$ acts affinely on hom-sets. Fix objects Y, Z and write $\Omega_Z = \mathcal{C}(I, Z)$. Suppose there are affine maps

$$\sigma_Y : \mathcal{C}(E, Y) \rightarrow \Omega_Z, \quad \rho_Y : \Omega_Z \rightarrow \mathcal{C}(E, Y).$$

Then:

- (a) If $\rho_Y \circ \sigma_Y = \text{id}_{\mathcal{C}(E, Y)}$, then $\mathcal{C}(E, Y)$ is affinely isomorphic to an affine retract of Ω_Z .
- (b) If, in addition, σ_Y is surjective, then σ_Y and ρ_Y are inverse affine bijections, so $\mathcal{C}(E, Y) \cong \Omega_Z$. This supplies only the tensor-unit component of a possible adjunction.
- (c) A functor $R : \mathcal{C} \rightarrow \mathcal{C}$ is right adjoint to T_E precisely when there are bijections

$$\Phi_{X,Y} : \mathcal{C}(X \otimes E, Y) \xrightarrow{\cong} \mathcal{C}(X, RY)$$

natural in X and Y . Under the hypotheses of Proposition 2.3, these bijections are automatically affine.

Proof. For (a), set

$$r_Y = \sigma_Y \circ \rho_Y : \Omega_Z \rightarrow \Omega_Z.$$

Then

$$r_Y^2 = \sigma_Y \rho_Y \sigma_Y \rho_Y = \sigma_Y (\rho_Y \sigma_Y) \rho_Y = \sigma_Y \rho_Y = r_Y.$$

Thus r_Y is an affine idempotent. Let

$$S_Y = \sigma_Y(\mathcal{C}(E, Y)).$$

For every $z \in \Omega_Z$, one has $r_Y(z) = \sigma_Y(\rho_Y(z)) \in S_Y$, so $\text{im}(r_Y) \subseteq S_Y$. Conversely, for $s = \sigma_Y(x) \in S_Y$,

$$r_Y(s) = \sigma_Y \rho_Y \sigma_Y(x) = \sigma_Y(x) = s,$$

so $S_Y \subseteq \text{im}(r_Y)$. Hence $\text{im}(r_Y) = S_Y$.

The map $\sigma_Y : \mathcal{C}(E, Y) \rightarrow S_Y$ is injective because $\rho_Y \sigma_Y = \text{id}$, and it is surjective by definition of S_Y . Its inverse is $\rho_Y|_{S_Y}$. Both maps are affine, so $\mathcal{C}(E, Y)$ and the affine retract S_Y are affinely isomorphic.

For (b), let $z \in \Omega_Z$. Surjectivity gives $z = \sigma_Y(x)$ for some $x \in \mathcal{C}(E, Y)$. Then

$$\sigma_Y \rho_Y(z) = \sigma_Y \rho_Y \sigma_Y(x) = \sigma_Y(x) = z.$$

Thus $\sigma_Y \rho_Y = \text{id}_{\Omega_Z}$, while $\rho_Y \sigma_Y = \text{id}_{\mathcal{C}(E, Y)}$ by hypothesis. Hence the two affine maps are inverse affine bijections.

For (c), natural hom-set bijections are the standard definition of an adjunction. Explicitly, the unit and counit induced by Φ are

$$\eta_X = \Phi_{X, T_E X}(\text{id}_{T_E X}), \quad \epsilon_Y = \Phi_{R_Y, Y}^{-1}(\text{id}_{R_Y}).$$

Naturality of Φ gives the triangle identities, so $T_E \dashv R$. Conversely, every right adjoint supplies these natural bijections. If T_E and composition are affine, Proposition 2.3 shows that the bijections are affine. $\square$

Remark 5.8 (Programmable processors). The Nielsen–Chuang no-programming theorem concerns exact implementation of distinct unitary channels by a fixed processor and implies orthogonality of the corresponding program states [18]. By contrast, the state-space obstruction concerns natural representability of hom-functors by a right adjoint. The two statements are logically independent. An operational codec-to-adjunction theorem requires encoder–evaluator data that extend to natural bijections for every ancillary test object; the companion work on process storage and evaluation develops such a model [1].

6 Deterministic Quantum Channels: Arithmetic Obstruction

The profile and state-space tests are first applied to ungraded deterministic processes. Quantum channels fail at the arithmetic object-dimension layer and, independently, at the affine extreme-boundary layer. Let **Chan** have nonzero finite-dimensional complex Hilbert spaces as objects and CPTP maps as morphisms [38]; categorical universal properties of quantum channels are studied in [22]. Put $a = d_A$ and $b = d_B$.

Proposition 6.1 (Affine dimension of channels).

$$\dim_{\text{aff}} \mathbf{Chan}(A, B) = a^2(b^2 - 1).$$

Proof. Under the Choi isomorphism [13, 38], Hermiticity-preserving linear maps $A \rightarrow B$ correspond to Hermitian operators on $B \otimes A^*$. This real vector space has dimension

$$\dim_{\mathbb{R}} \text{Herm}(B \otimes A^*) = (ab)^2 = a^2 b^2.$$

Trace preservation is the affine equation

$$\text{Tr}_B J = I_A.$$

The linear map

$$\mathrm{Tr}_B : \mathrm{Herm}(B \otimes A^*) \rightarrow \mathrm{Herm}(A)$$

is surjective: for every $H \in \mathrm{Herm}(A)$,

$$\mathrm{Tr}_B \left(\frac{I_B}{b} \otimes H \right) = H.$$

Its rank is therefore a^2 , and rank-nullity gives

$$\dim \ker \mathrm{Tr}_B = a^2 b^2 - a^2.$$

Every nonempty fiber of a linear map is a translate of its kernel, so the trace-preserving affine fiber has this dimension.

It remains to determine the affine hull after imposing positivity. The operator

$$J_0 = \frac{I_B}{b} \otimes I_A$$

is positive definite and satisfies $\mathrm{Tr}_B J_0 = I_A$. Positive definiteness is open in the Hermitian topology, so a sufficiently small neighborhood of J_0 within the trace-preserving affine fiber remains positive. Hence the CPTP Choi set has nonempty relative interior in that fiber and has the same affine hull. Its affine dimension is therefore $a^2(b^2 - 1)$. $\square$

The profile is

$$p_B = (b^2 - 1)f, \quad f(A) = a^2,$$

and $T_E^* f = e^2 f$, where $e = d_E$. If a right adjoint existed, automatic affinity and Theorem 3.1 would require an integer $r = d_{R(B)}$ with

$$r^2 = e^2(b^2 - 1) + 1. \tag{3}$$

This is also the object-dimension condition obtained from Theorem 5.2, since the Hermitian state spaces of E and B have real dimensions e^2 and b^2 .

Theorem 6.2 (Complete arithmetic classification). Fix an integer $e > 1$. Positive integer solutions (b, r) of (3) correspond exactly to ordered factor pairs (u, v) satisfying

$$1 \leq u < v, \quad uv = e^2 - 1,$$

and

$$u + v \equiv 0 \pmod{2e}, \quad v - u \equiv 0 \pmod{2},$$

through

$$b = \frac{u + v}{2e}, \quad r = \frac{v - u}{2}.$$

Every solution satisfies

$$b \leq \frac{e}{2}.$$

In particular, $b = e$ is never a solution. The bound is sharp for every even $e = 2k$, for which

$$(b, r) = (k, 2k^2 - 1)$$

is a solution.

Proof. Equation (3) is

$$(eb)^2 - r^2 = e^2 - 1.$$

Since $e > 1$, $r < eb$. Set $u = eb - r$ and $v = eb + r$. Since $r > 0$, $1 \leq u < v$; moreover $uv = e^2 - 1$, and the displayed reconstruction formulas and congruences follow. Conversely, every such factor pair reconstructs a solution.

For positive integers u, v ,

$$u + v \leq uv + 1$$

because $(u - 1)(v - 1) \geq 0$. Therefore

$$2eb = u + v \leq uv + 1 = e^2,$$

which gives $b \leq e/2$. Hence $b = e$ is impossible. Finally, if $e = 2k$, then

$$e^2(k^2 - 1) + 1 = 4k^2(k^2 - 1) + 1 = (2k^2 - 1)^2.$$

□

Theorem 6.3 (Affine channel-state obstruction). Let E and B be quantum systems with $e = d_E > 1$ and $b = d_B > 1$. The set of extreme points of $\mathbf{Chan}(E, B)$ is not closed. Consequently, $\mathbf{Chan}(E, B)$ is not affinely isomorphic to the state space $\mathbf{Chan}(I, R)$ of any quantum object R .

Proof. Choose orthonormal vectors $f_0, f_1 \in B$ and an orthonormal basis $\{e_i\}_{i=1}^e$ of E . For $t \geq 0$, define unit vectors

$$v_1(t) = f_0, \quad v_2(t) = \frac{tf_0 + f_1}{\sqrt{1+t^2}}, \quad v_i(t) = \frac{f_0 + f_1}{\sqrt{2}} \quad (3 \leq i \leq e),$$

where the third family is absent when $e = 2$. Put

$$K_i(t) = |v_i(t)\rangle\langle e_i| : E \longrightarrow B.$$

Since every $v_i(t)$ is a unit vector,

$$\sum_{i=1}^e K_i(t)^\dagger K_i(t) = \sum_{i=1}^e |e_i\rangle\langle e_i| = I_E.$$

Hence

$$\Phi_t(X) = \sum_{i=1}^e K_i(t) X K_i(t)^\dagger$$

is a channel. The Kraus operators are linearly independent: if $\sum_i \alpha_i K_i(t) = 0$, evaluation at e_j gives $\alpha_j v_j(t) = 0$, and therefore $\alpha_j = 0$ for every j . Thus the Kraus representation is minimal.

The Choi extremality criterion for channels states that a channel with a minimal Kraus family $\{K_i\}$ is extreme if and only if the operators $\{K_i^\dagger K_j\}_{i,j}$ are linearly independent [13, 38]. In the present family,

$$K_i(t)^\dagger K_j(t) = \langle v_i(t), v_j(t) \rangle |e_i\rangle\langle e_j|.$$

For $t > 0$, every inner product in this formula is nonzero. The resulting operators are nonzero scalar multiples of the matrix units and are therefore linearly independent. Thus Φ_t is extreme for every $t > 0$. At $t = 0$,

$$\langle v_1(0), v_2(0) \rangle = \langle f_0, f_1 \rangle = 0,$$

so the Kraus-product family contains the zero operator and is not linearly independent. The same criterion shows that Φ_0 is not extreme. Since $\Phi_t \rightarrow \Phi_0$ in the finite-dimensional Choi topology as $t \downarrow 0$, the extreme-point set of $\mathbf{Chan}(E, B)$ is not closed.

It remains to compare quantum state spaces. The extreme points of $\mathbf{Chan}(I, R)$ are exactly the rank-one density operators. Indeed, a density operator of rank at least two can be perturbed with opposite signs along two of its positive eigenspaces while preserving positivity and trace, so it is not extreme. Conversely, if a rank-one density operator is a nontrivial convex combination of two density operators, positivity forces both summands to have support in the same one-dimensional subspace, and the trace condition then makes both summands equal to the original state. The rank-one density operators form the continuous image of the unit sphere under $\psi \mapsto |\psi\rangle\langle\psi|$ and hence form a compact, in particular closed, set.

An affine isomorphism between finite-dimensional compact convex sets extends to an affine isomorphism of their affine hulls by Lemma 2.1; it is therefore a homeomorphism. It also maps extreme points bijectively, because an affine decomposition transports in both directions. Such a homeomorphism cannot map a nonclosed extreme-point set onto a closed one. Hence no affine isomorphism $\mathbf{Chan}(E, B) \cong \mathbf{Chan}(I, R)$ exists. $\square$

Corollary 6.4 (Complete affine channel-state classification). Fix $e = d_E > 1$. For quantum objects B and R , there is an affine isomorphism

$$\mathbf{Chan}(E, B) \cong \mathbf{Chan}(I, R)$$

if and only if $d_B = d_R = 1$; in that case both convex sets are singletons.

Proof. If $d_B > 1$, Theorem 6.3 excludes such an affine isomorphism for every R . If $d_B = 1$, there is exactly one channel $E \rightarrow B$, namely discarding. The quantum state space $\mathbf{Chan}(I, R)$ is a singleton exactly when $d_R = 1$. This proves both directions. $\square$

Example 6.5 (Individual realizable profiles). The dimension equation can hold for particular codomains even when the environment is nontrivial. For example, if $e = 4$ and $b = 2$, then

$$e^2(b^2 - 1) + 1 = 4^2(2^2 - 1) + 1 = 49 = 7^2.$$

Thus a 7-dimensional candidate has the required affine dimension for this single codomain. Despite this equality, Corollary 6.4 shows that no quantum state space is affinely isomorphic to $\mathbf{Chan}(E, B)$. A right adjoint must also satisfy the dimension equation for every codomain; the choice $B = E$ is excluded by Theorem 6.2.

Corollary 6.6 (No environmental right adjoint for quantum channels). For every quantum environment E with $d_E > 1$, the functor $-\otimes E : \mathbf{Chan} \rightarrow \mathbf{Chan}$ has no ordinary right adjoint.

Proof. Suppose that $-\otimes E$ had an ordinary right adjoint R . Tensoring and composition are affine in $\mathbf{Chan}$, so Proposition 2.3 makes the adjunction transpositions affine. The dimension-profile theorem then gives, for every codomain B ,

$$d_{R(B)}^2 = e^2(b^2 - 1) + 1.$$

Set $B = E$, so $b = e$. Theorem 6.2 shows that every positive integer solution of this equation satisfies $b \leq e/2$. The equality $b = e$ is incompatible with that bound for $e > 1$. Equivalently, Theorem 5.2 assigns the causal internal-hom candidate the dimension

$$\dim W(E, E) = e^2(e^2 - 1) + 1 = e^4 - e^2 + 1,$$

whereas the Hermitian operator space of a quantum object has dimension r^2 . The arithmetic classification shows that no such r exists. Therefore the right adjoint cannot exist. $\square$

Corollary 6.7 (Exact arithmetic channel defect). Give the one-dimensional profile space the coefficient norm $\|\alpha f\| = |\alpha|$. For $B = E$ and $e > 1$,

$$\text{dist}(T_{EP}^*, \mathfrak{P}_{\mathbf{Chan}}) = e^2 - 1.$$

Proof. The required square is $N = e^4 - e^2 + 1$. It lies between the consecutive squares $(e^2 - 1)^2$ and e^4 , with gaps

$$N - (e^2 - 1)^2 = e^2, \quad e^4 - N = e^2 - 1.$$

The second is smaller. Since channel profile coefficients are $m^2 - 1$, the coefficient distance equals the distance from N to the nearest square. $\square$

7 Deterministic Classical Channels: Polytopal Obstruction

The classical case separates dimension from convex shape. Unlike quantum channels, transformed affine dimensions are always realizable; the obstruction appears only when the full state polytope is compared. Let **FinStoch** have nonempty finite sets as objects and stochastic matrices as morphisms; this is the standard finite Markov category [12, 37]. If $|A| = a$ and $|B| = b$, then

$$\mathbf{FinStoch}(A, B) \cong (\Delta_{b-1})^a, \quad \dim_{\text{aff}} \mathbf{FinStoch}(A, B) = a(b-1),$$

where Δ_{b-1} is the standard probability simplex. Its profile is $p_B = (b-1)f$ with $f(A) = a$. For an environment of size e , $T_E^* f = ef$, and the transformed dimension is always realizable by an object of size

$$r = e(b-1) + 1.$$

This is exactly the dimension of $W(E, B)$ from Theorem 5.2. Thus the dimension-profile condition alone does not obstruct currying.

Theorem 7.1 (Classical polytopal obstruction). If $e > 1$, the functor $- \times E : \mathbf{FinStoch} \rightarrow \mathbf{FinStoch}$ has no ordinary right adjoint.

Proof. Choose any B with $b > 1$. By Corollary 5.3 and Proposition 2.3, a right adjoint would give an affine isomorphism

$$\mathbf{FinStoch}(E, B) \cong \mathbf{FinStoch}(I, RB).$$

The left side is $(\Delta_{b-1})^e$. Its dimension is the sum of the dimensions of its e factors, namely $e(b-1)$. A point of a Cartesian product of polytopes is extreme exactly when each component is extreme; since Δ_{b-1} has b vertices, the product has b^e vertices. The right side is the simplex Δ_{r-1} , of dimension $r-1$ and with r vertices.

An affine isomorphism bijects extreme points. Indeed, if the image of an extreme point were a nontrivial convex combination, applying the affine inverse would express the original point as a nontrivial convex combination; the converse follows symmetrically. Dimension preservation therefore forces

$$r = 1 + e(b-1).$$

Vertex preservation forces $r = b^e$. But, for $e, b > 1$,

$$b^e = (1 + (b-1))^e > 1 + e(b-1)$$

by the binomial theorem. This is a contradiction. $\square$

Remark 7.2. The obstruction occurs at the affine face-lattice level even though the transformed dimensions are realizable.

8 Graded Convex Categories and Full Grade Rigidity

For instruments, varying outcome numbers prevent the total hom-set from being a single convex set. This structure is modeled by a graded convex category.

Definition 8.1 (Positive-integer graded convex category). A category $\mathcal{C}$ is graded convex if

$$\mathcal{C}(X, Y) = \coprod_{n \geq 1} \mathcal{C}_n(X, Y),$$

each stratum $\mathcal{C}_n(X, Y)$ is convex, identities lie in grade 1, composition sends grades (m, n) to grade mn and is separately affine on each pair of strata. The grade of f is denoted $\deg f$.

Theorem 8.1 (Full graded-adjunction rigidity). Let $\mathcal{C}$ be a positive-integer graded convex category, let T preserve grades, and suppose $T \dashv R$ as an ordinary adjunction. Assume that for every Y there are an X and a grade-1 map $f : TX \rightarrow Y$. Then:

- (a) $\deg \epsilon_Y = 1$ for every Y ;
- (b) both transposition maps preserve every grade;
- (c) $\deg \eta_X = 1$ for every X ;
- (d) R preserves grades.

If T and composition are affine on strata, every restricted transposition

$$\mathcal{C}_n(TX, Y) \cong \mathcal{C}_n(X, RY)$$

is an affine isomorphism.

Proof. Choose a grade-1 map $f : TX \rightarrow Y$. Surjectivity of inverse transposition gives $g : X \rightarrow RY$ with

$$f = \epsilon_Y \circ T(g).$$

Thus

$$1 = \deg f = (\deg \epsilon_Y)(\deg g).$$

Both factors are positive integers, so both equal 1. This proves (a). It follows that

$$\deg(\epsilon_Y \circ T(g)) = \deg g,$$

so inverse transposition preserves grade. Bijectivity gives (b).

The triangle identity

$$\epsilon_{TX} \circ T(\eta_X) = \text{id}_{TX}$$

then gives $1 = 1 \cdot \deg \eta_X$, proving (c).

For $h : Y \rightarrow Z$, consider

$$h \circ \epsilon_Y : T(RY) \rightarrow Z.$$

Its forward transpose is

$$\begin{aligned} R(h \circ \epsilon_Y) \circ \eta_{RY} &= R(h) \circ R(\epsilon_Y) \circ \eta_{RY} \\ &= R(h), \end{aligned}$$

where the second equality is the triangle identity $R(\epsilon_Y) \circ \eta_{RY} = \text{id}_{RY}$. Since $\deg \epsilon_Y = 1$, the untransposed map has grade

$$\deg(h \circ \epsilon_Y) = \deg h.$$

Part (b) shows that forward transposition preserves grade; hence $\deg R(h) = \deg h$. This proves (d).

On a fixed stratum, inverse transposition is affine because it is $g \mapsto \epsilon_Y \circ T(g)$. Its inverse is affine by Lemma 2.2. $\square$

Corollary 8.2 (Graded state-space representability). Let $\mathcal{C}$ have tensor unit I , let $T_E = - \otimes E$ preserve grades and be grade-wise affine, and suppose T_E has an ordinary right adjoint R . If every object Y admits a grade-1 map $E \rightarrow Y$, then for every n there is an affine bijection

$$\mathcal{C}_n(E, Y) \cong \mathcal{C}_n(I, RY),$$

where $E \cong I \otimes E$ through the left unitor.

Proof. The grade-1 maps $E = T_E I \rightarrow Y$ verify the hypothesis of Theorem 8.1; hence transposition preserves every grade. Setting $X = I$ in the adjunction restricts the hom-bijection to the displayed strata. The last assertion of Theorem 8.1 makes this restricted bijection affine. $\square$

Remark 8.3 (Outcome grade and causality). Grade 1 denotes one formally labelled outcome; causality is defined with respect to a chosen discarding family. Full graded rigidity makes the unit and counit one-outcome deterministic channels, makes R grade preserving, and implies that an instrument adjunction restricts to the grade-1 channel subcategory. These outcome-grade conclusions are independent of the causal hypotheses. If the counit is also normalized, Proposition 14.4(i) shows that inverse transposition preserves causal morphisms. If, in addition, the unit is normalized and R preserves causal morphisms, Proposition 14.4(ii) restricts the grade-preserving adjunction bijections to the causal hom-sets.

9 Quantum and Classical Instruments

A strict convention for outcome labels is fixed before treating the quantum and classical cases.

Definition 9.1 (Strict formal outcome labels). An n -outcome instrument carries the finite ordinal $[n] = \{1, \dots, n\}$ as part of its morphism data. If an m -outcome instrument is followed by an n -outcome instrument, the composite label $(j, i) \in [n] \times [m]$ is flattened to

$$(j - 1)m + i \in [nm].$$

For three instruments this gives $(k - 1)nm + (j - 1)m + i$ independently of parenthesization, so composition of labels is strictly associative and outcome cardinality is multiplicative. Zero CP maps are permitted. Padding along an injection of finite ordinals inserts zero maps on the new labels and produces a distinct morphism; it is not an identification of grades.

9.1 Quantum instruments

An n -outcome quantum instrument $A \rightarrow B$ is an n -tuple of CP maps whose sum is trace preserving, in the finite-outcome specialization of the Davies–Lewis–Ozawa instrument formalism [16, 17]. If $\Phi = (\Phi_i)_{i=1}^m$ and $\Psi = (\Psi_j)_{j=1}^n$ are composable, their composite has branches $\Psi_j \circ \Phi_i$ with the flattened labels fixed above. The one-outcome identity instrument supplies identities. Thus the formally labelled instruments form a positive-integer graded convex category, and tensoring by a system acts branchwise without changing the grade.

Proposition 9.1 (Quantum-instrument affine dimension). For nonzero finite-dimensional Hilbert spaces,

$$\dim_{\text{aff}} \mathbf{Instr}_n(A, B) = na^2b^2 - a^2 = a^2(nb^2 - 1).$$

Proof. Each Choi matrix belongs to the a^2b^2 -dimensional real vector space $\text{Herm}(B \otimes A^*)$. Thus the ambient real vector space of n -tuples has dimension na^2b^2 . Define

$$Q(J_1, \dots, J_n) = \sum_{j=1}^n \text{Tr}_B J_j.$$

This map is surjective onto $\text{Herm}(A)$. Indeed, for any $H \in \text{Herm}(A)$, the tuple

$$\left(\frac{I_B}{b} \otimes H, 0, \dots, 0\right)$$

is mapped to H . Consequently, $\text{rank } Q = a^2$ and

$$\dim \ker Q = na^2b^2 - a^2.$$

The normalization equation for an instrument is $Q(J_1, \dots, J_n) = I_A$; its solution set is a nonempty affine translate of $\ker Q$ and therefore has dimension $na^2b^2 - a^2$.

Finally, the tuple

$$J_j = \frac{1}{nb} I_B \otimes I_A, \quad j = 1, \dots, n,$$

is positive definite and satisfies the normalization equation. Positive definiteness is open, so a relative neighborhood of this tuple inside the affine fiber remains positive. The instrument set therefore has the entire normalization fiber as its affine hull, proving the formula. $\square$

Tensoring by E preserves outcome labels. Every relevant hom-set contains a grade-1 replacement channel, so Theorem 8.1 applies to an ordinary adjunction.

Theorem 9.2 (Quantum-instrument environmental no-go). If $d_E > 1$, the grade-preserving functor $- \otimes E : \mathbf{Instr} \rightarrow \mathbf{Instr}$ has no ordinary right adjoint.

Proof. Let $b = d_B$ and $r = d_{R(B)}$. A grade-1 replacement channel $E \rightarrow Y$ exists for every quantum codomain Y , so Corollary 8.2 gives

$$\dim_{\text{aff}} \mathbf{Instr}_n(E, B) = \dim_{\text{aff}} \mathbf{Instr}_n(I, RB)$$

for every n . Thus

$$e^2(nb^2 - 1) = nr^2 - 1$$

for every positive integer n . At $n = 1$ and $n = 2$ this gives

$$e^2(b^2 - 1) = r^2 - 1, \quad e^2(2b^2 - 1) = 2r^2 - 1.$$

Subtracting the first equation from the second yields $e^2b^2 = r^2$. Substitution into the first equation then gives

$$e^2b^2 - e^2 = e^2b^2 - 1,$$

so $e^2 = 1$, contrary to $e > 1$. $\square$

Equivalently, define

$$f_1(A, n) = na^2, \quad f_2(A, n) = -a^2.$$

Then

$$p_B = b^2 f_1 + f_2,$$

and the coefficient functional of f_2 has constant value 1 on every profile but is rescaled by e^2 under environmental pullback. This is the affine-hyperplane form of the same proof.

9.2 Classical instruments

An n -outcome classical instrument $A \rightarrow B$ is an n -tuple of substochastic matrices whose sum is stochastic. Branchwise matrix composition with the same flattened labels, together with the one-outcome identity, gives the corresponding positive-integer graded convex category.

Proposition 9.3 (Classical-instrument affine dimension). For nonempty finite sets,

$$\dim_{\text{aff}} \mathbf{Instr}_n^{\text{cl}}(A, B) = nab - a = a(nb - 1).$$

Proof. Index an instrument by input $x \in A$, output $y \in B$, and outcome $j \in [n]$. Its ambient real vector space has one coordinate $p(j, y \mid x)$ for each triple and therefore has dimension nab . Define the linear normalization map

$$Q : \mathbb{R}^{nab} \rightarrow \mathbb{R}^a, \quad Q(p)_x = \sum_{j=1}^n \sum_{y \in B} p(j, y \mid x).$$

The map is surjective: for each input x , a vector supported on one coordinate with that input maps to the x th standard basis vector of $\mathbb{R}^a$. Hence $\text{rank } Q = a$ and

$$\dim \ker Q = nab - a.$$

The stochastic normalization equations are $Q(p) = (1, \dots, 1)$, so their solution set is an affine translate of $\ker Q$ of dimension $nab - a$.

The point

$$p(j, y \mid x) = \frac{1}{nb}$$

is strictly positive and satisfies every normalization equation. A sufficiently small relative neighborhood of this point in the affine fiber remains nonnegative. Thus imposing nonnegativity does not reduce the affine hull, and the affine dimension is $a(nb - 1)$. $\square$

Theorem 9.4 (Classical-instrument environmental no-go). If $|E| > 1$, Cartesian product by E has no ordinary right adjoint in the formally labelled classical-instrument category.

Proof. Let $e = |E|$, $b = |B|$, and $r = |R(B)|$. A grade-1 replacement stochastic channel $E \rightarrow Y$ exists for every nonempty finite codomain Y . Corollary 8.2 therefore gives affine bijections on every grade and hence

$$e(nb - 1) = nr - 1$$

for every positive integer n . At $n = 1$ and $n = 2$, respectively,

$$e(b - 1) = r - 1, \quad e(2b - 1) = 2r - 1.$$

Subtracting the first equation from the second gives $eb = r$. Substituting this value into $e(b - 1) = r - 1$ gives

$$eb - e = eb - 1,$$

so $e = 1$, contrary to $|E| > 1$. $\square$

The same obstruction has the affine-hyperplane form of Corollary 3.2. Define, on the test pairs (A, n) ,

$$f_1(A, n) = n|A|, \quad f_2(A, n) = -|A|.$$

Then

$$p_B = |B|f_1 + f_2.$$

The functions f_1, f_2 are linearly independent because n can vary while A is fixed. If ℓ extracts the f_2 coefficient, then

$$\ell(p_B) = 1$$

for every codomain B . Cartesian product with an environment of cardinality e gives

$$T_E^* f_1 = e f_1, \quad T_E^* f_2 = e f_2,$$

so $\ell \circ T_E^* = e\ell$. The affine-hyperplane corollary therefore forces $e = 1$.

9.3 Grade-by-grade isolated representability

The instrument no-go theorems use all outcome grades simultaneously. This subsection classifies when a single grade can be represented in isolation and identifies the compatibility condition preventing isolated representations from assembling into a graded adjunction. In the quantum statements write

$$e = d_E, \quad b = d_B, \quad r = d_{R(B)}.$$

In the classical statements, the same letters denote the corresponding set cardinalities.

Proposition 9.5 (Quantum fixed-grade dimension classification). Fix integers $e > 1$ and $n \geq 1$. Positive integer solutions (b, r) of

$$e^2(nb^2 - 1) = nr^2 - 1 \tag{4}$$

exist only if $n \mid e^2 - 1$. When $n \mid e^2 - 1$, put

$$k = \frac{e^2 - 1}{n}.$$

Then solutions correspond bijectively to ordered factor pairs (u, v) satisfying

$$1 \leq u < v, \quad uv = k, \quad u \equiv v \pmod{2}, \quad u + v \equiv 0 \pmod{2e},$$

through

$$b = \frac{u + v}{2e}, \quad r = \frac{v - u}{2}.$$

Every solution satisfies

$$b \leq \frac{k + 1}{2e} = \frac{e^2 + n - 1}{2en}.$$

Proof. Equation (4) is equivalent to

$$e^2 nb^2 - e^2 = nr^2 - 1,$$

and therefore to

$$n(e^2 b^2 - r^2) = e^2 - 1.$$

Thus $n \mid e^2 - 1$. With $k = (e^2 - 1)/n$, factorization gives

$$(eb - r)(eb + r) = k.$$

Set

$$u = eb - r, \quad v = eb + r.$$

Since $r > 0$, $u < v$. Since $uv = k > 0$ and $v > 0$, also $u > 0$. Moreover,

$$u + v = 2eb, \quad v - u = 2r,$$

which gives the parity and divisibility conditions.

Conversely, suppose (u, v) satisfies the stated conditions and define

$$b = \frac{u+v}{2e}, \quad r = \frac{v-u}{2}.$$

The congruence condition makes b integral, the parity condition makes r integral, and $1 \leq u < v$ makes both positive. Finally,

$$(eb - r)(eb + r) = uv = \frac{e^2 - 1}{n},$$

which rearranges to (4).

For the bound, $(u-1)(v-1) \geq 0$ gives

$$u + v \leq uv + 1 = k + 1.$$

Since $u + v = 2eb$, one obtains

$$2eb \leq k + 1,$$

which is the stated inequality. $\square$

Corollary 9.6 (Grade-assembly obstruction). Let $e > 1$. No single quantum object dimension r can represent two distinct grades $n \neq m$ for the same codomain dimension b at the state-space dimension level.

Proof. If the same b, r satisfy (4) at grades n and m , then

$$e^2 b^2 - r^2 = \frac{e^2 - 1}{n}$$

and

$$e^2 b^2 - r^2 = \frac{e^2 - 1}{m}.$$

Since $e^2 - 1 > 0$, equality gives $1/n = 1/m$ and hence $n = m$, a contradiction. By Corollary 8.2, a graded adjunction would use the single object $R(B)$ at every grade, so the isolated representations cannot assemble into a graded adjunction. $\square$

Remark 9.7 (Relation to the channel classification). At $n = 1$, Proposition 9.5 reduces to Theorem 6.2. For $n > 1$, its bound is strictly stronger because

$$\frac{e^2 + n - 1}{2en} < \frac{e}{2}$$

is equivalent to $(n-1)(e^2-1) > 0$. If $n > e^2 - 1$, the divisibility condition fails. If $n = e^2 - 1$, then $k = 1$ and the only factor pair has $u = v = 1$, which would give $r = 0$; hence no positive-dimensional solution exists.

The proposition classifies dimension compatibility for an isolated quantum grade. The following theorem gives the complete affine classification.

Theorem 9.8 (Quantum fixed-grade affine obstruction). Let $e = d_E > 1$, $n \geq 1$, $b = d_B \geq 1$, and $r = d_R \geq 1$. There is an affine isomorphism

$$\mathbf{Instr}_n(E, B) \cong \mathbf{Instr}_n(I, R)$$

if and only if $n = b = r = 1$. In that case both convex sets are singletons. Thus every nontrivial fixed-grade quantum process space fails the affine state-space test, including every nontrivial solution of (4).

Proof. First consider the grade- n state space

$$\mathcal{S}_{n,r} = \mathbf{Instr}_n(I, R) = \left\{ (\rho_1, \dots, \rho_n) : \rho_j \geq 0, \sum_{j=1}^n \text{Tr } \rho_j = 1 \right\}.$$

Its extreme points are exactly the tuples having one rank-one density operator in one component and zero in every other component. To prove this, suppose first that at least two components are nonzero. Choose a nonzero component ρ_j whose trace s satisfies $0 < s < 1$. Then the tuple is the nontrivial convex combination, with weights s and $1 - s$, of the normalized tuple supported only in component j and the normalized tuple formed by all remaining components. It is therefore not extreme. If exactly one component is nonzero, extremality reduces to extremality in the ordinary quantum state space, whose extreme points are precisely the rank-one density operators, as proved in Theorem 6.3. Consequently,

$$\text{Ext}(\mathcal{S}_{n,r}) \cong \prod_{j=1}^n \mathbb{CP}^{r-1}.$$

This is a closed set. If $r > 1$, it has no isolated points; if $r = 1$, it consists of exactly n isolated points.

Suppose next that $b > 1$. Theorem 6.3 supplies extreme channels $\Phi_t : E \rightarrow B$ for $t > 0$ converging to a nonextreme channel Φ_0 . Embed a channel as the first component of an instrument:

$$j(\Phi) = (\Phi, 0, \dots, 0).$$

If Φ is extreme among channels, then $j(\Phi)$ is extreme among instruments. Indeed, in any convex decomposition of $j(\Phi)$, positivity of the Choi operators forces every component numbered $2, \dots, n$ in both summands to vanish. The first components of the summands are then channels, and extremality of Φ makes them equal to Φ . Conversely, a nontrivial channel decomposition of Φ gives a nontrivial instrument decomposition of $j(\Phi)$. Hence $j(\Phi_t)$ are extreme instruments converging to the nonextreme instrument $j(\Phi_0)$. The extreme-point set of $\mathbf{Instr}_n(E, B)$ is therefore not closed. It cannot be affinely isomorphic to $\mathcal{S}_{n,r}$, whose extreme-point set is closed, because an affine isomorphism is a homeomorphism and preserves extreme points.

It remains to consider $b = 1$. Then $\mathbf{Instr}_n(E, B)$ is the convex set of n -outcome POVMs

$$\mathcal{P}_{n,e} = \{(M_1, \dots, M_n) : M_j \geq 0, \sum_j M_j = I_E\}.$$

For $n > 1$, the deterministic POVM

$$P_1 = (I_E, 0, \dots, 0)$$

is an isolated point of $\text{Ext}(\mathcal{P}_{n,e})$. It is extreme because positivity forces every zero component to remain zero in any convex decomposition. To prove isolation, let $(M_1, \dots, M_n)$ be an extreme POVM satisfying

$$\|M_1 - I_E\|_{\text{op}} < 1.$$

Then M_1 is positive definite. If some $M_j \neq 0$ with $j > 1$, choose $0 < s < 1$ sufficiently small that $M_1 - sM_j \geq 0$. Replacing the pair (M_1, M_j) by

$$(M_1 + sM_j, (1 - s)M_j) \quad \text{and} \quad (M_1 - sM_j, (1 + s)M_j)$$

produces two distinct POVMs whose midpoint is the original one, contrary to extremality. Hence all M_j with $j > 1$ vanish, and normalization gives $M_1 = I_E$. This proves that P_1 is isolated among the extreme points.

If $r > 1$, the state-space extreme set has no isolated points, so no affine isomorphism is possible. If $r = 1$, affine dimensions give

$$\dim_{\text{aff}} \mathcal{P}_{n,e} = e^2(n-1) > n-1 = \dim_{\text{aff}} \mathcal{S}_{n,1}$$

for $e > 1$ and $n > 1$, again excluding an affine isomorphism. Finally, when $n = b = 1$, the process space is a singleton, while $\mathcal{S}_{1,r}$ is a singleton exactly when $r = 1$. This establishes the claimed equivalence. $\square$

Proposition 9.9 (Classical fixed-grade dimension classification). Fix integers $e > 1$ and $n \geq 1$. Positive integer solutions (b, r) of

$$e(nb - 1) = nr - 1 \tag{5}$$

exist if and only if $n \mid e - 1$. In that case, for every $b \geq 1$,

$$r = eb - \frac{e-1}{n}$$

is a positive integer solution.

Proof. Equation (5) is equivalent to

$$enb - e = nr - 1,$$

and hence to

$$n(eb - r) = e - 1.$$

Thus $n \mid e - 1$ is necessary. Conversely, if $n \mid e - 1$, define

$$r = eb - \frac{e-1}{n}.$$

This is an integer, and since $b \geq 1$ and $(e-1)/n \leq e-1$,

$$r \geq e - (e-1) = 1.$$

Substitution verifies (5). $\square$

Theorem 9.10 (Classical fixed-grade affine obstruction). Let $e > 1$. A grade- n classical instrument process space $\mathbf{Instr}_n^{\text{cl}}(E, B)$ is affinely isomorphic to a grade- n classical state space if and only if $n = b = 1$. In that case $r = 1$ and both spaces are singletons. For every $nb > 1$, no affine state-space representation exists.

Proof. For each of the e inputs, an instrument chooses a probability distribution over the nb outcome-output pairs. Therefore

$$\mathbf{Instr}_n^{\text{cl}}(E, B) \cong (\Delta_{nb-1})^e.$$

This polytope has affine dimension $e(nb-1)$ and $(nb)^e$ vertices. A grade- n state on an r -point object is the simplex Δ_{nr-1} , with dimension $nr-1$ and nr vertices.

An affine isomorphism preserves dimensions and extreme points. Dimension equality gives

$$nr = 1 + e(nb-1),$$

while equality of vertex counts gives

$$nr = (nb)^e.$$

Consequently,

$$(nb)^e = 1 + e(nb-1).$$

If $nb > 1$, then the binomial theorem gives

$$(nb)^e = (1 + (nb - 1))^e > 1 + e(nb - 1),$$

a contradiction. Hence $nb = 1$, which forces $n = b = 1$. The dimension equation then gives $r = 1$, and both process spaces are singletons. Conversely, those singleton spaces are affinely isomorphic. $\square$

Remark 9.11 (Consequences for grade assembly). Isolated quantum grades can pass the dimension test only under the arithmetic conditions of Proposition 9.5, and no two distinct grades can use the same representing dimension. Theorem 9.8 shows more: every nontrivial isolated quantum grade fails at the affine level. The classical analogue fails by vertex counts. Thus the quantum and classical graded adjunctions are already obstructed grade by grade, while Corollary 9.6 gives the additional incompatibility of the quantum dimension equations across grades.

10 Deterministic Comb States

This section concerns the convex set of deterministic comb states with a fixed causal order. Affine and semidefinite characterizations of channels, instruments, combs, and higher-order maps are developed systematically in [23]. The categorical superchannel construction is treated separately in the following section.

10.1 Recursive dimension

The recursive Choi constraints used below are standard in the theory of quantum combs and higher-order quantum maps [32, 21]. Let

$$\mathbf{C} = ((A_1, B_1), \dots, (A_N, B_N)),$$

where $a_k = d_{A_k}$ and $b_k = d_{B_k}$. Put

$$P_0 = 1, \quad P_k = \prod_{i=1}^k a_i^2 b_i^2.$$

Let $\mathbf{Comb}(\mathbf{C})$ be the convex set of positive Choi operators R_N for which there are operators $R_{N-1}, \dots, R_0 = 1$ satisfying

$$\mathrm{Tr}_{B_k} R_k = I_{A_k} \otimes R_{k-1} \quad (k = 1, \dots, N), \quad (6)$$

with the slots ordered from innermost to outermost.

Theorem 10.1 (Affine dimension of deterministic comb states). Let

$$D_k = \dim_{\mathrm{aff}} \mathbf{Comb}((A_1, B_1), \dots, (A_k, B_k)), \quad D_0 = 0.$$

Then

$$D_k = P_k - a_k^2 P_{k-1} + D_{k-1} \quad (7)$$

$$= a_k^2 P_{k-1} (b_k^2 - 1) + D_{k-1}, \quad (8)$$

and hence

$$D_N = \sum_{k=1}^N \left(\prod_{i=1}^{k-1} a_i^2 b_i^2 \right) a_k^2 (b_k^2 - 1). \quad (9)$$

Proof. At stage k , the ambient Hermitian space has real dimension P_k . The partial trace in (6) is onto the Hermitian space on A_k and all previous wires, of dimension $a_k^2 P_{k-1}$. For fixed R_{k-1} , its fiber therefore has dimension $P_k - a_k^2 P_{k-1}$. The allowed right-hand sides $I_{A_k} \otimes R_{k-1}$ form an affine set of dimension D_{k-1} , so the preimage of this affine set has dimension

$$P_k - a_k^2 P_{k-1} + D_{k-1}.$$

Positivity does not reduce the affine hull. Starting with $R_0 = 1$, choose recursively

$$R_k = \frac{I_{B_k}}{b_k} \otimes I_{A_k} \otimes R_{k-1}.$$

Every R_k is positive definite and satisfies (6); it is therefore a relative-interior point of the positive feasible set. This proves the recurrence, and iteration proves (9). $\square$

Corollary 10.2 (Prepended innermost slot). Let $L = ((A, B))$ be prepended as the innermost slot of $\mathbf{C}$. Then

$$\dim_{\text{aff}} \mathbf{Comb}(L \star \mathbf{C}) = a^2 b^2 (1 + \dim_{\text{aff}} \mathbf{Comb}(\mathbf{C})) - a^2.$$

Proof. The contribution of L is $a^2(b^2 - 1)$. Every later summand in (9) acquires the preceding factor $a^2 b^2$. Thus

$$D(L \star \mathbf{C}) = a^2(b^2 - 1) + a^2 b^2 D(\mathbf{C}) = a^2 b^2 (1 + D(\mathbf{C})) - a^2.$$

$\square$

Remark 10.3 (Dependence on the higher-order convention). Theorem 10.1 concerns deterministic comb states with one fixed causal order. A higher-order hom-set additionally depends on source dualization, remote-connection conditions, and wire order. The following result isolates the conclusions valid for any separately specified higher-map category with the stated profile; the concrete categorical results thereafter concern deterministic superchannels.

Theorem 10.4 (Conditional arbitrary-order profile obstructions). Let $\mathcal{H}$ be a category with nonempty finite-dimensional convex hom-sets and separately affine composition, and write $D(X, Y) = \dim_{\text{aff}} \mathcal{H}(X, Y)$. Let

$$\mathfrak{S} = \{m^2 : m \in \mathbb{N}_{>0}\},$$

and suppose $\mathcal{H}$ has source objects $X_{u,v}$ indexed by $(u, v) \in \mathfrak{S}^2$ and a class of targets satisfying

$$D(X_{u,v}, Y) = \alpha_Y uv + \beta_Y u - \beta_Y, \quad \alpha_Y, \beta_Y > 0. \quad (10)$$

Let $\Delta : \mathcal{H} \rightarrow \mathcal{H}$ act affinely on hom-sets and suppose that, on the source family,

$$\Delta X_{u,v} = X_{su,tv}, \quad s > 1, \quad t > 0,$$

where $s\mathfrak{S} \subseteq \mathfrak{S}$ and $t\mathfrak{S} \subseteq \mathfrak{S}$. Then:

- (a) Δ has no right adjoint whose values remain in the target profile class of (10).
- (b) Fix a target Y and suppose

$$D(X_{u,v}, \Delta Y) = \alpha_{\Delta Y} uv + s\beta_Y u - s\beta_Y, \quad \alpha_{\Delta Y} > 0.$$

There is no left adjoint $L \dashv \Delta$ for which $LX_{u,v}$ belongs to the same two-parameter family with parameter functions

$$U(u, v) = au + bv + c, \quad V(u, v) = du + gv + f$$

affine on $\mathfrak{S}^2$.

Proof. For (a), suppose $\Delta \dashv R$. Proposition 2.3 makes the adjunction transpositions affine. Thus, for every $(u, v) \in \mathfrak{S}^2$,

$$D(\Delta X_{u,v}, Y) = D(X_{u,v}, RY).$$

Using (10) gives

$$st\alpha_Y uv + s\beta_Y u - \beta_Y = \alpha_{RY} uv + \beta_{RY} u - \beta_{RY}.$$

The two sides are polynomials in u, v that agree on the product of two infinite subsets of $\mathbb{R}$, hence have identical coefficients. The constant terms give $\beta_{RY} = \beta_Y$, while the u -coefficients give $\beta_{RY} = s\beta_Y$. Since $\beta_Y > 0$, this forces $s = 1$, a contradiction.

For (b), suppose $L \dashv \Delta$. Forward transposition is

$$h : LX \rightarrow Y \mapsto \Delta(h) \circ \eta_X : X \rightarrow \Delta Y.$$

It is affine because Δ acts affinely and composition is separately affine. Its inverse is affine by Lemma 2.2; therefore adjunction gives equality of affine dimensions. With $A = \alpha_Y > 0$ and $B = \beta_Y > 0$, this equality is

$$AUV + BU - B = \alpha_{\Delta Y} uv + sBu - sB \tag{11}$$

on $\mathfrak{S}^2$. It is consequently a polynomial identity.

Expanding

$$UV = (au + bv + c)(du + gv + f)$$

shows that the u^2 and v^2 coefficients in the left side of (11) are Aad and Abg . Hence

$$ad = 0, \quad bg = 0.$$

There are four cases.

If $a = b = 0$, then $U = c$, so the left side has zero uv -coefficient, contrary to $\alpha_{\Delta Y} > 0$. If $a = g = 0$, positivity of the uv -coefficient gives $bd \neq 0$. The vanishing v -coefficient is

$$b(Af + B) = 0,$$

so $Af + B = 0$. The constant term on the left is then

$$c(Af + B) - B = -B,$$

whereas the right-hand constant is $-sB$; thus $s = 1$, a contradiction.

If $d = b = 0$, positivity of the uv -coefficient gives $ag \neq 0$. The vanishing v -coefficient is $Acg = 0$, so $c = 0$. The left-hand constant is again $-B$, forcing $s = 1$. Finally, if $d = g = 0$, then $V = f$, so the left side again has zero uv -coefficient, contrary to $\alpha_{\Delta Y} > 0$. Every case contradicts the hypotheses, proving (b). $\square$

11 Deterministic Superchannels

Unlike the comb-state recurrence, the results in this section concern a category whose objects are channel types and whose morphisms transform channels. The category and its environmental endofunctor are specified before the adjoint obstructions are derived.

11.1 Category, Choi representation, and dimension

Definition 11.1 (Superchannel category). The category **SuperChan** is defined as follows.

- (i) An object is a channel type (A, B) , with A, B nonzero finite-dimensional Hilbert spaces.
- (ii) A morphism

$$\Phi : (A, B) \longrightarrow (C, D)$$

is a complex-linear completely positive map

$$\Phi : \mathcal{L}(B \otimes A) \longrightarrow \mathcal{L}(D \otimes C)$$

that sends every deterministic channel Choi operator

$$J \geq 0, \quad \text{Tr}_B J = I_A,$$

to a deterministic channel Choi operator

$$\Phi(J) \geq 0, \quad \text{Tr}_D \Phi(J) = I_C.$$

- (iii) Composition is ordinary composition of linear maps, and the identity on (A, B) is the identity map on $\mathcal{L}(B \otimes A)$.

Lemma 11.1 (**SuperChan** is a convex category). The preceding data define a category. Every hom-set is convex, and composition is separately affine.

Proof. Compositions and identities preserve complete positivity and preserve the set of deterministic channel Choi operators. Associativity and the unit laws are those of ordinary composition of linear maps. Convex combinations of completely positive channel-preserving maps are again completely positive and channel preserving. Finally, composition of linear maps is bilinear and hence separately affine. $\square$

The standard parallel tensor product is

$$(A, B) \otimes (C, D) = (A \otimes C, B \otimes D)$$

on objects and the tensor product of the representing supermaps on morphisms, with the canonical permutation of Choi factors. Its tensor unit is $I = (\mathbb{C}, \mathbb{C})$.

The map-based definition also implies complete channel preservation, as recorded before passing to the Choi representation.

Lemma 11.2 (Complete channel preservation). A morphism of **SuperChan** remains channel preserving after tensoring with the identity superchannel on an arbitrary ancillary channel type $(R_{\text{in}}, R_{\text{out}})$.

Proof. Write $b = d_B$. Let

$$L = \text{Tr}_D \circ \Phi : \mathcal{L}(B \otimes A) \longrightarrow \mathcal{L}(C).$$

The Choi operator

$$J_0 = \frac{I_B}{b} \otimes I_A$$

is positive definite and channel normalized. If $K = K^\dagger$ satisfies $\text{Tr}_B K = 0$, then $J_0 \pm tK$ is positive and channel normalized for all sufficiently small $t > 0$. Hence

$$L(J_0 + tK) = L(J_0 - tK) = I_C,$$

so $L(K) = 0$. This proves annihilation of the Hermitian part of $\ker \text{Tr}_B$; complex linearity extends it to the full complex kernel. Thus L factors through partial trace:

$$L = M \circ \text{Tr}_B, \quad M(H) = L\left(\frac{I_B}{b} \otimes H\right).$$

The preparation map $H \mapsto I_B \otimes H/b$ and L are completely positive, so M is completely positive. Channel preservation gives $M(I_A) = I_C$.

Let J be the Choi operator of a channel on an ancillary extension of the input type. Applying the ampliation of the factorization and tracing the output ancillary wire gives

$$\text{Tr}_{DR_{\text{out}}}[(\Phi \otimes \text{id}_{(R_{\text{in}}, R_{\text{out}})})(J)] = (M \otimes \text{id}_{R_{\text{in}}})(\text{Tr}_{BR_{\text{out}}} J) = I_C \otimes I_{R_{\text{in}}}.$$

Complete positivity of $\Phi \otimes \text{id}_{(R_{\text{in}}, R_{\text{out}})}$ preserves positivity. Therefore the extended output is again a deterministic channel Choi operator. $\square$

The following is the standard finite-dimensional Choi characterization of deterministic superchannels; see [19, 32, 20]. The convention below uses unnormalized Choi operators and the wire order (D, B, A, C) .

Theorem 11.3 (Superchannel Choi characterization). The Choi correspondence gives an affine bijection between morphisms $\Phi : (A, B) \rightarrow (C, D)$ in **SuperChan** and positive operators

$$R \in \text{Pos}(D \otimes B \otimes A \otimes C)$$

for which there is a positive operator $R_{CA} \in \text{Pos}(A \otimes C)$ satisfying

$$\text{Tr}_D R = I_B \otimes R_{CA}, \quad \text{Tr}_A R_{CA} = I_C. \quad (12)$$

Writing $b = d_B$, the auxiliary operator is unique and equals

$$R_{CA} = \frac{1}{b} \text{Tr}_{DB} R.$$

Proof. The equivalence between deterministic superchannels and the positive Choi operators satisfying (12) is the standard superchannel Choi theorem in the cited finite-dimensional framework. It remains only to verify the normalization convention for R_{CA} . Tracing the first equation in (12) over B gives

$$\text{Tr}_{DB} R = \text{Tr}_B(I_B \otimes R_{CA}) = bR_{CA},$$

which proves the displayed formula and uniqueness. $\square$

Theorem 11.4 (Affine dimension of deterministic superchannels). Let $a = d_A$, $b = d_B$, $c = d_C$, and $d = d_D$. Then

$$\dim_{\text{aff}} \mathbf{SuperChan}((A, B), (C, D)) = c^2(a^2 - 1) + c^2 a^2 b^2 (d^2 - 1).$$

Proof. The ambient Hermitian space for R_{CA} has dimension $a^2 c^2$. The map Tr_A is onto the c^2 -dimensional Hermitian space on C , so the fiber $\text{Tr}_A R_{CA} = I_C$ has dimension

$$a^2 c^2 - c^2 = c^2(a^2 - 1).$$

For fixed feasible R_{CA} , the map

$$\text{Tr}_D : \text{Herm}(D \otimes B \otimes A \otimes C) \longrightarrow \text{Herm}(B \otimes A \otimes C)$$

is surjective. Its fiber over $I_B \otimes R_{CA}$ has dimension

$$d^2 b^2 a^2 c^2 - b^2 a^2 c^2 = c^2 a^2 b^2 (d^2 - 1).$$

Adding the dimension of the allowed R_{CA} gives the stated formula.

Positivity does not reduce the affine hull. The operators

$$R_{CA}^0 = \frac{I_A}{a} \otimes I_C, \quad R^0 = \frac{I_D}{d} \otimes I_B \otimes R_{CA}^0$$

are positive definite and satisfy (12). Thus the positive feasible set has nonempty relative interior in the affine solution space. $\square$

11.2 Fixed-type decoration and the superchannel no-go

Fix a channel type $K = (F, G)$ and write

$$k = d_F, \quad \ell = d_G.$$

Definition 11.2 (Fixed-type decoration on superchannels). On objects, define

$$T_K(A, B) = (A \otimes F, B \otimes G).$$

On morphisms, define $T_K(\Phi)$ to be the parallel tensor product with the identity superchannel on K :

$$T_K(\Phi) = \Phi \otimes \text{id}_K,$$

using the canonical permutation of Choi factors. Diagonal environmental extension is the specialization $K = (E, E)$ and is denoted by T_E .

Proposition 11.5 (Fixed-type decoration is an affine endofunctor). For every channel type K , the operation T_K preserves deterministic superchannels, identities, and composition, and it is affine on every hom-set.

Proof. Complete positivity is preserved under tensor products. By Lemma 11.2, tensoring a deterministic superchannel with the identity superchannel on the ancillary type K maps every channel on the product source type to a channel on the product target type. Hence $T_K(\Phi)$ is a morphism of **SuperChan**.

The interchange law gives

$$(\Psi \circ \Phi) \otimes \text{id}_K = (\Psi \otimes \text{id}_K) \circ (\Phi \otimes \text{id}_K),$$

and the tensor product of identity superchannels is the identity on the product channel type. Thus T_K preserves composition and identities. Tensoring a linear map with the fixed map id_K is linear, hence affine on each hom-set. $\square$

Theorem 11.6 (Complete fixed-type right-adjoint classification). Let $K = (F, G)$ be a finite-dimensional channel type. The affine endofunctor

$$T_K : \mathbf{SuperChan} \longrightarrow \mathbf{SuperChan}$$

has an ordinary right adjoint if and only if

$$d_F = d_G = 1.$$

Consequently, every nontrivial input-only, output-only, or diagonal environmental decoration of deterministic superchannels has no ordinary right adjoint.

Proof. If $d_F = d_G = 1$, then K is isomorphic to the tensor unit. Hence T_K is naturally isomorphic to the identity functor and, by transport of the identity adjunction, has a right adjoint.

Conversely, suppose $T_K \dashv \mathcal{R}$. By Proposition 2.3, every adjunction transposition is affine. Fix a target (C, D) and write

$$\mathcal{R}(C, D) = (P, Q), \quad p = d_P, \quad q = d_Q.$$

For every source (A, B) , Theorem 11.4 and preservation of affine dimension give

$$\begin{aligned} c^2(k^2a^2 - 1) + c^2k^2\ell^2a^2b^2(d^2 - 1) \\ = p^2(a^2 - 1) + p^2a^2b^2(q^2 - 1). \end{aligned} \quad (13)$$

The functions 1, a^2 , and a^2b^2 are linearly independent on positive integer pairs (a, b) . Indeed, if

$$\alpha + \beta a^2 + \gamma a^2b^2 = 0$$

for all positive integers a, b , evaluation at $(1, 1)$ and $(1, 2)$ gives $3\gamma = 0$. With $\gamma = 0$, evaluation at $a = 1$ and $a = 2$ gives $3\beta = 0$, and then $\alpha = 0$.

Comparison of the constant and a^2 coefficients in (13) gives

$$p^2 = c^2, \quad p^2 = c^2k^2.$$

Since $c, k > 0$, it follows that $k = 1$. Comparison of the a^2b^2 coefficients then gives

$$q^2 = \ell^2(d^2 - 1) + 1.$$

This integer equation must hold for every target output dimension d . If $\ell > 1$, choose $d = \ell$. The required square becomes

$$q^2 = \ell^4 - \ell^2 + 1,$$

but

$$(\ell^2 - 1)^2 < \ell^4 - \ell^2 + 1 < \ell^4.$$

The two bounding squares are consecutive, so no integer q exists. Hence $\ell = 1$. Thus existence of a right adjoint forces $d_F = d_G = 1$, completing the classification. $\square$

11.3 Tensor-unit state-space components

The superchannel no-go theorem uses all source channel types. The weaker tensor-unit test asks whether

$$\mathbf{SuperChan}(T_E I, Y)$$

can be represented by a superchannel state space for an individual target Y . For the parallel tensor specified above, $T_E I = (E, E)$.

For $Z = (P, Q)$, with $p = d_P$ and $q = d_Q$, Theorem 11.4 gives

$$\dim_{\text{aff}} \Omega_Z = \dim_{\text{aff}} \mathbf{SuperChan}(I, Z) = p^2(q^2 - 1).$$

For $Y = (C, D)$, with $c = d_C$, $d = d_D$, and $e = d_E$,

$$\dim_{\text{aff}} \mathbf{SuperChan}(T_E I, Y) = c^2(e^2 - 1 + e^4(d^2 - 1)).$$

Proposition 11.7 (Complete tensor-unit dimension classification). Fix positive integers $e > 1, c, d$ and put

$$N = c^2(e^2 - 1 + e^4(d^2 - 1)).$$

Positive integer solutions (p, q) of

$$p^2(q^2 - 1) = N \quad (14)$$

correspond bijectively to factor pairs (u, v) satisfying

$$1 \leq u < v, \quad uv = N, \quad u \equiv v \pmod{2}, \quad v - u \mid u + v,$$

through

$$p = \frac{v - u}{2}, \quad q = \frac{u + v}{v - u}.$$

Equivalently, the dimension test passes exactly when N has a square divisor p^2 for which $N/p^2 + 1$ is a perfect square.

Proof. Suppose (p, q) satisfies (14). Since $N > 0$, one has $q > 1$. Define

$$u = p(q - 1), \quad v = p(q + 1).$$

Then $1 \leq u < v$, $uv = N$, and

$$v - u = 2p, \quad u + v = 2pq.$$

Thus u and v have the same parity, $v - u$ divides $u + v$, and the displayed reconstruction formulas hold.

Conversely, let (u, v) satisfy the factor-pair conditions and define p, q by those formulas. The parity condition makes p a positive integer, and the divisibility condition makes q an integer. Moreover, $u + v > v - u > 0$, so $q > 1$. Direct calculation gives

$$p(q - 1) = u, \quad p(q + 1) = v,$$

and hence

$$p^2(q^2 - 1) = uv = N.$$

The two constructions are inverse.

Finally, (14) is equivalent to

$$q^2 = \frac{N}{p^2} + 1.$$

This gives the square-divisor formulation. □

Proposition 11.8 (A tensor-unit arithmetic obstruction). Let $e = 2$ and $Y = (\mathbb{C}, \mathbb{C}^2)$. Then

$$\dim_{\text{aff}} \mathbf{SuperChan}(T_E I, Y) = 51,$$

and this convex set is not affinely isomorphic to the state space of any object of **SuperChan**.

Proof. For $e = 2$, $c = 1$, and $d = 2$,

$$\dim_{\text{aff}} \mathbf{SuperChan}((E, E), (C, D)) = (2^2 - 1) + 2^4(2^2 - 1) = 3 + 48 = 51.$$

If this space were affinely isomorphic to the state space of $Z = (P, Q)$, preservation of affine dimension would give

$$p^2(q^2 - 1) = 51.$$

Since $51 = 3 \cdot 17$ is squarefree, the square p^2 can divide 51 only when $p = 1$. The equation would then imply $q^2 = 52$, impossible for an integer q . Hence no superchannel state space has the required affine dimension. □

Proposition 11.9 (Representable tensor-unit components). For every finite-dimensional C and E ,

$$\mathbf{SuperChan}((E, E), (C, I)) \cong \mathbf{SuperChan}(I, (C, E))$$

affinely. Both convex sets are canonically affinely isomorphic to **Chan** (C, E) .

Proof. A superchannel Choi operator for a morphism

$$(E, E) \longrightarrow (C, I)$$

has, by Theorem 11.3, an auxiliary operator $R_{EC} \geq 0$ satisfying

$$\mathrm{Tr}_E R_{EC} = I_C.$$

Because the target output is one-dimensional, the first superchannel constraint reduces to

$$R = I_E \otimes R_{EC}.$$

Thus the morphism is uniquely determined by R_{EC} , and R_{EC} is exactly the Choi operator of a channel $C \rightarrow E$. This gives an affine isomorphism

$$\mathbf{SuperChan}((E, E), (C, I)) \cong \mathbf{Chan}(C, E).$$

A state of the object (C, E) is a morphism

$$I = (\mathbb{C}, \mathbb{C}) \longrightarrow (C, E).$$

For this source type, the auxiliary Choi operator is forced to be I_C , and the remaining constraint on the full Choi operator is

$$R \geq 0, \quad \mathrm{Tr}_E R = I_C.$$

Hence the state space is again canonically affinely isomorphic to $\mathbf{Chan}(C, E)$. Composing these two identifications proves the result. $\square$

Lemma 11.10 (Polygonal sections of spectrahedra). Let $\mathcal{A}$ be an affine subspace of $\mathrm{Herm}(\mathbb{C}^n)$ containing a positive-definite operator, and put

$$K = \mathcal{A} \cap \mathrm{Pos}(\mathbb{C}^n).$$

If a two-dimensional affine plane $L \subseteq \mathcal{A}$ contains a positive-definite point and $K \cap L$ is a bounded polygon with m edges, then

$$m \leq n.$$

Proof. Choose affine coordinates on L in which its positive-definite point is M_0 and write

$$M(x, y) = M_0 + xH_1 + yH_2.$$

The polynomial

$$f(x, y) = \det M(x, y)$$

has degree at most n and is not identically zero because $f(0, 0) = \det M_0 > 0$.

Let an edge of $K \cap L$ lie on the affine line $\ell(x, y) = 0$. Every point in the relative interior of that edge is a boundary point of the positive-semidefinite cone within L and is therefore singular. Hence f vanishes on a nondegenerate line segment contained in $\ell = 0$. Restricting f to that line gives a one-variable polynomial with infinitely many zeros, so it vanishes identically; equivalently, the linear polynomial ℓ divides f in $\mathbb{R}[x, y]$. Distinct edges of a bounded convex polygon have distinct supporting lines. Their distinct linear polynomials are pairwise relatively prime, so their product divides f . Consequently, the number of edges is at most $\deg f \leq n$. $\square$

Theorem 11.11 (Complete tensor-unit affine classification). Fix a quantum environment E with $e = d_E > 1$, and let $Y = (C, D)$ be a superchannel target. The process body

$$\mathbf{SuperChan}((E, E), (C, D))$$

is affinely isomorphic to the state space of an object of $\mathbf{SuperChan}$ if and only if $d_D = 1$. When $d_D = 1$, the canonical representing object is (C, E) .

Proof. If $d_D = 1$, Proposition 11.9 gives the canonical affine isomorphism

$$\mathbf{SuperChan}((E, E), (C, I)) \cong \mathbf{SuperChan}(I, (C, E)).$$

It remains to exclude every $d = d_D > 1$.

Write $c = d_C$ and let

$$\mathcal{K} = \mathbf{SuperChan}((E, E), (C, D)).$$

Its Choi operators act on a space of dimension

$$n_{\mathcal{K}} = de^2c.$$

The positive-definite operator

$$R^0 = \frac{1}{de} I_{D \otimes E \otimes E \otimes C}$$

is feasible, with auxiliary operator

$$R_{EC}^0 = \frac{1}{e} I_E \otimes I_C.$$

Put $g = e^2c$ and index a basis of the three factors $E \otimes E \otimes C$ by $r = 1, \dots, g$. For every r , choose a rotated regular d -gon of unit vectors

$$w_{r,1}, \dots, w_{r,d} \in \mathbb{R}^2$$

such that all gd vectors are distinct. Each group satisfies

$$\sum_{\delta=1}^d w_{r,\delta} = 0.$$

When $d = 2$, choose at least two nonparallel antipodal pairs; this is possible because $g = e^2c \geq 4$. The rotations can be chosen to avoid all coincidences. For each normal w , the functional $z \mapsto w \cdot z$ is uniquely maximized at w among the chosen unit vectors, so every normal is an exposed vertex of their joint convex hull. If $d \geq 3$, each regular polygon contains the origin in its interior; if $d = 2$, two nonparallel antipodal pairs already do so. Hence the origin lies in the interior of the joint convex hull.

For $s = 1, 2$, define the diagonal Hermitian operator

$$H_s = \sum_{r=1}^g \sum_{\delta=1}^d (w_{r,\delta})_s |\delta, r\rangle \langle \delta, r|.$$

The group-sum condition gives

$$\mathrm{Tr}_D H_s = 0.$$

Hence H_s belongs to the direction space of the superchannel affine hull, with zero auxiliary variation. The two-dimensional section

$$R(x, y) = R^0 + xH_1 + yH_2$$

is feasible exactly when

$$\frac{1}{de} + (x, y) \cdot w_{r,\delta} \geq 0 \quad (r = 1, \dots, g; \delta = 1, \dots, d).$$

It is bounded because the origin is interior to the convex hull of the normals. Every inequality defines an edge: at

$$(x, y) = -\frac{1}{de} w_{r,\delta}$$

the corresponding diagonal entry is zero, while every other entry is strictly positive because distinct unit vectors have inner product strictly less than one. Thus $\mathcal{K}$ has a two-dimensional polygonal section with exactly

$$n_{\mathcal{K}} = de^2c$$

edges.

Suppose, for contradiction, that

$$\mathcal{K} \cong \mathbf{SuperChan}(I, (P, Q)) \cong \mathbf{Chan}(P, Q)$$

affinely, and put $p = d_P$, $q = d_Q$, and $n_{\mathcal{S}} = pq$. Equality of affine dimensions gives

$$p^2(q^2 - 1) = c^2(e^2 - 1 + e^4(d^2 - 1)). \quad (15)$$

The right-hand side is at least 51, so $q > 1$ and $(p, q) \neq (1, 2)$.

The channel body $\mathbf{Chan}(P, Q)$ also has a polygonal section with exactly $pq = n_{\mathcal{S}}$ edges. Indeed, index diagonal entries by an input index $i = 1, \dots, p$ and an output index $j = 1, \dots, q$. If $q \geq 3$, choose for each i a rotated regular q -gon of distinct unit normals $z_{i,1}, \dots, z_{i,q}$. If $q = 2$, then $p \geq 2$; choose distinct nonparallel antipodal pairs. In both cases,

$$\sum_{j=1}^q z_{i,j} = 0$$

for every i , all pq normals are distinct vertices of their joint convex hull, and the origin lies in its interior. The diagonal directions

$$L_s = \sum_{i=1}^p \sum_{j=1}^q (z_{i,j})_s |j, i\rangle \langle j, i| \quad (s = 1, 2)$$

satisfy $\text{Tr}_Q L_s = 0$. Therefore the section

$$\frac{1}{q} I_{Q \otimes P} + x L_1 + y L_2$$

is, by the same argument, a bounded polygon with exactly pq edges.

An affine isomorphism of compact convex sets extends to their affine hulls, maps relative interiors to relative interiors, and carries the intersection with an affine plane onto the intersection with the image plane. In an affine slice of a positive-semidefinite cone that contains a positive-definite point, the relative-interior points are precisely the positive-definite feasible points. Indeed, positive definiteness is relatively open; if a feasible M is singular and M_0 is positive definite, then for a nonzero $v \in \ker M$ and every $t > 0$,

$$v^\dagger (M + t(M - M_0))v = -t v^\dagger M_0 v < 0,$$

so M is a relative-boundary point. Thus the image sections satisfy the hypothesis of Lemma 11.10.

Applying that lemma to the image of the $n_{\mathcal{K}}$ -edge section inside the $n_{\mathcal{S}} \times n_{\mathcal{S}}$ channel spectrahedron gives

$$n_{\mathcal{K}} \leq n_{\mathcal{S}}.$$

Applying the same lemma to the inverse image of the $n_{\mathcal{S}}$ -edge channel section inside the $n_{\mathcal{K}} \times n_{\mathcal{K}}$ superchannel spectrahedron gives the reverse inequality. Hence

$$pq = de^2c. \quad (16)$$

Squaring (16) and subtracting (15) yields

$$p^2 = c^2(e^4 - e^2 + 1).$$

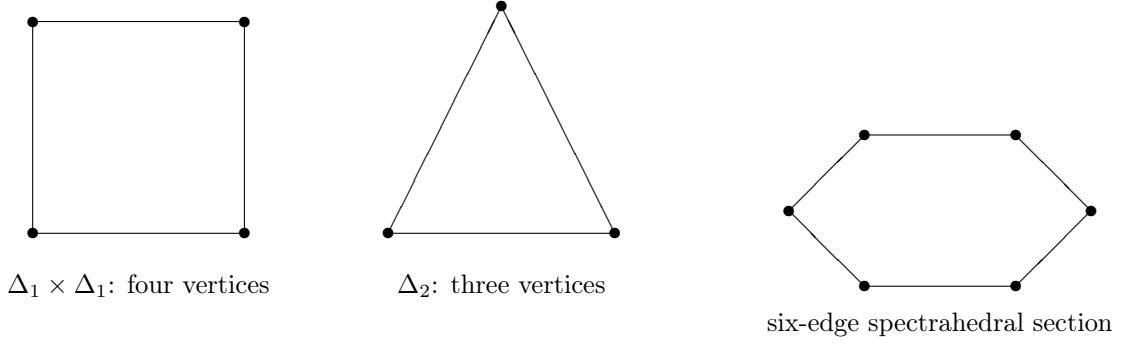


Figure 2: Affine-shape obstructions beyond dimension. The square and triangle have the same affine dimension but different extreme-point counts. The final panel represents a six-edge polygonal section; the determinant-degree argument bounds the number of distinct edge lines by the matrix size of a spectrahedral representation. Only the combinatorial data are used.

Thus the integer $e^4 - e^2 + 1$ is a square in $\mathbb{Q}$. An integer that is a rational square is an integer square: if a/b is in lowest terms and $(a/b)^2$ is integral, then $b^2 \mid a^2$, so $b = 1$. But

$$(e^2 - 1)^2 < e^4 - e^2 + 1 < e^4$$

for $e > 1$, placing this integer strictly between consecutive squares. This contradiction excludes every $d > 1$ and completes the proof. $\square$

Example 11.12 (A dimension-compatible component). For $e = 4$, $c = 1$, and $d = 2$,

$$\dim_{\text{aff}} \mathbf{SuperChan}((E, E), (C, D)) = 4^2 - 1 + 4^4(2^2 - 1) = 783 = 28^2 - 1.$$

The factor pair $(u, v) = (27, 29)$ in Proposition 11.7 gives $(p, q) = (1, 28)$. Thus this process space has the same affine dimension as the state space of $(\mathbb{C}, \mathbb{C}^{28})$. Nevertheless, Theorem 11.11 excludes an affine isomorphism because $d = 2$. This gives a higher-order instance in which dimension compatibility is strictly weaker than affine state-space representability.

Remark 11.13 (Scope of the tensor-unit test). The tensor-unit affine problem is completely classified for every nontrivial environment. The test succeeds exactly for targets (C, I) , represented by (C, E) , and fails for every target with $d_D > 1$. Proposition 11.7 separately classifies the weaker dimension test and exhibits dimension-compatible false positives. Thus Theorem 11.11 already supplies a tensor-unit proof of the diagonal environmental no-adjoint result, while the all-source theorem 11.6 classifies every fixed channel-type decoration. Its input-decoration case is a coefficient obstruction; its pure output-decoration case uses the discrete output-dimension spectrum.

Table 1 collects the concrete obstruction mechanisms proved above. Figure 2 isolates the affine invariants that remain after dimension matching.

12 Ambient Transfer and Finite Presheaf Realization

12.1 Relative reflection and presheaf completion

The transfer requires less than a global coreflector. The exact relative criterion is stated first.

Theorem 12.1 (Relative right-adjoint reflection). Let $\mathcal{C}$ and $\mathcal{H}$ be ordinary categories, let $I : \mathcal{C} \rightarrow \mathcal{H}$ be fully faithful, and let $T : \mathcal{C} \rightarrow \mathcal{C}$ and $T^{\mathcal{H}} : \mathcal{H} \rightarrow \mathcal{H}$ satisfy a natural isomorphism

$$IT \cong T^{\mathcal{H}}I.$$

Table 1: Comparison of the principal environmental obstructions. Here a, b, c, d, e denote the corresponding finite system dimensions and n an instrument grade.

Process theory	Affine hom-space dimension	Dimension/profile result	Finer invariant	Classification
Quantum channels	$a^2(b^2 - 1)$	At $B = E$, the required object size obeys $r^2 = e^4 - e^2 + 1$, which is not a square.	Dimension-compatible fixed codomains have a nonclosed extreme-point set.	No right adjoint for $e > 1$; a channel body is a state space only in the singleton case.
Classical channels	$a(b - 1)$	The transformed dimension is realized by $r = 1 + e(b - 1)$.	$(\Delta_{b-1})^e$ has b^e vertices and is not the simplex Δ_{r-1} when $e, b > 1$.	No right adjoint for a nontrivial environment.
Quantum instruments	$a^2(nb^2 - 1)$	All grades violate the profile intercept; an isolated grade must satisfy $e^2(nb^2 - 1) = nr^2 - 1$.	Channel extreme-boundary topology for $b > 1$; isolated deterministic POVM extremes for $b = 1$.	The only affinely representable fixed grade has $n = b = r = 1$.
Classical instruments	$a(nb - 1)$	All grades violate the profile intercept; an isolated grade must satisfy $e(nb - 1) = nr - 1$.	Product/simplex vertex counts disagree whenever $nb > 1$.	A fixed grade is affinely representable only when $n = b = r = 1$.
Deterministic superchannels	$c^2(a^2 - 1) + c^2a^2b^2(d^2 - 1)$	Input decoration fails coefficient comparison; output decoration fails the object-dimension spectrum.	Tensor-unit components with nontrivial target output fail the polygonal section invariant.	Fixed-type decoration has a right adjoint exactly for the tensor unit.

Suppose

$$T^{\mathcal{H}} \dashv R^{\mathcal{H}}.$$

Then T has a right adjoint if and only if there are objects $S_Y \in \mathcal{C}$ and bijections

$$\Theta_{X,Y} : \mathcal{H}(IX, R^{\mathcal{H}}IY) \xrightarrow{\cong} \mathcal{C}(X, S_Y) \quad (17)$$

natural in $X \in \mathcal{C}$. No functorial action on $Y \mapsto S_Y$ and no naturality in Y need be assumed; these are forced by universality.

Proof. Assume first that $T \dashv S$. Naturality of every step in the following chain gives the required family:

$$\begin{aligned} \mathcal{H}(IX, R^{\mathcal{H}}IY) &\cong \mathcal{H}(T^{\mathcal{H}}IX, IY) \\ &\cong \mathcal{H}(I(TX), IY) \\ &\cong \mathcal{C}(TX, Y) \\ &\cong \mathcal{C}(X, SY). \end{aligned}$$

The first and last bijections are the two adjunctions, the second is induced by the commuting natural isomorphism, and the third is full faithfulness of I .

Conversely, suppose the objects S_Y and the X -natural bijections Θ are given. For every X, Y ,

compose

$$\begin{aligned}
\mathcal{C}(TX, Y) &\cong \mathcal{H}(I(TX), IY) \\
&\cong \mathcal{H}(T^{\mathcal{H}}IX, IY) \\
&\cong \mathcal{H}(IX, R^{\mathcal{H}}IY) \\
&\cong \mathcal{C}(X, S_Y).
\end{aligned}$$

This family is natural in X . Let its inverse send id_{S_Y} to $\epsilon_Y : T(S_Y) \rightarrow Y$. Naturality in X implies that the inverse sends every $g : X \rightarrow S_Y$ to $\epsilon_Y \circ T(g)$. Hence (S_Y, ϵ_Y) is a T -universal evaluator. Theorem 5.6 assembles these objectwise data uniquely into a functor S and an adjunction $T \dashv S$. $\square$

Example 12.2 (The relative criterion is strictly weaker than coreflectivity). Let $\mathcal{C}$ be the terminal one-object category, let $\mathcal{H}$ be the discrete two-object category, and let I include one of its objects. Take T , $T^{\mathcal{H}}$, and $R^{\mathcal{H}}$ to be identity functors. The only relative hom-set

$$\mathcal{H}(I*, R^{\mathcal{H}}I*)$$

is a singleton and is represented by the sole object of $\mathcal{C}$; hence Theorem 12.1 applies. However, I has no right adjoint: for the other object $h \in \mathcal{H}$, the set $\mathcal{H}(I*, h)$ is empty, whereas every hom-set in $\mathcal{C}$ is a singleton. Thus relative representability on the objects $R^{\mathcal{H}}IY$ does not imply a global coreflector.

A global coreflector supplies the relative representers automatically, giving the following corollary.

Theorem 12.3 (Coreflector inheritance). Let $\mathcal{H}$ be an ordinary category containing **SuperChan** as a full subcategory through a fully faithful functor

$$I : \mathbf{SuperChan} \hookrightarrow \mathcal{H}.$$

Suppose that:

- (i) $\mathcal{H}$ has an endofunctor $T_E^{\mathcal{H}}$ satisfying

$$T_E^{\mathcal{H}}I = IT_E;$$

- (ii) I has a right adjoint $K : \mathcal{H} \rightarrow \mathbf{SuperChan}$, equivalently natural bijections

$$\mathcal{H}(IX, Z) \cong \mathbf{SuperChan}(X, KZ).$$

Then a right adjoint to $T_E^{\mathcal{H}}$ would induce a right adjoint to T_E on **SuperChan**. Consequently, if $d_E > 1$, then $T_E^{\mathcal{H}}$ has no ordinary right adjoint.

Proof. Suppose $T_E^{\mathcal{H}} \dashv R^{\mathcal{H}}$ and define

$$R^{\mathbf{SuperChan}} = KR^{\mathcal{H}}I.$$

For $X, Y \in \mathbf{SuperChan}$, natural bijections give

$$\begin{aligned}
\mathbf{SuperChan}(X, R^{\mathbf{SuperChan}}Y) &= \mathbf{SuperChan}(X, KR^{\mathcal{H}}IY) \\
&\cong \mathcal{H}(IX, R^{\mathcal{H}}IY) \\
&\cong \mathcal{H}(T_E^{\mathcal{H}}IX, IY) \\
&= \mathcal{H}(I(T_EX), IY) \\
&\cong \mathbf{SuperChan}(T_EX, Y).
\end{aligned}$$

Thus $R^{\mathbf{SuperChan}}$ would be right adjoint to T_E , contradicting Theorem 11.6. $\square$

Theorem 12.4 (Presheaf-completion transfer boundary). Let $\mathcal{C}$ be a small category, let $T : \mathcal{C} \rightarrow \mathcal{C}$, and write

$$\widehat{\mathcal{C}} = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$$

for its presheaf category with Yoneda embedding $y : \mathcal{C} \rightarrow \widehat{\mathcal{C}}$. Define

$$T^*G = G \circ T^{\text{op}}$$

and define $T_! : \widehat{\mathcal{C}} \rightarrow \widehat{\mathcal{C}}$ pointwise by

$$(T_!F)(c) = \int^{x \in \mathcal{C}} F(x) \times \mathcal{C}(c, Tx). \quad (18)$$

Then:

(a) $T_! \dashv T^*$;

(b) there is a natural isomorphism

$$T_! yX \cong y(TX);$$

(c) for every $X, Y \in \mathcal{C}$,

$$\widehat{\mathcal{C}}(yX, T^*yY) \cong \mathcal{C}(TX, Y)$$

naturally in X, Y ;

(d) the presheaves T^*yY are representable by objects yS_Y for every Y if and only if T has a right adjoint. Thus an ambient right adjoint to $T_!$ need not restrict to the representable objects.

In particular, take the standard small skeleton of **SuperChan** and a fixed channel type K . The cocontinuous extension $T_{K,!}$ of fixed-type decoration always has the ambient right adjoint T_K^* on the presheaf category. If K is not isomorphic to the tensor unit, at least one presheaf T_K^*yY is not representable by a deterministic-superchannel object.

Proof. A natural transformation $T_!F \rightarrow G$ is, by the coend universal property, a family of maps

$$F(x) \times \mathcal{C}(c, Tx) \longrightarrow G(c)$$

natural in c and dinatural in x . For fixed x , the Yoneda lemma identifies natural maps

$$\mathcal{C}(-, Tx) \longrightarrow G$$

with elements of $G(Tx)$. The preceding family is therefore equivalent to a family of maps

$$F(x) \longrightarrow G(Tx) = (T^*G)(x)$$

natural in x . Hence

$$\widehat{\mathcal{C}}(T_!F, G) \cong \widehat{\mathcal{C}}(F, T^*G)$$

naturally in F, G , proving (a).

For a representable presheaf, (18) and the co-Yoneda lemma give

$$\begin{aligned} (T_!yX)(c) &= \int^x \mathcal{C}(x, X) \times \mathcal{C}(c, Tx) \\ &\cong \mathcal{C}(c, TX) = (y(TX))(c), \end{aligned}$$

naturally in c, X . This proves (b). Ordinary Yoneda then gives

$$\begin{aligned}\widehat{\mathcal{C}}(yX, T^*yY) &\cong (T^*yY)(X) \\ &= (yY)(TX) \\ &= \mathcal{C}(TX, Y),\end{aligned}$$

which proves (c).

If $T \dashv S$, then

$$T^*yY(X) = \mathcal{C}(TX, Y) \cong \mathcal{C}(X, SY) = y(SY)(X),$$

so $T^*yY \cong y(SY)$ naturally in X . Conversely, suppose every T^*yY is represented by some S_Y . Part (c) yields bijections

$$\mathcal{C}(TX, Y) \cong \mathcal{C}(X, S_Y)$$

natural in X . Theorem 5.6 assembles these objectwise representers uniquely into a functor S with $T \dashv S$. This proves (d).

The standard skeletal superchannel category is small, and part (b) gives

$$T_{K,!}y \cong yT_K.$$

Part (a) supplies $T_{K,!} \dashv T_K^*$. If every T_K^*yY were representable, part (d) would give a right adjoint to T_K . For nonunit K this contradicts Theorem 11.6. Hence at least one ambient right-adjoint value lies outside the representable image of **SuperChan**. $\square$

Let **Conv** denote the category of possibly empty real convex spaces and affine maps. Its tensor product $P \otimes_{\text{cvx}} Q$ represents maps out of $P \times Q$ that are affine in each variable separately; **Conv** is complete, cocomplete, and symmetric monoidal closed [10, 12]. A category with convex hom-sets and separately affine composition is therefore a **Conv**-enriched category.

Theorem 12.5 (Convex-causal presheaf transfer boundary). Let $\mathcal{C}$ be a small **Conv**-enriched category and let $T : \mathcal{C} \rightarrow \mathcal{C}$ act affinely on every hom-space. Write

$$\widehat{\mathcal{C}}_{\text{cvx}} = [\mathcal{C}^{\text{op}}, \mathbf{Conv}]$$

for the enriched presheaf category and $y_{\text{cvx}} : \mathcal{C} \rightarrow \widehat{\mathcal{C}}_{\text{cvx}}$ for enriched Yoneda. Define enriched precomposition T^* and the enriched left Kan extension

$$(T_!F)(c) = \int_{\mathbf{Conv}}^{x \in \mathcal{C}} F(x) \otimes_{\text{cvx}} \mathcal{C}(c, Tx).$$

Then:

- (a) $T_! \dashv T^*$ as a **Conv**-enriched adjunction;
- (b) $T_!y_{\text{cvx}}X \cong y_{\text{cvx}}(TX)$ naturally;
- (c) there are natural affine isomorphisms

$$\widehat{\mathcal{C}}_{\text{cvx}}(y_{\text{cvx}}X, T^*y_{\text{cvx}}Y) \cong \mathcal{C}(TX, Y);$$

- (d) every $T^*y_{\text{cvx}}Y$ is representable if and only if T has a **Conv**-enriched right adjoint. Under the hypotheses of Proposition 2.3, this is equivalent to existence of an ordinary right adjoint.

For the standard small skeletons of **FinStoch** and **SuperChan**, representable presheaves take values in the actual stochastic or positive-semidefinite causal process bodies, so positivity, normalization, and probabilistic mixing are retained pointwise. Nevertheless:

- (i) if $|E| > 1$, the convex-presheaf extension of $- \times E : \mathbf{FinStoch} \rightarrow \mathbf{FinStoch}$ has an ambient right adjoint, but for every B with $|B| > 1$ the presheaf $T_E^* y_{\text{cvx}} B$ is not representable by a finite classical object;
- (ii) for a nonunit channel type K , the convex-presheaf extension of $T_K : \mathbf{SuperChan} \rightarrow \mathbf{SuperChan}$ has an ambient right adjoint, but at least one $T_K^* y_{\text{cvx}} Y$ is not representable by a deterministic-superchannel object.

Proof. The enriched coend–hom adjunction gives

$$\begin{aligned} \widehat{\mathcal{C}}_{\text{cvx}}(T_! F, G) &\cong \int_{x \in \mathcal{C}} \mathbf{Conv}(F(x), G(Tx)) \\ &= \widehat{\mathcal{C}}_{\text{cvx}}(F, T^* G) \end{aligned}$$

as an isomorphism of convex spaces, naturally in F, G . This proves (a). The enriched co-Yoneda lemma gives

$$\begin{aligned} (T_! y_{\text{cvx}} X)(c) &= \int_{\mathbf{Conv}}^x \mathcal{C}(x, X) \otimes_{\text{cvx}} \mathcal{C}(c, Tx) \\ &\cong \mathcal{C}(c, TX), \end{aligned}$$

proving (b). Enriched Yoneda then yields

$$\begin{aligned} \widehat{\mathcal{C}}_{\text{cvx}}(y_{\text{cvx}} X, T^* y_{\text{cvx}} Y) &\cong (T^* y_{\text{cvx}} Y)(X) \\ &= \mathcal{C}(TX, Y), \end{aligned}$$

which proves (c).

If $T \dashv R$ enrichedly, the enriched adjunction isomorphism identifies $T^* y_{\text{cvx}} Y$ with $y_{\text{cvx}} RY$. Conversely, enriched representers for all these presheaves give enriched universal evaluators; the enriched Yoneda construction, equivalently the argument of Theorem 5.6 with affine structure retained at every step, assembles them into an enriched right adjoint. This proves the first assertion of (d). If an ordinary right adjoint exists, Proposition 2.3 makes every transposition affine. The action of the right adjoint on a morphism is the affine forward transpose of its composite with the counit, so the right adjoint itself acts affinely on hom-spaces. It is therefore **Conv**-enriched, proving the final assertion.

For **FinStoch**, the enriched representable at B evaluates on X as the channel polytope $\mathbf{FinStoch}(X, B)$. If $T_E^* y_{\text{cvx}} B$ were representable by an object R , evaluation at the tensor unit would give an affine isomorphism

$$(\Delta_{b-1})^{|E|} \cong \Delta_{|R|-1}.$$

The dimension and vertex argument of Theorem 7.1 excludes this whenever $|E|, b > 1$. The ambient enriched adjoint exists by (a).

For **SuperChan**, representable values are exactly the convex sets of positive Choi operators satisfying the deterministic superchannel constraints, and presheaf action is affine composition. Part (a) again supplies the ambient right adjoint. If all its values on representables were representable, part (d) would give a right adjoint to T_K , contradicting Theorem 11.6 for nonunit K . This proves (i) and (ii). $\square$

Remark 12.6 (Operational scope of the convex completion). The enriched completion preserves the actual convex positive and causal process bodies on representable objects and preserves affine mixing under composition. It does not assert that every convex presheaf is realizable as a finite laboratory system; the theorem identifies representability as exactly the missing descent condition.

12.2 Finite presheaf realization and autonomous control

Theorem 12.7 (Finite classical laboratory realization). Let $\mathcal{C}$ be a finite category and let $T : \mathcal{C} \rightarrow \mathcal{C}$. Write

$$\widehat{\mathcal{C}}_{\text{fin}} = [\mathcal{C}^{\text{op}}, \mathbf{FinSet}].$$

Then:

- (a) precomposition T^* has a left adjoint $T_!$ within $\widehat{\mathcal{C}}_{\text{fin}}$, given by the finite coend

$$(T_!F)(c) = \left(\coprod_{x \in \mathcal{C}} F(x) \times \mathcal{C}(c, Tx) \right) / \sim;$$

moreover $T_!yX \cong y(TX)$;

- (b) every finite presheaf F has a direct finite classical realization on the context-labelled register

$$\Omega_F = \coprod_{c \in \text{Ob } \mathcal{C}} \{c\} \times F(c).$$

Its normalized state space is the finite simplex $\Delta(\Omega_F)$, and each presheaf action map extends to a deterministic stochastic map between the corresponding context faces;

- (c) every T^*yY is representable if and only if T has a right adjoint in $\mathcal{C}$.

There are finite examples in which the ambient right adjoint exists and every ambient value has the laboratory realization in part (b), but descent to an object of $\mathcal{C}$ fails. Explicitly, let $\mathcal{V}$ be the three-object poset category

$$0 < a, \quad 0 < b,$$

with a, b incomparable, and let T be constant at 0. Then $T_! \dashv T^*$ on $\widehat{\mathcal{V}}_{\text{fin}}$, while T^*y0 is the constant singleton presheaf and is not representable in $\mathcal{V}$.

Proof. Because $\mathcal{C}$ has finitely many objects and morphisms and every $F(x)$ is finite, the displayed coproduct is finite and its coend equivalence relation has finitely many classes. Thus $T_!F$ is again a finite presheaf. The usual coend universal property gives

$$\widehat{\mathcal{C}}_{\text{fin}}(T_!F, G) \cong \widehat{\mathcal{C}}_{\text{fin}}(F, T^*G),$$

so $T_! \dashv T^*$. The finite co-Yoneda calculation gives

$$(T_!yX)(c) = \int^x \mathcal{C}(x, X) \times \mathcal{C}(c, Tx) \cong \mathcal{C}(c, TX),$$

proving (a).

The set Ω_F is finite. Probability distributions on it form a finite classical simplex. For an arrow $u : c \rightarrow d$, the presheaf function

$$F(u) : F(d) \rightarrow F(c)$$

acts deterministically from the face supported on $\{d\} \times F(d)$ to the face supported on $\{c\} \times F(c)$. Linear extension on probability distributions is a column-stochastic map, so positivity and normalization are preserved. This proves (b).

Yoneda gives

$$\widehat{\mathcal{C}}_{\text{fin}}(yX, T^*yY) \cong \mathcal{C}(TX, Y).$$

If $T \dashv R$, this presheaf is $y(RY)$. Conversely, representers for every T^*yY give universal evaluators and assemble into a right adjoint by Theorem 5.6. This proves (c).

For the stated poset, $T(x) = 0$ for all x , so

$$(T^*y0)(x) = \mathcal{V}(0, 0)$$

is a singleton for each of the three objects. A representable $y(z)$ is supported on the principal downset of z , which is one of

$$\{0\}, \quad \{0, a\}, \quad \{0, b\}.$$

None is supported on all three objects. Hence T^*y0 is not representable, although it has the three-state laboratory register $\Omega_{T^*y0} = \{(0, *), (a, *), (b, *)\}$. Part (a) supplies the ambient right adjoint, completing the example. $\square$

Remark 12.8 (Finite laboratory scope). The theorem realizes every ambient finite presheaf as a finite classical context-labelled register with stochastic control between context faces. It does not identify all such controllers with single objects of the original process category; that stronger identification is exactly representable descent.

Theorem 12.9 (Compilation into a single finite controller). Let $\mathcal{C}$ be a finite category and let $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{FinSet}$ be a finite presheaf. Define the single finite state set

$$\Omega_F^\perp = \{\perp\} \sqcup \coprod_{c \in \text{Ob } \mathcal{C}} \{c\} \times F(c),$$

where $\perp$ is an absorbing error state, and let the finite control alphabet be $U = \text{Mor } \mathcal{C}$. For a control $u : c \rightarrow d$, define $P_F^u : \Omega_F^\perp \rightarrow \Omega_F^\perp$ by

$$P_F^u(d, s) = (c, F(u)s), \quad P_F^u(z) = \perp \quad \text{if } z \notin \{d\} \times F(d), \quad P_F^u(\perp) = \perp.$$

Equivalently, the family is one deterministic processor

$$P_F : U \times \Omega_F^\perp \longrightarrow \Omega_F^\perp.$$

Then:

- (a) every P_F^u is a deterministic classical channel, and on the valid context face it realizes exactly the presheaf action $F(u)$;
- (b) for composable arrows $u : c \rightarrow d$ and $v : d \rightarrow e$,

$$P_F^u P_F^v(e, s) = P_F^{v \circ u}(e, s) \quad (s \in F(e)),$$

and identity controls act identically on their matching context sectors;

- (c) every natural transformation $\eta : F \rightarrow G$ compiles to the deterministic map

$$\hat{\eta} : \Omega_F^\perp \rightarrow \Omega_G^\perp, \quad \hat{\eta}(c, s) = (c, \eta_c(s)), \quad \hat{\eta}(\perp) = \perp,$$

which intertwines every control. Conversely, every error-preserving, context-preserving deterministic map intertwining all controls arises uniquely in this way;

- (d) applying probability distributions turns the processor and compiled maps into stochastic channels on the single finite simplexes $\Delta(\Omega_F^\perp)$. Hence the finite-presheaf Kan adjunction of Theorem 12.7 is realizable by single finite controlled devices, not merely by an external family of context faces.

Proof. Each P_F^u is a function on a finite set, and therefore its linear action on probability vectors is a deterministic column-stochastic matrix. Its restriction to the d -sector is exactly $F(u)$ with the output relabelled by c , proving (a).

For $u : c \rightarrow d$ and $v : d \rightarrow e$, contravariant functoriality gives

$$F(v \circ u) = F(u) \circ F(v).$$

Starting from (e, s) , first applying P_F^v and then P_F^u therefore gives

$$(e, s) \mapsto (d, F(v)s) \mapsto (c, F(u)F(v)s) = (c, F(v \circ u)s),$$

which is $P_F^{v \circ u}(e, s)$. The identity statement follows from $F(\text{id}_c) = \text{id}_{F(c)}$, proving (b).

If $\eta : F \rightarrow G$ is natural, then on a valid d -sector

$$\begin{aligned} \hat{\eta}P_F^u(d, s) &= (c, \eta_c F(u)s) \\ &= (c, G(u)\eta_d(s)) \\ &= P_G^u \hat{\eta}(d, s). \end{aligned}$$

On a mismatched sector, both composites produce $\perp$, so the maps intertwine globally. Conversely, an error- and context-preserving map restricts to functions $\eta_c : F(c) \rightarrow G(c)$. Intertwining P_F^u and P_G^u is exactly the naturality equation

$$\eta_c F(u) = G(u)\eta_d.$$

This correspondence is unique and respects identities and composition, proving (c).

Finally, every finite function induces a deterministic stochastic channel, and composition and intertwining identities survive linear extension to probability distributions. The constructions of $T_!$ and T^* in Theorem 12.7 remain finite presheaves, so their compiled registers and processors are finite. This proves (d). $\square$

Theorem 12.10 (Autonomous finite-comb internalization of control). Retain the finite category $\mathcal{C}$, finite presheaf F , register $\Omega_F^\perp$, and control maps P_F^u of Theorem 12.9. Form the finite set

$$\mathbf{M}_{\mathcal{C}} = \{\mathbf{1}, \mathbf{0}\} \sqcup \text{Mor } \mathcal{C}$$

as a disjoint union. Define a product $\star$ by taking $\mathbf{1}$ as a global unit, $\mathbf{0}$ as a zero, and, for arrows $u : c \rightarrow d$ and $v : a \rightarrow b$, setting

$$u \star v = \begin{cases} v \circ u, & a = d, \\ \mathbf{0}, & a \neq d. \end{cases}$$

Let $Z_F : \Omega_F^\perp \rightarrow \Omega_F^\perp$ be the constant map with value $\perp$, and define

$$\rho_F(\mathbf{1}) = \text{id}_{\Omega_F^\perp}, \quad \rho_F(\mathbf{0}) = Z_F, \quad \rho_F(u) = P_F^u.$$

Then:

- (a) $(\mathbf{M}_{\mathcal{C}}, \star)$ is a finite monoid and

$$\rho_F(m \star n) = \rho_F(m) \circ \rho_F(n) \quad (m, n \in \mathbf{M}_{\mathcal{C}}).$$

Consequently every finite control word, including a noncomposable word, reduces to one element of the fixed finite program monoid; noncomposable words reduce to $\mathbf{0}$.

- (b) The relation

$$m \sim_F n \iff \rho_F(m) = \rho_F(n)$$

is a monoid congruence. The quotient program monoid $\overline{\mathbf{M}}_F = \mathbf{M}_{\mathcal{C}} / \sim_F$ has exactly $|\rho_F(\mathbf{M}_{\mathcal{C}})|$ states and is the smallest classical program simplex, measured by its number of perfectly distinguishable basis states, for exact realization of all distinct compiled control maps. This minimum remains valid when mixed program states are allowed.

- (c) A single fixed deterministic processor

$$\mathfrak{U}_F : \mathbf{M}_C \times \Omega_F^\perp \longrightarrow \mathbf{M}_C \times \Omega_F^\perp, \quad \mathfrak{U}_F(m, z) = (m, \rho_F(m)z),$$

internalizes the control label as a physical classical program state. It has a deterministic two-stage classical-comb realization: the first stage stores m in a finite memory and the second applies $\rho_F(m)$ to the data register. Equivalently, with $H = \{\text{run}, \text{halt}\}$, the single time-homogeneous channel on

$$S_F = \mathbf{M}_C \times \Omega_F^\perp \times H$$

defined by

$$\mathfrak{A}_F(m, z, \text{run}) = (m, \rho_F(m)z, \text{halt}), \quad \mathfrak{A}_F(m, z, \text{halt}) = (m, z, \text{halt})$$

executes the chosen program in one autonomous step and then halts. The quotient program register gives the program-minimal product realization with

$$|S_F^{\text{pmin}}| = 2|\rho_F(\mathbf{M}_C)| \left(1 + \sum_{c \in \text{Ob } C} |F(c)| \right).$$

No time-dependent external control is used after preparation of the initial program state.

- (d) If $\eta : F \rightarrow G$ is natural, then

$$\tilde{\eta}(m, z, h) = (m, \hat{\eta}(z), h)$$

intertwines the autonomous channels:

$$\tilde{\eta} \mathfrak{A}_F = \mathfrak{A}_G \tilde{\eta}.$$

Conversely, every intertwiner of this displayed program- and phase-preserving form whose data map is error preserving and context preserving arises from a unique natural transformation.

- (e) Linear extension makes the two-stage comb and autonomous update deterministic stochastic channels. A probability distribution μ on the program register induces the randomized control channel

$$p \longmapsto \sum_{m \in \mathbf{M}_C} \mu(m) (\rho_F(m))_* p.$$

The construction does not alter the Kan adjunction or the representable-descent boundary. For the branching-poset example of Theorem 12.7, the nonrepresentable constant singleton presheaf has $|\Omega_F^\perp| = 4$, seven distinct program maps, and therefore a 56-state program-minimal product realization despite having no representer in the original category.

Thus the external family of arrow-labelled controls compiles into one fixed finite classical comb and, after initial program preparation, one autonomous finite-state channel. Autonomy here means fixed dynamics with a physical program register; it does not identify distinct desired controls with one fixed initial program state.

Proof. The only nontrivial associativity check for $\star$ involves three arrow elements. Let

$$u : c \rightarrow d, \quad v : a \rightarrow b, \quad w : r \rightarrow s.$$

The product $(u \star v) \star w$ is nonzero exactly when $a = d$ and $r = b$. These are also exactly the conditions under which $u \star (v \star w)$ is nonzero. Under them, categorical associativity gives

$$(u \star v) \star w = w \circ (v \circ u) = (w \circ v) \circ u = u \star (v \star w).$$

If either condition fails, both sides are $\mathbf{0}$. Products involving $\mathbf{1}$ or $\mathbf{0}$ are associative by definition, proving that $\mathbf{M}_C$ is a monoid.

It remains to verify the representation identity on arrow elements. Suppose $u : c \rightarrow d$ and $v : d \rightarrow e$. On the e -sector,

$$\begin{aligned} P_F^u P_F^v(e, s) &= P_F^u(d, F(v)s) \\ &= (c, F(u)F(v)s) \\ &= (c, F(v \circ u)s) \\ &= P_F^{v \circ u}(e, s). \end{aligned}$$

Every other context sector and $\perp$ are sent to $\perp$ by both sides, so the maps agree globally. If instead $v : a \rightarrow b$ has $a \neq d$, then P_F^v either produces $\perp$ or produces a point in the a -sector; in both cases subsequent application of P_F^u produces $\perp$. Hence

$$P_F^u P_F^v = Z_F = \rho_F(\mathbf{0}).$$

The identity and zero cases are immediate because every P_F^u fixes $\perp$. Thus ρ_F is a monoid homomorphism. Induction on word length proves the control-word reduction in (a), with the empty word represented by $\mathbf{1}$.

The kernel relation of any monoid homomorphism is a congruence, so $\sim_F$ is one. The induced map

$$\bar{\rho}_F : \bar{\mathbf{M}}_F \longrightarrow \rho_F(\mathbf{M}_C)$$

is a bijective monoid homomorphism. It therefore supplies one classical basis program state for each distinct compiled transformation. Conversely, a fixed basis program state in any classical programmable processor determines one stochastic data channel. A mixed program produces a convex combination of those basis-program channels. Each deterministic channel is extreme in the stochastic-channel polytope: column by column, a pure output distribution can be a convex combination only when every positively weighted column equals that same pure distribution. Hence a mixed program can realize a specified deterministic channel only from basis states that already realize that channel. Distinct deterministic targets require disjoint nonempty sets of basis programs, so a register realizing N such targets has at least N basis states. Taking $N = |\rho_F(\mathbf{M}_C)|$ proves both minimality and (b).

The map $\mathfrak{U}_F$ is a function between finite sets. A two-stage memory realization is explicit: with $I = \{*\}$, use

$$L : \mathbf{M}_C \rightarrow I \times \mathbf{M}_C, \quad L(m) = (*, m),$$

followed by

$$E : \mathbf{M}_C \times \Omega_F^\perp \rightarrow (\mathbf{M}_C \times \Omega_F^\perp) \times I, \quad E(m, z) = ((m, \rho_F(m)z), *).$$

Linking the memory output of L to the memory input of E yields $\mathfrak{U}_F$. Both stages are total deterministic channels, so this is a deterministic classical two-stage comb satisfying causal normalization. The map $\mathfrak{A}_F$ is likewise a single function on the finite set S_F ; its second clause makes every halted state fixed. Starting from (m, z, run) , one iteration gives $(m, \rho_F(m)z, \text{halt})$. Replacing the program monoid by its quotient preserves this action, and counting the program, data, and two phase states gives the displayed formula. This proves (c).

If $\eta : F \rightarrow G$ is natural, Theorem 12.9 gives

$$\hat{\eta} P_F^u = P_G^u \hat{\eta}$$

for every arrow u . The same equality holds for $\mathbf{1}$, and it holds for $\mathbf{0}$ because $\hat{\eta}(\perp) = \perp$. Hence

$$\hat{\eta} \rho_F(m) = \rho_G(m) \hat{\eta} \quad (m \in \mathbf{M}_C),$$

which is exactly the autonomous intertwining equation in each phase. Conversely, evaluate that equation on (u, z, run) for every arrow program u . It gives

$$\widehat{\eta}P_F^u = P_G^u\widehat{\eta}.$$

The converse clause of Theorem 12.9 then produces the unique natural transformation, proving (d).

Every finite function extends to a column-stochastic matrix on probability vectors. If the program and data are initially independent with distributions μ and p , marginalizing the retained program output of $\mathfrak{U}_F$ gives

$$\sum_m \mu(m)(\rho_F(m))_*p,$$

which is positive and normalized. The compilation changes neither the presheaf F nor the functors $T!, T^*$, so representability is unchanged. For the branching category $0 < a, 0 < b$, there are three identity arrows and two nonidentity arrows. Together with $\mathbf{1}$ and $\mathbf{0}$, these give seven program elements. On the four data states

$$\perp, (0, *), (a, *), (b, *),$$

the global identity, the constant-error map, the three context projections, and the two maps from the a - or b -sector to the 0 -sector are pairwise distinct. Thus the quotient still has seven states and the program-minimal product construction has

$$2 \cdot 7 \cdot 4 = 56.$$

Theorem 12.7 proves that the underlying presheaf is not representable. This establishes (e) and the final representability boundary. $\square$

13 Universal Evaluators, Tomography, and Symmetry

Pointwise representation does not by itself supply categorical coherence. The results in this section pass from that boundary to universal evaluators, finite experimental criteria, and symmetry-generated tomography.

13.1 Naturality and evaluator assembly

Theorem 13.1 (Pointwise complete order does not imply naturality). Let X be a finite set with $|X| > 1$, and let $\mathcal{M}_X$ be the one-object category whose endomorphism monoid is

$$\mathcal{M}_X(*, *) = \mathbf{FinStoch}(X, X),$$

with stochastic-matrix composition. Define an affine endofunctor $Q : \mathcal{M}_X \rightarrow \mathcal{M}_X$ by

$$Q(*) = *, \quad Q(f) = \text{id}_X \quad \text{for every } f.$$

Then:

- (a) the sole pointwise process bodies

$$\mathcal{M}_X(Q*, *) \quad \text{and} \quad \mathcal{M}_X(*, *)$$

are identical. Their identity identification is affine and is compatible with every inherited matrix-order level in the diagonal Choi realization;

- (b) Q has no ordinary right adjoint.

Consequently, even complete-order-compatible pointwise representation of every hom-body does not imply naturality, composition compatibility, or an adjunction.

Proof. The category is concrete: its morphisms are the deterministic classical channels on X . Composition is bilinear and preserves stochasticity, so the hom-set is convex and composition is separately affine. The assignment Q preserves the identity and composition because

$$Q(g \circ f) = \text{id}_X = \text{id}_X \circ \text{id}_X = Q(g) \circ Q(f),$$

and it acts affinely because it is constant on the hom-set.

Since Q fixes the sole object, both displayed hom-bodies are literally the same stochastic polytope. In the diagonal Choi realization, their generated $*$ -spaces and every inherited ancillary cone are also identical. The identity identification is therefore a complete-order isomorphism, proving (a).

Suppose $Q \dashv R$. There is only one object, so the adjunction supplies a bijection

$$\Phi : \mathcal{M}_X(Q*, *) \longrightarrow \mathcal{M}_X(*, *)$$

natural in the source object. For any stochastic maps $a, f : X \rightarrow X$, source naturality gives

$$\Phi(f \circ Q(a)) = \Phi(f) \circ a.$$

Since $Q(a) = \text{id}_X$, this becomes

$$\Phi(f) = \Phi(f) \circ a \quad \text{for every } a, f.$$

Surjectivity of Φ permits a choice of f with $\Phi(f) = \text{id}_X$. Hence

$$\text{id}_X = \text{id}_X \circ a = a$$

for every stochastic a , which is false because $|X| > 1$ admits nonidentity channels. Thus no right adjoint exists, proving (b). $\square$

Remark 13.2 (Coherence threshold). The counterexample isolates the missing datum. If pointwise complete-order isomorphisms are induced by evaluators

$$\epsilon_Y : T(R_Y) \rightarrow Y, \quad g \longmapsto \epsilon_Y \circ T(g),$$

with unique factorization for every test object, then Theorem 5.6 forces functoriality and naturality. Complete order controls ancillary positivity; evaluator factorization supplies categorical coherence.

Theorem 13.3 (Environmental evaluator assembly and module coherence). Let $\mathcal{C}$ be a symmetric monoidal category with convex hom-spaces and separately affine composition and tensor product. Fix an object E and put $T_E = - \otimes E$. Suppose that for every Y there are an object R_Y and an evaluator

$$\epsilon_Y : R_Y \otimes E \rightarrow Y$$

such that, for every X , the map

$$\Psi_{X,Y} : \mathcal{C}(X, R_Y) \circ \mathcal{C}(X \otimes E, Y), \quad \Psi_{X,Y}(g) = \epsilon_Y \circ (g \otimes \text{id}_E),$$

is bijective. Then:

- (a) the objects R_Y assemble uniquely into a right adjoint $T_E \dashv R$, with counit ϵ and affine natural transpositions $\Psi_{X,Y}$;

(b) R carries a canonical lax left $\mathcal{C}$ -module structure

$$\sigma_{A,Y} : A \otimes RY \longrightarrow R(A \otimes Y),$$

uniquely characterized, up to the monoidal coherence isomorphisms, by

$$\epsilon_{A \otimes Y} \circ (\sigma_{A,Y} \otimes \text{id}_E) = \text{id}_A \otimes \epsilon_Y. \quad (19)$$

The maps $\sigma_{A,Y}$ are natural in A, Y and satisfy the unit and associativity axioms for a lax module functor;

- (c) if the homogeneous extensions of all $\Psi_{X,Y}$ are complete-order isomorphisms for specified concrete matrix orderings, then the assembled adjunction is complete-order compatible at every ancillary level. By Theorem 18.2, fixed concrete embeddings require only the level $\max\{m, n\}$ tests;
- (d) if every evaluator ϵ_Y is normalized, inverse transposition preserves normalized maps. If, in addition, the unit is normalized and R preserves normalized maps, the adjunction restricts to the causal hom-spaces and every $\sigma_{A,Y}$ is normalized.

Thus all-test evaluator factorization forces ordinary naturality, monoidal module coherence, ancillary complete order, and causal restriction under the respective explicit hypotheses.

Proof. The pair (R_Y, ϵ_Y) is a T_E -universal evaluator for every Y . Theorem 5.6 therefore defines, for each $h : Y \rightarrow Z$, the unique morphism $R(h) : R_Y \rightarrow R_Z$ satisfying

$$\epsilon_Z \circ (R(h) \otimes \text{id}_E) = h \circ \epsilon_Y,$$

and proves functoriality, naturality, and $T_E \dashv R$. Each inverse transposition is affine because tensoring and postcomposition are affine; its inverse is affine by Lemma 2.2. This proves (a).

For fixed A, Y , apply universality of $(R(A \otimes Y), \epsilon_{A \otimes Y})$ at the test object $A \otimes RY$ to the morphism

$$(A \otimes RY) \otimes E \cong A \otimes (RY \otimes E) \xrightarrow{\text{id}_A \otimes \epsilon_Y} A \otimes Y.$$

Its unique transpose is $\sigma_{A,Y}$ and satisfies (19). Naturality in A and Y follows because both sides of each naturality square have the same inverse transpose. Likewise, the two composites from $(A \otimes B) \otimes RY$ to $R((A \otimes B) \otimes Y)$ occurring in the module associativity axiom both inverse-transpose to

$$\text{id}_A \otimes \text{id}_B \otimes \epsilon_Y.$$

Uniqueness makes them equal. The module unit axiom follows by the same argument from the fact that both sides inverse-transpose to ϵ_Y . This proves (b).

Under the hypothesis of (c), the affine hom-isomorphisms produced in (a) are, by assumption, cone isomorphisms at every matrix level. Naturality and the module maps in (b) are assembled from these same transpositions, so the complete-order structure is coherent with their functorial action. The finite level reduction is exactly Theorem 18.2.

The first assertion of (d) is Proposition 14.5. Under the additional unit and functor hypotheses, Corollary 14.7 gives full causal restriction. The untransposed map defining $\sigma_{A,Y}$ is $\text{id}_A \otimes \epsilon_Y$, which is normalized when ϵ_Y is. Forward causal preservation therefore makes $\sigma_{A,Y}$ normalized. This proves (d). $\square$

13.2 Tomographic evaluator criteria

Theorem 13.4 (Tomographic codec-to-adjunction criterion). Retain the environmental setting and evaluator maps of Theorem 13.3, but do not assume unique factorization. For each X, Y , write

$$\Gamma_{X,Y} : \mathcal{C}(X, R_Y) \circ \mathcal{C}(X \otimes E, Y), \quad \Gamma_{X,Y}(g) = \epsilon_Y \circ (g \otimes \text{id}_E).$$

Suppose that:

- (a) $\Gamma_{X,Y}$ is surjective: every process admits at least one program;
- (b) there are affine program statistics

$$s_{X,Y,j} : \mathcal{C}(X, R_Y) \rightarrow \mathbb{R} \quad (1 \leq j \leq m_{X,Y})$$

that jointly separate programs, and affine evaluated-process statistics

$$\bar{s}_{X,Y,j} : \mathcal{C}(X \otimes E, Y) \rightarrow \mathbb{R}$$

such that

$$s_{X,Y,j} = \bar{s}_{X,Y,j} \circ \Gamma_{X,Y} \quad \text{for every } j.$$

Then every $\Gamma_{X,Y}$ is bijective. Consequently the evaluators assemble uniquely into $T_E \dashv R$, with the affine naturality and lax module coherence of Theorem 13.3.

If

$$d_{X,Y} = \dim_{\text{aff}} \mathcal{C}(X, R_Y),$$

then at least $d_{X,Y}$ scalar affine statistics are required by any affine tomography of this program body, and $d_{X,Y}$ suffice abstractly. Thus the finite-dimensional scalar tomography budget is exact. Complete-order and causal compatibility of the resulting adjunction follow under the additional hypotheses of parts (c) and (d) of Theorem 13.3.

Proof. Fix X, Y and suppose

$$\Gamma_{X,Y}(g_1) = \Gamma_{X,Y}(g_2).$$

For every j , factorization of the program statistic through the evaluated process gives

$$\begin{aligned} s_{X,Y,j}(g_1) &= \bar{s}_{X,Y,j}(\Gamma_{X,Y}(g_1)) \\ &= \bar{s}_{X,Y,j}(\Gamma_{X,Y}(g_2)) \\ &= s_{X,Y,j}(g_2). \end{aligned}$$

Joint separation implies $g_1 = g_2$. Hence $\Gamma_{X,Y}$ is injective; it is surjective by hypothesis and therefore bijective. Since this holds for every test object and target, the evaluators are universal. Theorem 13.3 then gives the adjunction and module coherence.

Let $P = \mathcal{C}(X, R_Y)$ have affine dimension d . A family of m affine scalar statistics is a single affine map

$$S : P \rightarrow \mathbb{R}^m.$$

If it separates points, S is injective. The convex set P has nonempty relative interior in its affine hull; hence a nonzero vector in the kernel of the linear part would give two nearby relative-interior points with the same image. The linear part is therefore injective on the d -dimensional direction space of $\text{aff}(P)$, so its rank is d and $m \geq d$. Conversely, an affine coordinate isomorphism $\text{aff}(P) \cong \mathbb{R}^d$ supplies d separating affine statistics on P . Once $\Gamma_{X,Y}$ is bijective, each such statistic factors through it by

$$\bar{s} = s \circ \Gamma_{X,Y}^{-1},$$

and the inverse is affine by Lemma 2.2. This proves the exact tomography count. The final complete-order and causal assertions are precisely the corresponding clauses of Theorem 13.3. $\square$

Theorem 13.5 (Finite experimental tester-frame criterion). Retain the environmental setting of Theorem 13.4. For fixed X, Y , write

$$P = \mathcal{C}(X, R_Y), \quad Q = \mathcal{C}(X \otimes E, Y), \quad \Gamma = \Gamma_{X,Y} : P \rightarrow Q,$$

and assume that Γ is surjective and that $d = \dim_{\text{aff}} P < \infty$. Choose physical programs $p_0, \dots, p_d \in P$ forming an affine basis of $\text{aff}(P)$. Let

$$t_\alpha : Q \longrightarrow [0, 1] \quad (1 \leq \alpha \leq m)$$

be a finite family of evaluated-process tester-outcome probabilities, and put

$$\mathbf{M}(q) = (t_1(q), \dots, t_m(q))^T, \quad q_0 = \Gamma(p_0).$$

The calibrated response-difference matrix is

$$C_{\alpha j} = t_\alpha(\Gamma(p_j)) - t_\alpha(\Gamma(p_0)) \quad \begin{matrix} 1 \leq \alpha \leq m, \\ 1 \leq j \leq d. \end{matrix} \quad (20)$$

Then the following conditions are equivalent:

- (i) the concrete data map $\mathbf{M}\Gamma$ separates all points of P ;
- (ii) $\text{rank } C = d$;
- (iii) the linear parts of the pullback statistics $t_\alpha \circ \Gamma$ span $(\text{dir } \text{aff}(P))^*$;
- (iv) for one, equivalently every, positive diagonal matrix W , the tester frame operator $C^T W C$ is positive definite.

For $d = 0$, these conditions have their empty-matrix interpretation and surjectivity already makes Γ bijective. For $d > 0$, whenever the conditions hold, write the affine coordinates of $p \in P$ as

$$p = p_0 + \sum_{j=1}^d \xi_j(p)(p_j - p_0).$$

They are recovered from the concrete evaluated-process data by

$$\bar{\xi}(q) = (C^T W C)^{-1} C^T W (\mathbf{M}(q) - \mathbf{M}(q_0)), \quad \xi = \bar{\xi} \circ \Gamma. \quad (21)$$

Consequently every affine program statistic factors through an affine postprocessing of these outcome probabilities. The map Γ is bijective, and if the tester-frame condition holds for every X, Y , then $T_E \dashv R$ with the naturality and module coherence of Theorem 13.3.

The criterion has the following exact refinements.

- (a) If the outcomes are grouped into settings $a = 1, \dots, r$, with k_a outcomes satisfying

$$\sum_{b=1}^{k_a} t_{a,b}(q) = 1,$$

then

$$d \leq \sum_{a=1}^r (k_a - 1).$$

Whenever the frame condition holds, a nonsingular $d \times d$ row minor of C selects d scalar outcome probabilities that already separate P . Thus the exact count concerns independent probabilities, not raw normalized outcomes.

- (b) Put

$$\kappa = \sigma_{\min}(W^{1/2}C) > 0.$$

If the measured data are perturbed by $\eta \in \mathbb{R}^m$, the decoder in (21) obeys the calibrated stability estimate

$$\|\hat{\xi} - \xi(p)\|_2 \leq \kappa^{-1} \|W^{1/2}\eta\|_2.$$

This bound uses the displayed experimental frame and chosen coordinate norm; it is not a consequence of affine dimension alone.

(c) If M is informationally complete on Q , then, under surjectivity of Γ ,

$$\text{rank } C = d \iff \dim_{\text{aff}} P = \dim_{\text{aff}} Q.$$

Thus, under surjective programming, an informationally complete evaluated-process tester together with the corresponding affine-dimension equality supplies the factorable program tomography required in Theorem 13.4.

Conversely, if Γ has a nontrivial fiber, every experiment whose statistics depend only on the evaluated process is constant on that fiber. No evaluated-process tester, including one with an ancillary probe, can then supply program tomography.

Proof. Fix X, Y . Since $p_0, \dots, p_d$ are an affine basis, every point of $\text{aff}(P)$ has a unique coordinate vector ξ in

$$p = p_0 + \sum_{j=1}^d \xi_j(p)(p_j - p_0).$$

By Lemma 2.1, Γ and every tester statistic extend uniquely to the relevant affine hulls. Their affinity then gives the central calibration identity

$$M(\Gamma(p)) - M(q_0) = C\xi(p). \quad (22)$$

If C has full column rank, equal data imply equal coordinate vectors and therefore equal programs. Conversely, suppose that $0 \neq v \in \ker C$. Choose $p_* \in \text{relint } P$. The vector

$$u = \sum_{j=1}^d v_j(p_j - p_0)$$

is nonzero because the displayed differences form a basis of $\text{dir aff}(P)$. For sufficiently small $s > 0$, both $p_* + su$ and $p_* - su$ lie in P . Their coordinate vectors differ by $2sv$, so (22) gives identical tester data. Hence (i) and (ii) are equivalent.

In the chosen affine basis, the rows of C are precisely the coordinate vectors of the linear parts of $t_\alpha \circ \Gamma$. They span the dual direction space exactly when C has column rank d , proving equivalence with (iii). For every positive diagonal W and every $z \in \mathbb{R}^d$,

$$z^\top C^\top W C z = \|W^{1/2} C z\|_2^2.$$

This quadratic form is positive away from zero exactly when C has trivial kernel. This proves (iv) and also shows that “one” and “every” positive W are equivalent.

Under these conditions the weighted left inverse

$$C_W^+ = (C^\top W C)^{-1} C^\top W$$

satisfies $C_W^+ C = I_d$. Applying it to (22) proves (21). Every affine statistic on P has the form

$$s(p) = a_0 + a^\top \xi(p).$$

Defining

$$\bar{s}(q) = a_0 + a^\top \bar{\xi}(q)$$

gives the concrete factorization $s = \bar{s} \circ \Gamma$. The data map $M\Gamma$ separates P , so Γ itself is injective; together with the assumed surjectivity it is bijective. Applying this conclusion for every X, Y proves universality of all evaluators, and Theorem 13.3 supplies the adjunction and coherence.

For a normalized setting a , summing the corresponding rows of C gives the zero row. Its row space therefore has dimension at most $k_a - 1$. Summing over settings gives

$$\text{rank } C \leq \sum_a (k_a - 1),$$

which proves the outcome bound. A rank- d matrix has a nonsingular $d \times d$ row minor, and the selected rows retain an injective calibration map. This proves (a).

For perturbed data, the coordinate error is $C_W^+ \eta$. If $B = W^{1/2}C$, then

$$C_W^+ \eta = B^+ W^{1/2} \eta,$$

where $B^+ = (B^\top B)^{-1} B^\top$. The operator norm of B^+ is $\sigma_{\min}(B)^{-1} = \kappa^{-1}$, proving (b).

For (c), let $U = \text{dir aff}(P)$ and $V = \text{dir aff}(Q)$. Surjectivity on the process bodies gives

$$\Gamma(\text{aff}(P)) = \text{aff}(\Gamma(P)) = \text{aff}(Q),$$

so the linear part $G : U \rightarrow V$ of Γ is surjective. Let $A : V \rightarrow \mathbb{R}^m$ be the linear part of $\mathbf{M}$. Informational completeness of $\mathbf{M}$ implies that A is injective: otherwise a nonzero kernel direction would produce two nearby relative-interior points of Q with identical data. In the selected basis, C represents AG . Therefore

$$\text{rank } C = \text{rank}(AG) = \text{rank } G = \dim V = \dim_{\text{aff}} Q.$$

Since $d = \dim U$, full column rank is equivalent to $\dim_{\text{aff}} P = \dim_{\text{aff}} Q$.

Finally, if $p_1 \neq p_2$ satisfy $\Gamma(p_1) = \Gamma(p_2)$, then every statistic t on Q satisfies

$$t(\Gamma(p_1)) = t(\Gamma(p_2)).$$

This includes every ancillary experiment whose probability is a function of the evaluated process. Hence no such tester separates the fiber, proving the last assertion. $\square$

Corollary 13.6 (Faithful-ancilla quantum evaluator tomography). Let A, B be finite-dimensional Hilbert spaces, let $A' \cong A$, and let $Q \subseteq \mathbf{Chan}(A, B)$ be a convex process body. Prepare the normalized full-Schmidt-rank probe

$$\omega_A = |\Omega_A\rangle\langle\Omega_A|, \quad |\Omega_A\rangle = \frac{1}{\sqrt{d_A}} \sum_{i=1}^{d_A} |i\rangle_A |i\rangle_{A'},$$

apply $\Phi \in Q$ to the A system, and measure a finite family of effects F_α on $B \otimes A'$ whose real span is $\text{Herm}(B \otimes A')$. Then the outcome probabilities

$$t_\alpha(\Phi) = \text{Tr}[F_\alpha(\Phi \otimes \text{id}_{A'})](\omega_A)$$

are informationally complete on Q . Such a spanning family always exists as finitely many binary POVM settings.

Consequently, for every surjective affine evaluator $\Gamma : P \rightarrow Q$, the finite response matrix (20) has full column rank exactly when

$$\dim_{\text{aff}} P = \dim_{\text{aff}} Q.$$

Either the rank test itself or this dimension equality therefore certifies the explicit factorization (21). If these conditions hold for every test object and target in a quantum realization of the environmental setting, the evaluators assemble into the right adjoint $T_E \dashv R$.

Proof. Suppose that two channels $\Phi, \Psi \in Q$ give the same probabilities for all α . Their ancillary output difference

$$\Delta = (\Phi \otimes \text{id}_{A'}) (\omega_A) - (\Psi \otimes \text{id}_{A'}) (\omega_A)$$

is Hermitian and satisfies $\text{Tr}(F_\alpha \Delta) = 0$ for every α . Because the F_α span the full real Hermitian space, nondegeneracy of the Hilbert–Schmidt pairing gives $\Delta = 0$. In the convention used here, the Choi operator obeys

$$J(\Phi) = d_A (\Phi \otimes \text{id}_{A'}) (\omega_A).$$

Thus $J(\Phi) = J(\Psi)$, and injectivity of the Choi correspondence gives $\Phi = \Psi$. The tester is informationally complete.

For existence, put $n = d_A d_B$ and choose a real Hermitian basis

$$I, H_1, \dots, H_{n^2-1}$$

with every H_j traceless. Rescale each H_j so that $\|H_j\|_\infty \leq 1$. The pairs

$$F_j^+ = \frac{I + H_j}{2}, \quad F_j^- = \frac{I - H_j}{2}$$

are binary POVMs, and their total real span contains I and every H_j . They therefore supply the required finite family. The remaining assertions are part (c) and the global conclusion of Theorem 13.5. $\square$

13.3 Finite and compact symmetry

Theorem 13.7 (Symmetry-orbit tester and isotypic rank criterion). Retain the finite-dimensional convex bodies P, Q and surjective affine evaluator $\Gamma : P \rightarrow Q$ of Theorem 13.5. Let a finite group G act by affine automorphisms

$$\alpha_g : P \rightarrow P, \quad \beta_g : Q \rightarrow Q,$$

and suppose that Γ is equivariant:

$$\Gamma \alpha_g = \beta_g \Gamma.$$

Let $\mathcal{E} \subseteq \text{Aff}(Q, [0, 1])$ be a physically accessible set of tester outcomes such that

$$t \in \mathcal{E} \implies t \circ \beta_{g^{-1}} \in \mathcal{E}.$$

Choose seed outcomes $t_1, \dots, t_r \in \mathcal{E}$.

Averaging any point of P gives a G -fixed point $p_\circ \in P$; put $q_\circ = \Gamma(p_\circ)$ and $U = \text{dir aff}(P)$. Let π be the dual direction representation on U^* , where $D\alpha_g$ denotes the linear part of α_g on U , defined by

$$(\pi(g)b)(u) = b(D\alpha_{g^{-1}}u),$$

and equip U^* with a G -invariant inner product. Denote by $b_j \in U^*$ the linear part of $t_j \circ \Gamma$. On the complexification, choose the isotypic decomposition

$$U_{\mathbb{C}}^* \cong \bigoplus_{\lambda \in \Lambda} W_\lambda \otimes M_\lambda, \quad \pi_{\mathbb{C}}(g) = \bigoplus_{\lambda} \pi_\lambda(g) \otimes I_{M_\lambda},$$

where the W_λ are pairwise inequivalent irreducible unitary representations and

$$d_\lambda = \dim_{\mathbb{C}} W_\lambda, \quad m_\lambda = \dim_{\mathbb{C}} M_\lambda.$$

For the λ -component $b_{j,\lambda} \in W_\lambda \otimes M_\lambda$, define the contraction map

$$A_{j,\lambda} : \overline{W_\lambda} \rightarrow M_\lambda, \quad A_{j,\lambda}(\overline{w}) = (\langle w, \cdot \rangle \otimes I) b_{j,\lambda},$$

and the multiplicity Gram operator

$$K_\lambda = \sum_{j=1}^r A_{j,\lambda} A_{j,\lambda}^*.$$

Then the following are equivalent:

(i) the finite orbit tester

$$\{t_j \circ \beta_{g^{-1}} : 1 \leq j \leq r, g \in G\}$$

separates every pair of programs in P after evaluation;

(ii) the orbit frame operator

$$S = \sum_{j=1}^r \sum_{g \in G} |\pi(g)b_j\rangle \langle \pi(g)b_j|$$

is positive definite on U^* ;

(iii) K_λ is positive definite on M_λ for every λ ;

(iv) for every λ ,

$$\sum_{j=1}^r \text{im } A_{j,\lambda} = M_\lambda.$$

More precisely, the complexified frame has the exact block form

$$S_{\mathbb{C}} = \bigoplus_{\lambda \in \Lambda} \frac{|G|}{d_\lambda} I_{W_\lambda} \otimes K_\lambda. \quad (23)$$

Consequently, full orbit tomography requires

$$rd_\lambda \geq m_\lambda \quad (\lambda \in \Lambda).$$

If the dual direction representation is multiplicity free, the criterion reduces to requiring that the seeds have a nonzero combined projection onto every irreducible summand.

Whenever these conditions hold, the response matrix formed from all orbit outcomes has full column rank. Its calibrated frame constant in orthonormal affine coordinates is

$$\kappa^2 = \lambda_{\min}(S),$$

so Theorem 13.5 gives explicit factorization, stability, and bijectivity of Γ . If the hypotheses hold for every test object and target, the evaluators assemble into $T_E \dashv R$.

Proof. The averaged point

$$p_\circ = \frac{1}{|G|} \sum_{g \in G} \alpha_g(p)$$

belongs to P by convexity and is fixed by every group element. Equivariance makes $q_\circ = \Gamma(p_\circ)$ fixed as well. The linear part of

$$(t_j \circ \beta_{g^{-1}}) \circ \Gamma = t_j \circ \Gamma \circ \alpha_{g^{-1}}$$

is exactly $\pi(g)b_j$. Thus the rows of the orbit-response matrix are the orbit covectors displayed in the theorem.

A finite family of affine functionals separates the convex body P if and only if its linear parts span U^* . Indeed, spanning immediately gives separation. Conversely, a nonzero common annihilator direction would give two nearby points $p_* \pm su$ in P about a relative-interior point p_* with identical values for every functional. The orbit covectors span U^* if and only if their frame operator is positive definite, because for every $x \in U^*$,

$$\langle x, Sx \rangle = \sum_{j,g} |\langle \pi(g)b_j, x \rangle|^2.$$

This proves equivalence of (i) and (ii).

It remains to compute the frame. For one vector $v \in W_\lambda \otimes M_\lambda$, finite-group twirling on the irreducible factor gives

$$\sum_{g \in G} |(\pi_\lambda(g) \otimes I)v\rangle\langle(\pi_\lambda(g) \otimes I)v| = \frac{|G|}{d_\lambda} I_{W_\lambda} \otimes \text{Tr}_{W_\lambda}(|v\rangle\langle v|).$$

This is Schur averaging: the left side commutes with $\pi_\lambda \otimes I$, its partial trace fixes the scalar, and intertwiners between inequivalent irreducible factors vanish. Standard finite-group representation theory supplies the same isotypic calculation in full generality [8]. By the definition of the contraction map,

$$\text{Tr}_{W_\lambda}(|b_{j,\lambda}\rangle\langle b_{j,\lambda}|) = A_{j,\lambda} A_{j,\lambda}^*.$$

Summing over j and over the isotypic blocks proves (23).

The block formula shows that S is positive definite exactly when every K_λ is. Moreover,

$$\ker K_\lambda = \bigcap_{j=1}^r \ker A_{j,\lambda}^*,$$

whose orthogonal complement is $\sum_j \text{im } A_{j,\lambda}$. Hence (iii) and (iv) are equivalent. Since each $A_{j,\lambda}$ has rank at most d_λ , positivity of K_λ implies

$$m_\lambda = \text{rank } K_\lambda \leq \sum_j \text{rank } A_{j,\lambda} \leq r d_\lambda.$$

When $m_\lambda = 1$, $K_\lambda > 0$ precisely when some seed has nonzero λ -component, proving the multiplicity-free clause.

Finally, (i) is the concrete separation hypothesis of Theorem 13.5. The frame matrix in orthonormal coordinates is $S = C^\top C$, so its least eigenvalue is κ^2 . That theorem supplies the decoder, noise estimate, bijectivity, and global adjunction conclusion. $\square$

Corollary 13.8 (Irreducible one-seed symmetry criterion). In the setting of Theorem 13.7, suppose that $U_\mathbb{C}^*$ is irreducible of complex dimension d and that one accessible seed outcome has nonzero pullback linear part $b \in U^*$. Then its finite symmetry orbit separates all programs and

$$S = \frac{|G|}{d} \|b\|^2 I_{U^*}, \quad \kappa = \sqrt{\frac{|G|}{d}} \|b\|.$$

Thus surjective programming is automatically unique, all affine program statistics factor through this one orbit of experiments, and the evaluator is universal. If this irreducible one-seed condition holds for every source and target, then $T_E \dashv R$.

Proof. There is one isotypic block and its multiplicity space is one-dimensional. The nonzero seed makes

$$K = AA^* = \|b\|^2 > 0.$$

Equation (23) therefore gives the displayed tight frame and its smallest singular value. The orbit criterion proves separation. Equivalently, the linear part of the equivariant evaluator is nonzero because $b \neq 0$; its kernel is an invariant subspace of the irreducible program direction space and hence is zero. Surjectivity then gives bijectivity, and the global statement follows from evaluator assembly. $\square$

Theorem 13.9 (Compact symmetry and finite orbit quadrature). Replace the finite group in Theorem 13.7 by a compact Hausdorff group K with normalized Haar probability measure μ . Assume that the affine actions on P, Q are continuous, that Γ is exactly equivariant, and that the

accessible tester outcomes are closed under the K -action. Let $U = \text{dir aff}(P)$ have real dimension d , choose a K -invariant inner product, and let $b_1, \dots, b_r \in U^*$ be the pullback linear parts of accessible seed outcomes. For the finite-dimensional continuous representation, write

$$U_{\mathbb{C}}^* \cong \bigoplus_{\lambda \in \Lambda} W_{\lambda} \otimes M_{\lambda}$$

and define $A_{j,\lambda}$ and K_{λ} exactly as in Theorem 13.7. Then:

- (a) the normalized Haar frame

$$S_H = \sum_{j=1}^r \int_K |\pi(g)b_j\rangle \langle \pi(g)b_j| d\mu(g)$$

has the exact isotypic form

$$(S_H)_{\mathbb{C}} = \bigoplus_{\lambda \in \Lambda} \frac{1}{d_{\lambda}} I_{W_{\lambda}} \otimes K_{\lambda}; \quad (24)$$

- (b) the compact orbit tester separates P after evaluation if and only if $K_{\lambda} > 0$ for every λ , equivalently

$$\sum_j \text{im } A_{j,\lambda} = M_{\lambda} \quad (\lambda \in \Lambda);$$

under these conditions its frame constant is $\kappa^2 = \lambda_{\min}(S_H)$;

- (c) if $S_H > 0$ and $d > 0$, there are

$$N \leq \frac{d(d+1)}{2}$$

seed-group pairs (j_k, g_k) and positive weights w_k such that

$$S_H = \sum_{k=1}^N w_k |\pi(g_k)b_{j_k}\rangle \langle \pi(g_k)b_{j_k}|. \quad (25)$$

Hence a finite weighted physical subtester has exactly the Haar frame, the same rank, and the same calibrated stability constant;

- (d) if $U_{\mathbb{C}}^*$ is irreducible and one seed b is nonzero, then

$$S_H = \frac{\|b\|^2}{d} I_{U^*}, \quad \kappa = \frac{\|b\|}{\sqrt{d}},$$

and a finite weighted orbit subtester with at most $d(d+1)/2$ outcomes already forces evaluator uniqueness.

Thus exact compact symmetry does not require an infinite experimental family: finite-dimensional orbit tomography always admits an exact finite weighted quadrature. If the criterion holds for every source and target, then $T_E \dashv R$.

Proof. Averaging any inner product on U^* against Haar measure makes the direction representation unitary. Orthogonal complements of invariant subspaces are invariant, so the finite-dimensional complexified representation decomposes into the displayed isotypic sum. Normalized Haar measure is left and right invariant; its standard existence and the corresponding compact-group Schur orthogonality are reviewed in [9].

For $v \in W_{\lambda} \otimes M_{\lambda}$, Haar twirling gives

$$\int_K |(\pi_{\lambda}(g) \otimes I)v\rangle \langle (\pi_{\lambda}(g) \otimes I)v| d\mu(g) = \frac{1}{d_{\lambda}} I_{W_{\lambda}} \otimes \text{Tr}_{W_{\lambda}}(|v\rangle \langle v|).$$

Indeed, invariance makes the integral an intertwiner; Schur's lemma removes cross terms between inequivalent irreducibles and makes the irreducible factor scalar, while the partial trace fixes that scalar. Since

$$\mathrm{Tr}_{W_\lambda} |b_{j,\lambda}\rangle\langle b_{j,\lambda}| = A_{j,\lambda} A_{j,\lambda}^*,$$

summing over the seeds proves (24). Positivity of the frame is therefore equivalent to positivity of every K_λ , and

$$(\ker K_\lambda)^\perp = \sum_j \mathrm{im} A_{j,\lambda}$$

gives the image criterion. As in Theorem 13.7, positivity is equivalent to separation of the program body. This proves (a) and (b).

For (c), put

$$D = \dim_{\mathbb{R}} \mathrm{Sym}(U^*) = \frac{d(d+1)}{2}, \quad T = \mathrm{Tr} S_H = \sum_j \|b_j\|^2.$$

Since $S_H > 0$, one has $T > 0$. Ignore zero seeds and consider the compact set of trace-one rank-one operators

$$\mathfrak{R} = \left\{ \frac{|\pi(g)b_j\rangle\langle\pi(g)b_j|}{\|b_j\|^2} : g \in \mathbf{K}, b_j \neq 0 \right\} \subseteq \mathrm{Sym}(U^*).$$

The normalized frame S_H/T is the barycenter of a probability measure on $\mathfrak{R}$ that chooses seed j with probability $\|b_j\|^2/T$ and then chooses g with Haar measure. The convex hull of $\mathfrak{R}$ is compact: ordinary Carathéodory reduction expresses each of its points using at most $D+1$ generators, so it is the continuous image of a compact finite product and simplex. The barycenter belongs to this convex hull; otherwise strict separation from that compact convex set would give an affine functional larger on the barycenter than on every point of $\mathfrak{R}$, contradicting integration.

All elements of $\mathfrak{R}$ lie in the affine hyperplane of trace-one symmetric operators, whose dimension is at most $D-1$. Carathéodory reduction therefore writes the barycenter as a convex combination of at most D points. If a convex representation uses more than D points, its affine dependence supplies coefficients summing to zero; moving the weights along that dependence until one becomes zero preserves the barycenter and removes one point. Iteration gives the bound. Thus there are $\theta_k > 0$, $\sum_k \theta_k = 1$, and orbit vectors $v_k = \pi(g_k)b_{j_k}$ such that

$$\frac{S_H}{T} = \sum_{k=1}^N \theta_k \frac{|v_k\rangle\langle v_k|}{\|b_{j_k}\|^2}.$$

Taking $w_k = T\theta_k/\|b_{j_k}\|^2$ proves (25). Every selected outcome is physically accessible by symmetry closure, and the positive weights are precisely a diagonal tester-frame weight matrix.

In the irreducible one-seed case, Schur averaging leaves one block and $K = \|b\|^2$, proving (d). The finite weighted subtester has a positive frame, so Theorem 13.5 gives exact factorization and bijectivity. Applying this at every source and target gives the adjunction. $\square$

Corollary 13.10 (Robustness under approximate orbit implementation). Let the finite weighted orbit subtester of Theorem 13.9 have ideal pullback vectors $v_k \in U^*$, weights $w_k > 0$, frame

$$S = \sum_k w_k |v_k\rangle\langle v_k|, \quad \gamma = \lambda_{\min}(S) > 0.$$

Suppose the actually implemented affine outcomes still factor through the actual evaluator and have pullback vectors

$$\tilde{v}_k = v_k + e_k, \quad \|e_k\| \leq \delta_k.$$

Define

$$\varepsilon = \sum_k w_k (2\|v_k\|\delta_k + \delta_k^2).$$

If $\varepsilon < \gamma$, then the actual tester frame satisfies

$$\tilde{S} = \sum_k w_k |\tilde{v}_k\rangle\langle\tilde{v}_k| \succeq (\gamma - \varepsilon)I,$$

so it remains informationally complete and has

$$\tilde{\kappa} \geq \sqrt{\gamma - \varepsilon}.$$

Consequently, surjective programming remains exactly unique and the actual outcomes still induce the adjunction when this condition holds for every source and target.

For the normalized-Haar irreducible one-seed frame, the quadrature weights may be chosen with $\sum_k w_k = 1$ and $\|v_k\| = \|b\|$. A uniform error bound $\delta_k \leq \delta$ is therefore sufficient whenever

$$2\|b\|\delta + \delta^2 < \frac{\|b\|^2}{d}.$$

Proof. For each k ,

$$|\tilde{v}_k\rangle\langle\tilde{v}_k| - |v_k\rangle\langle v_k| = |v_k\rangle\langle e_k| + |e_k\rangle\langle v_k| + |e_k\rangle\langle e_k|.$$

The operator norm is at most $2\|v_k\|\|e_k\| + \|e_k\|^2$. Summing with weights gives

$$\|\tilde{S} - S\| \leq \varepsilon.$$

For every unit vector x ,

$$\langle x, \tilde{S}x \rangle \geq \langle x, Sx \rangle - \varepsilon \geq \gamma - \varepsilon,$$

which proves the frame inequality and the bound on $\tilde{\kappa}$. The actual outcome rows therefore have full rank. Theorem 13.5 gives exact evaluator injectivity, not merely approximate recovery; surjectivity gives bijectivity and evaluator assembly gives the adjunction.

For one irreducible seed, the normalized Haar frame has $\gamma = \|b\|^2/d$. In the Carathéodory construction all orbit vectors have norm $\|b\|$, and tracing the quadrature identity gives $\sum_k w_k = 1$. Substitution into the general estimate proves the final condition. $\square$

14 Representability, Broadcasting, and Causal Restriction

The preceding no-go arguments concern representability of hom-functors. The fixed-codomain consequence is isolated first, followed by conditions under which an adjunction restricts to causal morphisms.

14.1 Hom-presheaf nonrepresentability

Definition 14.1 (Affine representability). The presheaf

$$X \mapsto \mathcal{C}(TX, Y_0)$$

is *affinely representable* if there is an object Z and a family of affine bijections

$$\mathcal{C}(TX, Y_0) \cong \mathcal{C}(X, Z)$$

natural in X .

Theorem 14.1 (Hom-presheaf obstruction). Fix Y_0 . Suppose Y_0 and every candidate representer Z belong to a class whose profiles satisfy

$$\ell(p_Z) = c \neq 0, \quad \ell \circ T^* = \lambda \ell, \quad \lambda \neq 1.$$

Then the presheaf $X \mapsto \mathcal{C}(TX, Y_0)$ is not affinely representable.

Proof. Suppose that Z affinely represents the presheaf. Then for every test object X there is an affine bijection

$$\mathcal{C}(TX, Y_0) \cong \mathcal{C}(X, Z).$$

Equality of affine dimensions gives

$$D(TX, Y_0) = D(X, Z)$$

for every X . Therefore the corresponding profiles agree pointwise:

$$T^*p_{Y_0} = p_Z.$$

Applying ℓ and using the two hypotheses gives

$$\lambda c = \lambda \ell(p_{Y_0}) = \ell(T^*p_{Y_0}) = \ell(p_Z) = c.$$

Since $c \neq 0$, this implies $\lambda = 1$, contradicting the assumption $\lambda \neq 1$. Hence no affine representer exists. $\square$

Remark 14.2 (Effect-presheaf terminology). When $Y_0 = I$, the functor

$$X \longmapsto \mathcal{C}(TX, I)$$

is the effect presheaf after precomposition by T . For arbitrary Y_0 , the term hom-presheaf is used. Theorem 14.1 applies to both cases.

Proposition 14.3 (Representability does not imply broadcasting). In the category of finite-dimensional Hilbert spaces and arbitrary completely positive maps, the presheaf

$$X \longmapsto \mathcal{C}(X \otimes E, I)$$

is representable for every finite-dimensional E . If $d_E > 1$, however, there is no channel

$$\mathcal{B} : E \longrightarrow E \otimes E$$

that broadcasts every density operator, in the sense that both reduced states of $\mathcal{B}(\rho)$ equal ρ for every ρ . Consequently, hom-presheaf representability does not entail a universal broadcasting map.

Proof. The category of arbitrary finite-dimensional completely positive maps has the standard Bell compact structure. Compact transposition therefore gives

$$\mathcal{C}(X \otimes E, I) \cong \mathcal{C}(X, \overline{E})$$

naturally in X , proving representability.

Suppose that a broadcasting channel $\mathcal{B}$ existed. If $\rho = |\psi\rangle\langle\psi|$ is pure, both marginals of $\mathcal{B}(\rho)$ are the same pure state. A bipartite positive operator with a pure marginal is supported on the corresponding one-dimensional tensor factor; applying this to both marginals gives

$$\mathcal{B}(|\psi\rangle\langle\psi|) = |\psi \otimes \psi\rangle\langle\psi \otimes \psi|.$$

Let $V : E \rightarrow E \otimes E \otimes K$ be a Stinespring isometry for $\mathcal{B}$. Since the reduced output above is pure, for every unit vector ψ there is a unit vector $a_\psi \in K$ such that

$$V\psi = \psi \otimes \psi \otimes a_\psi.$$

Choose distinct nonorthogonal unit vectors ψ, φ and put $c = |\langle \varphi, \psi \rangle|$, so $0 < c < 1$. Preservation of inner products by V gives

$$c = c^2 |\langle a_\varphi, a_\psi \rangle| \leq c^2,$$

which is impossible for $0 < c < 1$. Hence no universal broadcasting channel exists when $d_E > 1$. $\square$

14.2 Restriction to causal morphisms

Let $\mathcal{C}$ be symmetric monoidal and equipped with a monoidal discarding family, as in categorical probability, convex state–effect duality, and effectus-based treatments [11, 37, 33]:

$$\text{del}_X : X \rightarrow I, \quad \text{del}_{X \otimes Y} = \text{del}_X \otimes \text{del}_Y, \quad \text{del}_I = \text{id}_I.$$

A map $f : X \rightarrow Y$ is normalized if

$$\text{del}_Y \circ f = \text{del}_X.$$

Such a morphism is also called causal. In the concrete channel, instrument, comb, and superchannel categories, *deterministic* denotes this same normalization condition with the theory-specific discard map. Assume tensoring and composition are affine on hom-sets. The normalized hom-sets are convex because the normalization equation is affine.

Proposition 14.4 (Normalized adjunction restriction). Let $T_E = - \otimes E \dashv R$.

(i) If the counit is normalized, explicitly

$$\text{del}_Y \circ \epsilon_Y = \text{del}_{T_E(RY)} = \text{del}_{RY \otimes E},$$

then inverse transposition carries normalized maps to normalized maps.

(ii) If, in addition, the unit is normalized,

$$\text{del}_{R(T_EX)} \circ \eta_X = \text{del}_X,$$

and R carries normalized maps to normalized maps, then the adjunction restricts to affine isomorphisms

$$\mathcal{C}_{\text{norm}}(X \otimes E, Y) \cong \mathcal{C}_{\text{norm}}(X, RY).$$

Proof. Let $g : X \rightarrow RY$ be normalized. Then

$$\begin{aligned} \text{del}_Y \circ \epsilon_Y \circ (g \otimes \text{id}_E) &= \text{del}_{RY \otimes E} \circ (g \otimes \text{id}_E) \\ &= (\text{del}_{RY} \circ g) \otimes \text{del}_E \\ &= \text{del}_X \otimes \text{del}_E = \text{del}_{X \otimes E}. \end{aligned}$$

This proves (i).

Let $f : X \otimes E \rightarrow Y$ be normalized. If R preserves normalization, then

$$\text{del}_{RY} \circ R(f) = \text{del}_{R(X \otimes E)}.$$

Hence

$$\text{del}_{RY} \circ R(f) \circ \eta_X = \text{del}_{R(X \otimes E)} \circ \eta_X = \text{del}_X.$$

Thus forward transposition preserves normalization. The two restrictions are inverse bijections. They are affine by Proposition 2.3 and restriction to convex normalization fibers. $\square$

14.3 Automatic preservation of normalized morphisms

The preceding proposition gives sufficient conditions for restriction to the normalized subcategory. The inverse direction admits an exact characterization, and the forward direction yields corresponding necessary conditions.

Proposition 14.5 (Inverse normalized restriction). Let $T_E = - \otimes E \dashv R$. Inverse transposition preserves normalized morphisms for every X, Y if and only if the counit is normalized:

$$\mathrm{del}_Y \circ \epsilon_Y = \mathrm{del}_{RY \otimes E}.$$

Proof. If the counit is normalized, inverse preservation is Proposition 14.4(i).

Conversely, suppose inverse transposition preserves every normalized morphism. Set $X = RY$ and take $g = \mathrm{id}_{RY}$, which is normalized. Its inverse transpose is

$$\Phi_{RY, Y}^{-1}(\mathrm{id}_{RY}) = \epsilon_Y \circ T_E(\mathrm{id}_{RY}) = \epsilon_Y.$$

The hypothesis therefore makes ϵ_Y normalized for every Y . □

Proposition 14.6 (Necessary conditions for forward normalized restriction). Suppose $T_E = - \otimes E \dashv R$ and forward transposition preserves normalized morphisms for every X, Y .

(a) Every unit component is normalized:

$$\mathrm{del}_{R(T_EX)} \circ \eta_X = \mathrm{del}_X.$$

(b) If the counit is normalized, then R preserves normalized morphisms.

Proof. For (a), the identity id_{T_EX} is normalized, and its forward transpose is

$$\Phi_{X, T_EX}(\mathrm{id}_{T_EX}) = \eta_X.$$

Forward preservation therefore makes η_X normalized.

For (b), let $h : Y \rightarrow Z$ be normalized. If ϵ_Y is normalized, then

$$\begin{aligned} \mathrm{del}_Z \circ h \circ \epsilon_Y &= \mathrm{del}_Y \circ \epsilon_Y \\ &= \mathrm{del}_{T_E(RY)}, \end{aligned}$$

so $h \circ \epsilon_Y : T_E(RY) \rightarrow Z$ is normalized. Its forward transpose is

$$\begin{aligned} \Phi_{RY, Z}(h \circ \epsilon_Y) &= R(h \circ \epsilon_Y) \circ \eta_{RY} \\ &= R(h) \circ R(\epsilon_Y) \circ \eta_{RY} \\ &= R(h), \end{aligned}$$

where the last equality is the triangle identity. Forward preservation makes $R(h)$ normalized. □

Corollary 14.7 (Forward normalized restriction with normalized counit). Assume that the counit is normalized. Forward transposition preserves normalized morphisms for every X, Y if and only if the unit is normalized and R preserves normalized morphisms.

Proof. Assume first that forward transposition preserves normalized morphisms. For each X , the identity on T_EX is normalized and its forward transpose is the unit

$$\Phi_{X, T_EX}(\mathrm{id}_{T_EX}) = \eta_X,$$

so every η_X is normalized. Let $h : Y \rightarrow Z$ be normalized. Since the counit is normalized, so is $h \circ \epsilon_Y : T_E(RY) \rightarrow Z$. Functoriality of R and the triangle identity identify its forward transpose as

$$\Phi_{RY,Z}(h \circ \epsilon_Y) = R(h) \circ (R(\epsilon_Y) \circ \eta_{RY}) = R(h).$$

Forward preservation therefore makes $R(h)$ normalized. Thus normalization of the unit and preservation of normalized morphisms by R are necessary.

Conversely, suppose that the unit is normalized and that R preserves normalized morphisms. For a normalized map $f : T_EX \rightarrow Y$, its forward transpose is

$$\Phi_{X,Y}(f) = R(f) \circ \eta_X.$$

The map $R(f)$ is normalized by hypothesis, η_X is normalized, and composition of normalized morphisms is normalized. Hence every forward transpose of a normalized morphism is normalized, proving sufficiency. $\square$

Example 14.8 (Ordinary adjunctions need not preserve normalization). Let $\mathbf{Vect}_\Sigma$ have objects $\mathbb{R}^n$ for $n \geq 1$, all linear maps as morphisms, the usual tensor product with the standard identifications $\mathbb{R}^m \otimes \mathbb{R}^n \cong \mathbb{R}^{mn}$, and tensor unit $\mathbb{R}$. Define

$$\text{del}_n(x_1, \dots, x_n) = \sum_{i=1}^n x_i.$$

Then

$$\text{del}_{mn} = \text{del}_m \otimes \text{del}_n, \quad \text{del}_1 = \text{id}_{\mathbb{R}}.$$

A matrix is normalized precisely when every column sums to one. Positivity is not imposed.

Fix $E = \mathbb{R}^e$ with $e > 1$. The functor $T_E = - \otimes E$ has the standard right adjoint

$$R_0(Y) = E^* \otimes Y.$$

Its counit is evaluation,

$$\epsilon_Y((\varphi \otimes y) \otimes v) = \varphi(v)y.$$

For $Y = \mathbb{R}$, take

$$\varphi = (1, 0, \dots, 0) \in E^*, \quad v = (0, 1, 0, \dots, 0) \in E.$$

Both φ and v have coordinate sum one, but $\varphi(v) = 0$. Hence

$$\text{del}_Y \circ \epsilon_Y \neq \text{del}_{E^* \otimes Y \otimes E},$$

so the counit is not normalized.

The unit is

$$\eta_X(x) = \sum_{i=1}^e e_i^* \otimes x \otimes e_i$$

for dual bases (e_i^*) and (e_i) . Therefore

$$\text{del}_{R_0(T_EX)} \eta_X(x) = e \text{del}_X(x),$$

so the unit is not normalized when $e > 1$.

The normalized map

$$g : \mathbb{R} \rightarrow E^* \otimes \mathbb{R}, \quad g(1) = \varphi \otimes 1,$$

has inverse transpose

$$f : E \rightarrow \mathbb{R}, \quad f(v) = \varphi(v),$$

which is not normalized because $\varphi \neq \text{del}_E$. Thus inverse transposition fails to preserve normalization; forward preservation fails as well because the unit is not normalized.

The standard functor R_0 nevertheless preserves normalized maps. To show that preservation by a chosen right adjoint is not automatic, define, for $Y = \mathbb{R}^n$,

$$S_Y = n \text{ id}_{E^* \otimes Y}$$

and the naturally isomorphic functor

$$R_1(Y) = R_0(Y), \quad R_1(h) = S_Z R_0(h) S_Y^{-1}.$$

The family $S : R_0 \Rightarrow R_1$ is a natural isomorphism, so transporting the adjunction across S gives $T_E \dashv R_1$. If

$$h : \mathbb{R} \rightarrow \mathbb{R}^2, \quad h(1) = (1, 0),$$

then h is normalized but

$$R_1(h) = 2(\text{id}_{E^*} \otimes h)$$

has column sums equal to two and is not normalized. Thus ordinary adjointness alone forces none of the normalization properties appearing in Proposition 14.4.

15 Compact Closure and Causality

The preceding sections identify failures of environmental currying in normalized process categories. Compact closure supplies the complementary positive regime, while terminality shows why that regime cannot remain nontrivially fully causal. Compact closure and its coherence are treated in [24]; the categorical quantum-mechanical and completely positive settings are developed in [31, 25].

Proposition 15.1 (Environmental currying in compact-closed categories). In a symmetric compact-closed category, $- \otimes E$ has right adjoint $- \otimes E^*$. Explicitly,

$$\mathcal{C}(X \otimes E, Y) \cong \mathcal{C}(X, Y \otimes E^*)$$

naturally in X, Y . If the category is convex or linear enriched and the compact transposition respects that enrichment, then the isomorphism is affine or linear and

$$T_E^* p_Y = p_{Y \otimes E^*}.$$

Proof. Let

$$\eta_E : I \rightarrow E \otimes E^*, \quad \epsilon_E : E^* \otimes E \rightarrow I$$

be coevaluation and evaluation. For $f : X \otimes E \rightarrow Y$, define

$$\hat{f} = (f \otimes \text{id}_{E^*}) \circ (\text{id}_X \otimes \eta_E) : X \rightarrow Y \otimes E^*.$$

For $g : X \rightarrow Y \otimes E^*$, define

$$\tilde{g} = (\text{id}_Y \otimes \epsilon_E) \circ (g \otimes \text{id}_E) : X \otimes E \rightarrow Y.$$

Associators and unitors are suppressed by coherence. The two snake identities show that these constructions are inverse, and functoriality of tensor and composition gives naturality in X, Y . If compact transposition respects the convex or linear enrichment, these are affine or linear isomorphisms. Taking affine dimensions gives $T_E^* p_Y = p_{Y \otimes E^*}$. $\square$

The finite-dimensional completely positive realization makes the failure of causal restriction explicit; see also [6].

Proposition 15.2 (Bell compact currying does not restrict to channels). Let E be a nonzero finite-dimensional complex Hilbert space. In the category of finite-dimensional Hilbert spaces and arbitrary completely positive maps, the standard Bell compact structure gives

$$- \otimes E \dashv - \otimes \overline{E}.$$

This adjunction restricts to the trace-preserving subcategory **Chan** if and only if $d_E = 1$. If $d_E > 1$, no scalar multiple of its Bell counit is trace preserving.

Proof. Choose an orthonormal basis $(e_i)_{i=1}^{d_E}$ of E and put

$$|\Omega_E\rangle = \sum_{i=1}^{d_E} e_i \otimes \bar{e}_i, \quad w_E(\bar{e}_i \otimes e_j) = \delta_{ij}.$$

The unit component at H is the completely positive map

$$\eta_H(\rho) = \rho \otimes |\Omega_E\rangle\langle\Omega_E|.$$

Since $\langle\Omega_E | \Omega_E\rangle = d_E$, one has

$$\text{Tr } \eta_H(\rho) = d_E \text{Tr } \rho.$$

Thus the Bell unit is trace preserving only when $d_E = 1$.

The counit component at K has one Kraus operator

$$A = I_K \otimes w_E : K \otimes \overline{E} \otimes E \longrightarrow K, \quad \epsilon_K(\sigma) = A\sigma A^\dagger.$$

A one-Kraus completely positive map is trace preserving exactly when $A^\dagger A$ is the identity on its input space. Here

$$A^\dagger A = I_K \otimes w_E^\dagger w_E = I_K \otimes |\Theta_E\rangle\langle\Theta_E|, \quad |\Theta_E\rangle = \sum_{i=1}^{d_E} \bar{e}_i \otimes e_i.$$

The factor $|\Theta_E\rangle\langle\Theta_E|$ has rank one on the d_E^2 -dimensional space $\overline{E} \otimes E$. Hence, for $d_E > 1$, $A^\dagger A$ is not the identity, and multiplying $A^\dagger A$ by a scalar cannot repair the rank deficiency. The counit is therefore not trace preserving and admits no trace-preserving scalar rescaling.

Proposition 14.5 shows that a nonnormalized counit already prevents inverse transposition from restricting to channels; the trace-scaling unit independently violates the necessary forward restriction condition of Proposition 14.6. If $d_E = 1$, then $E \cong \mathbb{C}$, the two tensor functors are canonically the identity, and the Bell unit and counit are the unitors. They are trace preserving, so the adjunction restricts. This proves both directions. $\square$

Remark 15.3 (Weighted compact structures). Let $(e_i)_{i=1}^d$ be an orthonormal basis and let $\lambda_i > 0$ with $\sum_i \lambda_i = 1$. The normalized weighted cup and its snake-compatible inverse cap are

$$|\Omega_\lambda\rangle = \sum_{i=1}^d \sqrt{\lambda_i} e_i \otimes \bar{e}_i, \quad w_\lambda(\bar{e}_i \otimes e_j) = \frac{\delta_{ij}}{\sqrt{\lambda_i}}.$$

The Riesz vector of w_λ is $\sum_i \lambda_i^{-1/2} \bar{e}_i \otimes e_i$. Since its summands are orthogonal,

$$\|w_\lambda\|^2 = \sum_{i=1}^d \frac{1}{\lambda_i}.$$

Thus the functional norm is the square root of the reciprocal sum. For a countably infinite faithful weight sequence the reciprocal sum diverges, so this inverse cap is unbounded.

The finite-dimensional counit obstruction is stronger than this norm calculation and is independent of the weights. Any linear cap $w : \overline{E} \otimes E \rightarrow \mathbb{C}$ induces a one-Kraus completely positive counit with normalization operator

$$I_K \otimes w^\dagger w.$$

For $w \neq 0$, the factor $w^\dagger w$ has rank one; for $w = 0$, it has rank zero. Hence no weighted finite-dimensional cup–cap duality can have a trace-preserving counit when $d_E > 1$.

Proposition 15.4 (Sharp normalized weighted-cup defect). Let H be a δ -dimensional Hilbert space, let $\lambda_i > 0$ with $\sum_{i=1}^\delta \lambda_i = 1$, and use the weighted cup and inverse cap from the preceding remark. The cup induces a trace-preserving completely positive unit. Scale the completely positive counit at the superoperator level by $s \geq 0$ and require it to be trace nonincreasing. Then both triangle composites equal $s \text{id}$, and in every norm with $\|\text{id}\| = 1$ their common defect satisfies

$$\varepsilon_\Delta = \|s \text{id} - \text{id}\| \geq 1 - \frac{1}{\delta^2}.$$

The bound is sharp: equality is attained by uniform weights and $s = 1/\delta^2$. For $\delta > 1$, no scalar choice makes the counit trace preserving.

Proof. The cup vector has squared norm

$$\sum_i \lambda_i = 1,$$

so tensoring with its rank-one projector preserves trace. Put

$$|\Theta_\lambda\rangle = \sum_i \lambda_i^{-1/2} \bar{e}_i \otimes e_i.$$

For the one-Kraus counit operator $A = I \otimes w_\lambda$,

$$A^\dagger A = I \otimes |\Theta_\lambda\rangle\langle\Theta_\lambda|.$$

Its nonzero eigenvalue on the contracted factor is

$$\|\Theta_\lambda\|^2 = \sum_i \frac{1}{\lambda_i}.$$

The scaled counit is trace nonincreasing exactly when

$$sA^\dagger A \leq I,$$

which is equivalent to

$$s \sum_i \frac{1}{\lambda_i} \leq 1.$$

Cauchy–Schwarz gives

$$\left(\sum_i \lambda_i\right) \left(\sum_i \frac{1}{\lambda_i}\right) \geq \left(\sum_i 1\right)^2 = \delta^2.$$

Thus $0 \leq s \leq 1/\delta^2 \leq 1$.

The unscaled weighted cup and cap satisfy the snake identities. Scaling the counit superoperator by s therefore makes either triangle composite $s \text{id}$. Absolute homogeneity of the norm and $s \leq 1$ give

$$\varepsilon_\Delta = \|(s - 1) \text{id}\| = 1 - s \geq 1 - \frac{1}{\delta^2}.$$

For uniform weights $\lambda_i = 1/\delta$, the reciprocal sum is δ^2 , so $s = 1/\delta^2$ is admissible and attains equality. Exact trace preservation would require $sA^\dagger A = I$; for $\delta > 1$ the left side has rank one on a δ^2 -dimensional contracted factor, which is impossible. $\square$

Remark 15.5. For a cup on the environment object, $\delta = d_E$. For the pair object $\bar{E} \otimes E$, one has $\delta = d_E^2$, and the sharp bound becomes $1 - d_E^{-4}$ within this weighted cup-cap family.

Theorem 15.6 (Causal compact closure collapses). If a compact-closed category has terminal tensor unit I , then every hom-set is a singleton.

Proof. Compact closure gives natural bijections

$$\mathcal{C}(A, B) \cong \mathcal{C}(I, A^* \otimes B).$$

Duality also gives

$$\mathcal{C}(I, C) \cong \mathcal{C}(C^*, I).$$

Since I is terminal, $\mathcal{C}(C^*, I)$ is a singleton for every C . Hence $\mathcal{C}(I, C)$ and then $\mathcal{C}(A, B)$ are singletons. $\square$

Corollary 15.7 (Nontrivial causal process categories are not compact closed). A nontrivial category in which every morphism preserves discarding has terminal tensor unit and therefore cannot be compact closed.

Proof. If every $f : X \rightarrow I$ is causal, then $\text{del}_I \circ f = \text{del}_X$. Since $\text{del}_I = \text{id}_I$, one has $f = \text{del}_X$, so I is terminal. Theorem 15.6 then makes every hom-set a singleton, contrary to nontriviality. $\square$

Remark 15.8. Unnormalized linear categories such as finite-dimensional vector spaces may be compact closed because their tensor unit is not terminal. Normalized channel categories have terminal tensor unit and are not compact closed.

16 Quantitative Geometric Defects

The geometric separation associated with the preceding exact obstructions can be quantified without assigning it an operational error metric. Choose a norm on the finite-dimensional profile space U .

16.1 Profile and affine-rank defects

Definition 16.1 (Profile defect).

$$\delta_T(Y) = \text{dist}(T^*p_Y, \mathfrak{P}) = \inf_Z \|T^*p_Y - p_Z\|.$$

Proposition 16.1 (Quantitative profile bound). If $\delta_T(Y) > 0$, no object RY can satisfy

$$D(TX, Y) = D(X, RY) \quad \text{for every } X.$$

If all realizable profiles satisfy $\ell(p_Z) = c \neq 0$ and $\ell \circ T^* = \lambda\ell$, then

$$\delta_T(Y) \geq \frac{|(\lambda - 1)c|}{\|\ell\|_*}.$$

Proof. Suppose first that an object RY satisfied $D(TX, Y) = D(X, RY)$ for every X . Evaluating the two profiles at each test object gives

$$(T^*p_Y)(X) = D(TX, Y) = D(X, RY) = p_{RY}(X).$$

Hence $T^*p_Y = p_{RY} \in \mathfrak{P}$, and its distance from $\mathfrak{P}$ is zero. This contradicts $\delta_T(Y) > 0$.

For the lower bound, let Z be any realizable object. Since both Y and Z are realizable,

$$\ell(p_Y) = \ell(p_Z) = c.$$

Therefore

$$\begin{aligned}\ell(T^*p_Y - p_Z) &= (\ell \circ T^*)(p_Y) - \ell(p_Z) \\ &= \lambda \ell(p_Y) - \ell(p_Z) \\ &= (\lambda - 1)c.\end{aligned}$$

By definition of the dual norm,

$$|\ell(v)| \leq \|\ell\|_* \|v\| \quad (v \in U).$$

Taking $v = T^*p_Y - p_Z$ yields

$$\|T^*p_Y - p_Z\| \geq \frac{|(\lambda - 1)c|}{\|\ell\|_*}.$$

The inequality holds for every realizable Z ; taking the infimum over Z proves the stated bound. $\square$

Proposition 16.2 (Exact instrument profile defect). For quantum instruments, use the basis (f_1, f_2) from Section 9 and the norm

$$\|af_1 + bf_2\|_1 = |a| + |b|.$$

Then

$$\delta_{T_E}(B) = e^2 - 1.$$

Proof. Write $b = d_B$, $e = d_E$, and $z = d_Z$. Environmental pullback gives

$$T_E^*p_B = e^2b^2f_1 + e^2f_2,$$

whereas every realizable profile has the form

$$p_Z = z^2f_1 + f_2.$$

Their ℓ^1 distance is therefore

$$\|T_E^*p_B - p_Z\|_1 = |e^2b^2 - z^2| + |e^2 - 1|.$$

Since $e \geq 1$, the second term equals $e^2 - 1$, so every realizable Z satisfies

$$\|T_E^*p_B - p_Z\|_1 \geq e^2 - 1.$$

The object $Z = B \otimes E$ is available and has $z = be$. For this choice the first term vanishes, and the distance is exactly $e^2 - 1$. Hence the infimum over all realizable Z is $e^2 - 1$. $\square$

Let C_1, C_2 be nonempty finite-dimensional convex sets and let $\Phi : C_1 \rightarrow C_2$ be affine. By Lemma 2.1, Φ has a unique affine extension $\bar{\Phi}$ to the affine hulls. Let L be the linear part of $\bar{\Phi}$ and put

$$k(\Phi) = \dim \ker L, \quad c(\Phi) = \dim_{\text{aff}} C_2 - \text{rank } L.$$

Proposition 16.3 (Exact affine rank–nullity defect). If $d_i = \dim_{\text{aff}} C_i$ and $r = \text{rank } L$, then

$$\begin{aligned}k(\Phi) + c(\Phi) &= d_1 + d_2 - 2r \\ &= |d_1 - d_2| + 2(\min(d_1, d_2) - r) \\ &\geq |d_1 - d_2|.\end{aligned}$$

Proof. The linear part L has domain dimension d_1 and rank r . Rank-nullity gives

$$k(\Phi) = \dim \ker L = d_1 - r.$$

By definition, the codimension of its image in the codomain direction space is

$$c(\Phi) = d_2 - r.$$

Adding these expressions gives

$$k(\Phi) + c(\Phi) = d_1 + d_2 - 2r.$$

If $d_1 \leq d_2$, then

$$d_1 + d_2 - 2r = (d_2 - d_1) + 2(d_1 - r).$$

If $d_2 \leq d_1$, then

$$d_1 + d_2 - 2r = (d_1 - d_2) + 2(d_2 - r).$$

Both cases are summarized by

$$d_1 + d_2 - 2r = |d_1 - d_2| + 2(\min(d_1, d_2) - r).$$

Since $r \leq \min(d_1, d_2)$, the final term is nonnegative, proving the inequality. $\square$

For quantum instruments and the particular candidate $R = \text{id}$,

$$|D_n(A, B) - D_n(A \otimes E, B)| = a^2(e^2 - 1)(nb^2 - 1).$$

This is zero in the degenerate case $n = b = 1$ and otherwise gives the exact dimension lower bound in Proposition 16.3.

16.2 Metric approximate transposition

Theorem 16.4 (Metric approximate transposition and triangle bounds). Let $T : \mathcal{C} \rightarrow \mathcal{D}$ and $R : \mathcal{D} \rightarrow \mathcal{C}$ be functors, and equip every hom-set under consideration with a metric. Suppose there are maps

$$\Phi_{X,Y} : \mathcal{D}(TX, Y) \rightarrow \mathcal{C}(X, RY), \quad \Psi_{X,Y} : \mathcal{C}(X, RY) \rightarrow \mathcal{D}(TX, Y)$$

and constants $\delta_{\mathcal{D}}, \delta_{\mathcal{C}}, \nu_s, \nu_t \geq 0$ such that, for all correctly typed f, g, a, b ,

$$\begin{aligned} d_{\mathcal{D}}(\Psi\Phi(f), f) &\leq \delta_{\mathcal{D}}, \\ d_{\mathcal{C}}(\Phi\Psi(g), g) &\leq \delta_{\mathcal{C}}, \\ d_{\mathcal{D}}(\Psi(g \circ a), \Psi(g) \circ T(a)) &\leq \nu_s, \\ d_{\mathcal{C}}(\Phi(b \circ f), R(b) \circ \Phi(f)) &\leq \nu_t. \end{aligned}$$

Define

$$\eta_X = \Phi_{X, TX}(\text{id}_{TX}), \quad \epsilon_Y = \Psi_{RY, Y}(\text{id}_{RY}).$$

Then:

- (a) every transpose and inverse transpose is represented by the approximate unit and counit with bounds

$$\begin{aligned} d_{\mathcal{C}}(\Phi_{X,Y}(f), R(f) \circ \eta_X) &\leq \nu_t, \\ d_{\mathcal{D}}(\Psi_{X,Y}(g), \epsilon_Y \circ T(g)) &\leq \nu_s; \end{aligned}$$

(b) the two triangle composites satisfy

$$\begin{aligned} d_{\mathcal{D}}(\epsilon_{TX} \circ T(\eta_X), \text{id}_{TX}) &\leq \nu_s + \delta_{\mathcal{D}}, \\ d_{\mathcal{C}}(R(\epsilon_Y) \circ \eta_{RY}, \text{id}_{RY}) &\leq \nu_t + \delta_{\mathcal{C}}; \end{aligned}$$

(c) if all four defects vanish, then Φ and Ψ are mutually inverse natural transpositions and $T \dashv R$ with unit η and counit ϵ .

No Lipschitz property of composition is needed for these bounds; all metric control is contained in the four displayed hypotheses.

Proof. Fix $f : TX \rightarrow Y$. Apply approximate target naturality of Φ to $\text{id}_{TX} : TX \rightarrow TX$ and the target morphism $f : TX \rightarrow Y$. Since $f \circ \text{id}_{TX} = f$, this gives

$$d_{\mathcal{C}}(\Phi_{X,Y}(f), R(f) \circ \Phi_{X,TX}(\text{id}_{TX})) \leq \nu_t.$$

The second factor is $R(f) \circ \eta_X$, proving the first inequality in (a).

Fix $g : X \rightarrow RY$. Apply approximate source naturality of Ψ to $\text{id}_{RY} : RY \rightarrow RY$ and the source morphism $g : X \rightarrow RY$. Since $\text{id}_{RY} \circ g = g$, one obtains

$$d_{\mathcal{D}}(\Psi_{X,Y}(g), \Psi_{RY,Y}(\text{id}_{RY}) \circ T(g)) \leq \nu_s.$$

The second expression is $\epsilon_Y \circ T(g)$, completing (a).

For the first triangle, take $Y = TX$ and $g = \eta_X$ in the second inequality of (a). It gives

$$d_{\mathcal{D}}(\Psi_{X,TX}(\eta_X), \epsilon_{TX} \circ T(\eta_X)) \leq \nu_s.$$

By the definition of η_X and approximate inverse preservation,

$$d_{\mathcal{D}}(\Psi_{X,TX}(\eta_X), \text{id}_{TX}) = d_{\mathcal{D}}(\Psi\Phi(\text{id}_{TX}), \text{id}_{TX}) \leq \delta_{\mathcal{D}}.$$

The triangle inequality yields the first bound in (b).

For the second triangle, take $X = RY$ and $f = \epsilon_Y$ in the first inequality of (a):

$$d_{\mathcal{C}}(\Phi_{RY,Y}(\epsilon_Y), R(\epsilon_Y) \circ \eta_{RY}) \leq \nu_t.$$

The definition of ϵ_Y and the other approximate inverse bound give

$$d_{\mathcal{C}}(\Phi_{RY,Y}(\epsilon_Y), \text{id}_{RY}) = d_{\mathcal{C}}(\Phi\Psi(\text{id}_{RY}), \text{id}_{RY}) \leq \delta_{\mathcal{C}}.$$

A second application of the triangle inequality proves (b).

Suppose all defects are zero. The first two hypotheses make Φ and Ψ mutually inverse. The stated source naturality of Ψ and target naturality of Φ are exact. To obtain source naturality of Φ , let $a : X' \rightarrow X$. Both

$$\Phi_{X',Y}(f \circ T(a)) \quad \text{and} \quad \Phi_{X,Y}(f) \circ a$$

have inverse transpose $f \circ T(a)$ by exact inverse and source naturality of Ψ . Injectivity of Ψ makes them equal. Similarly, exact target naturality of Ψ follows by applying the injective map Φ and using target naturality of Φ . Thus the bijections are natural in both variables. The ordinary hom-set definition of an adjunction gives $T \dashv R$, and the definitions of η and ϵ are the usual unit and counit. This proves (c). $\square$

Theorem 16.5 (Inradius lower bound for approximate affine transposition). Let $P \subseteq p_0 + U$ and $Q \subseteq q_0 + V$ be nonempty finite-dimensional convex bodies with norms on their direction spaces. Suppose there are points $p_* \in P$, $q_* \in Q$ and radii $r, s > 0$ such that

$$p_* + rB_U \subseteq P, \quad q_* + sB_V \subseteq Q,$$

where B_U, B_V are the closed unit balls. Let

$$\Gamma : P \rightarrow Q, \quad \Delta : Q \rightarrow P$$

be affine and define the uniform reconstruction errors

$$E_P = \sup_{p \in P} \|\Delta \Gamma(p) - p\|_U,$$

$$E_Q = \sup_{q \in Q} \|\Gamma \Delta(q) - q\|_V.$$

Then:

- (a) if $E_P < r$, the linear part of Γ is injective and $\dim_{\text{aff}} P \leq \dim_{\text{aff}} Q$;
- (b) if $E_Q < s$, the linear part of Δ is injective and $\dim_{\text{aff}} Q \leq \dim_{\text{aff}} P$;
- (c) if both inequalities hold, then $\dim_{\text{aff}} P = \dim_{\text{aff}} Q$ and both linear parts are isomorphisms;
- (d) dimension mismatch forces the sharp directional lower bounds

$$\dim_{\text{aff}} P > \dim_{\text{aff}} Q \implies E_P \geq r, \quad \dim_{\text{aff}} Q > \dim_{\text{aff}} P \implies E_Q \geq s.$$

Applied to one hom-pair of a metric approximate transposition, these bounds show exactly which inradius hypotheses convert a profile mismatch into a uniform reconstruction obstruction. They do not follow from affine dimension alone.

Proof. Let $A : U \rightarrow V$ and $B : V \rightarrow U$ be the linear parts of Γ and Δ . For every unit vector $u \in U$, both $p_* + ru$ and $p_* - ru$ belong to P . Write

$$e_{\pm} = \Delta \Gamma(p_* \pm ru) - (p_* \pm ru), \quad \|e_{\pm}\|_U \leq E_P.$$

Subtracting the two affine reconstruction identities gives

$$2r(BA - I_U)u = e_+ - e_-.$$

Consequently,

$$\|(BA - I_U)u\|_U \leq \frac{E_P}{r}.$$

By homogeneity, this is the operator estimate

$$\|BA - I_U\| \leq \frac{E_P}{r}.$$

If $E_P < r$, then BA is injective: from $BAu = 0$ one obtains

$$\|u\| = \|(BA - I_U)u\| \leq \frac{E_P}{r}\|u\|,$$

which forces $u = 0$. Hence A is injective and $\dim U \leq \dim V$. This proves (a).

The same argument with Q , its inball, and the composite $\Gamma \Delta$ gives

$$\|AB - I_V\| \leq \frac{E_Q}{s}.$$

Thus $E_Q < s$ makes AB , and hence B , injective, proving (b). If both bounds hold, the two dimension inequalities are equalities. The injective maps A and B between equal finite dimensions are isomorphisms, proving (c).

If $\dim U > \dim V$, no linear map $A : U \rightarrow V$ is injective. The contrapositive of (a) therefore gives $E_P \geq r$. Interchanging P and Q gives the second lower bound in (d). Equality is attained in the first bound by taking $P = rB_U$, Q a singleton, Γ the unique map, and Δ the preparation of the origin; then $E_P = r$. The transposed construction attains the second bound. The role of the inradii is essential: without fixed positive r and s , Proposition 16.6 exhibits different-dimensional convex bodies at arbitrarily small Hausdorff distance. $\square$

16.3 Dimension-only metric boundary

Proposition 16.6 (No metric bound from dimension alone). Affine-dimension mismatch alone gives no uniform positive Hausdorff-distance bound between convex sets.

Proof. In $\mathbb{R}^2$, let

$$C = [0, 1] \times \{0\}, \quad D_\varepsilon = [0, 1] \times [0, \varepsilon].$$

Their affine dimensions are 1 and 2. Since $C \subseteq D_\varepsilon$, the directed distance from C to D_ε is zero. Every $(x, t) \in D_\varepsilon$ is at supremum-norm distance $t \leq \varepsilon$ from $(x, 0) \in C$, and equality is attained when $t = \varepsilon$. Thus the Hausdorff distance is exactly ε . Letting $\varepsilon \downarrow 0$ proves the claim. $\square$

Remark 16.7 (Norm dependence and operational recovery). Only the property $\delta_T(Y) > 0$ is invariant under changing norms in finite dimensions. For example,

$$\|af_1 + bf_2\|_\alpha = |a| + \alpha|b|$$

gives quantum-instrument defect $\alpha(e^2 - 1)$, which can be arbitrarily small or large. The profile and affine-rank defects are therefore geometric invariants rather than operational distances. Approximate-recovery results such as the Fawzi–Renner theorem [36] instead begin with metric or information-theoretic data and construct a recovery channel with a quantitative fidelity bound. Deriving such a bound from a dimension-profile defect would require a specified operational norm and continuity estimates relating that norm to the affine geometry of the hom-sets.

17 Ordered-Vector-Space and GPT Generalization

The channel calculations extend from classical and complex quantum systems to finite-dimensional ordered vector spaces in the generalized-probabilistic framework [28, 29, 30]. Let a GPT system X consist of a real vector space V_X , a proper generating cone V_X^+ , and a strictly positive deterministic effect u_X . The normalized base

$$\Omega_X = \{x \in V_X^+ : u_X(x) = 1\}$$

is compact. Let maximal deterministic transformations be all positive linear maps $F : V_X \rightarrow V_Y$ satisfying $u_Y F = u_X$.

Lemma 17.1 (Affine span of normalized states). The normalized base Ω_X has nonempty relative interior in the affine hyperplane $u_X = 1$. Consequently,

$$\dim_{\text{aff}} \Omega_X = \dim V_X - 1.$$

Proof. Because V_X^+ is generating and finite dimensional, its linear span is V_X , and therefore its interior is nonempty. Choose $x_0 \in \text{int}(V_X^+)$. Strict positivity of u_X gives $u_X(x_0) > 0$, so

$$\omega = \frac{x_0}{u_X(x_0)}$$

belongs to $\text{int}(V_X^+)$ and satisfies $u_X(\omega) = 1$.

Fix a norm on V_X . Since ω is interior, there is $\varepsilon > 0$ such that

$$\omega + B_\varepsilon(0) \subseteq V_X^+.$$

Let $H_X = \ker u_X$, the direction space of the affine hyperplane $u_X = 1$. For every $h \in H_X$ with $\|h\| < \varepsilon$,

$$\omega + h \in V_X^+, \quad u_X(\omega + h) = 1.$$

Thus Ω_X contains a relative neighborhood of ω in the hyperplane $u_X = 1$. Its affine hull is that hyperplane, whose dimension is $\dim V_X - 1$. $\square$

Proposition 17.2 (Maximal GPT transformation dimension). The maximal deterministic transformation set has nonempty relative interior in the affine normalization fiber and

$$\dim_{\text{aff}} \mathcal{C}_{\text{max}}(X, Y) = \dim V_X (\dim V_Y - 1).$$

Proof. Consider the linear map

$$Q : \text{Hom}(V_X, V_Y) \rightarrow V_X^*, \quad Q(F) = u_Y \circ F.$$

Choose $y_0 \in V_Y$ with $u_Y(y_0) = 1$. For each $\varphi \in V_X^*$, define

$$F_\varphi(x) = \varphi(x)y_0.$$

Then

$$Q(F_\varphi)(x) = u_Y(\varphi(x)y_0) = \varphi(x),$$

so Q is surjective and $\text{rank } Q = \dim V_X$. The ambient map space has dimension $\dim V_X \dim V_Y$. Therefore the affine normalization fiber $Q^{-1}(u_X)$ has dimension

$$\dim V_X \dim V_Y - \dim V_X = \dim V_X (\dim V_Y - 1).$$

It remains to show that positivity does not reduce this affine hull. By Lemma 17.1, choose an interior normalized state $\omega \in \Omega_Y$. Define

$$F_0(x) = u_X(x)\omega.$$

This map is positive and satisfies $u_Y \circ F_0 = u_X$. Fix norms on V_X and V_Y . Since ω is interior to V_Y^+ , there is $\varepsilon > 0$ such that

$$\omega + B_\varepsilon(0) \subseteq V_Y^+.$$

Compactness of Ω_X gives

$$M = \max_{x \in \Omega_X} \|x\| < \infty.$$

Since $u_X(x) = 1$ on Ω_X , the set does not contain 0, so $M > 0$. Let $G : V_X \rightarrow V_Y$ satisfy $u_Y \circ G = 0$ and $\|G\|_{\text{op}} < \varepsilon/M$. For $x \in \Omega_X$,

$$\|Gx\| \leq \|G\|_{\text{op}} M < \varepsilon,$$

so

$$(F_0 + G)(x) = \omega + Gx \in V_Y^+.$$

Every nonzero $v \in V_X^+$ can be written as $v = u_X(v)x$ with $x \in \Omega_X$ and $u_X(v) > 0$. Positive homogeneity then gives

$$(F_0 + G)(v) = u_X(v)(F_0 + G)(x) \in V_Y^+.$$

Thus every sufficiently small perturbation in the direction space $\ker Q$ remains positive. Hence F_0 is a relative-interior point of the positive normalization fiber, whose affine hull is all of $Q^{-1}(u_X)$. $\square$

Remark 17.3 (Ungraded and graded profile dimensions). For maximal ungraded transformations, Proposition 17.2 gives

$$p_Y(X) = \dim V_X (\dim V_Y - 1).$$

Thus the profile space is one-dimensional, generated by $f(X) = \dim V_X$. Writing the same formula as

$$\dim V_X \dim V_Y - \dim V_X$$

does not produce two independent source factors: the apparent bulk and normalization factors are proportional. Consequently, the two-coordinate affine-hyperplane argument is unavailable in the ungraded theory; obstruction may instead arise from the discrete object-dimension spectrum or from convex shape, as in the quantum and classical channel examples.

An independent grade parameter changes this structure. In the instrument examples, the bulk factor is multiplied by the outcome number while the normalization term is not, producing two linearly independent profile directions and the affine-hyperplane obstruction.

Under local tomography, Theorem 5.2 constructs the exact linear causal internal-hom candidate. Equip $W(E, Y)$ with the cone of positive maps $V_E^+ \rightarrow V_Y^+$ (or with the theory's specified completely positive cone) intersected with $W(E, Y)$, and normalize it by $u_W = 1$. This ordered vector space need not be an object of the GPT.

Theorem 17.4 (Homogenization of a restricted process body). Let $P \subseteq \text{Hom}(V_E, V_Y)$ be a nonempty compact convex set of normalized linear maps:

$$u_Y \circ G = u_E \quad (G \in P).$$

Define

$$H(P) = \text{span}_{\mathbb{R}} P, \quad H(P)^+ = \text{cone}(P).$$

Then:

- (a) $H(P)$ is a subspace of $W(E, Y)$, and the restriction of u_W to $H(P)$ is a nonzero linear functional;
- (b) $H(P)^+$ is a closed, pointed, generating cone on which u_W is strictly positive, and its normalized base is exactly P ;
- (c)
$$\text{aff}(P) = \{h \in H(P) : u_W(h) = 1\}, \quad \dim H(P) = \dim_{\text{aff}} P + 1;$$
- (d) for every finite-dimensional GPT object Z , an affine isomorphism $P \cong \Omega_Z$ exists if and only if it extends uniquely to a unital order isomorphism

$$(H(P), H(P)^+, u_W) \cong (V_Z, V_Z^+, u_Z).$$

Proof. Every $h \in H(P)$ has a representation

$$h = \sum_{i=1}^m \alpha_i G_i, \quad G_i \in P.$$

Normalization gives

$$u_Y \circ h = \left(\sum_i \alpha_i \right) u_E.$$

Thus $h \in W(E, Y)$ and $u_W(h) = \sum_i \alpha_i$. This value is independent of the chosen representation because $u_E \neq 0$, and it proves (a).

Every element of $H(P)^+$ has the form tG with $t \geq 0$ and $G \in P$: a finite positive combination can be written this way by dividing by the sum of its coefficients and using convexity of P . For $t > 0$,

$$u_W(tG) = t,$$

so u_W is strictly positive on the nonzero cone. The cone is pointed, because $h, -h \in H(P)^+$ would force both $u_W(h) > 0$ and $-u_W(h) > 0$ unless $h = 0$. It is generating because its linear span is $H(P)$ by definition.

To prove closedness, let $t_k G_k \rightarrow h$ with $t_k \geq 0$ and $G_k \in P$. Applying u_W gives

$$t_k = u_W(t_k G_k) \rightarrow u_W(h),$$

so (t_k) is bounded. Compactness of P supplies a convergent subsequence $G_{k_j} \rightarrow G \in P$, and along the same subsequence $t_{k_j} \rightarrow t \geq 0$. Hence $h = tG \in H(P)^+$. Finally,

$$\{h \in H(P)^+ : u_W(h) = 1\} = P,$$

which proves (b).

An affine combination of points of P has u_W -value one. Conversely, if $h = \sum_i \alpha_i G_i \in H(P)$ and $u_W(h) = 1$, then $\sum_i \alpha_i = 1$, so h is an affine combination of points of P . Therefore

$$\text{aff}(P) = \{h \in H(P) : u_W(h) = 1\}.$$

This is a nonempty affine hyperplane in $H(P)$, proving (c).

Let $\phi : P \rightarrow \Omega_Z$ be an affine isomorphism. Every nonzero $h \in H(P)^+$ has the unique form

$$h = u_W(h)G, \quad G \in P.$$

Define

$$\widehat{\phi}(0) = 0, \quad \widehat{\phi}(h) = u_W(h) \phi\left(\frac{h}{u_W(h)}\right) \quad (h \in H(P)^+ \setminus \{0\}).$$

For $h_i = t_i G_i$ with $t_i \geq 0$ and $t_1 + t_2 > 0$, affinity gives

$$\begin{aligned} \widehat{\phi}(h_1 + h_2) &= (t_1 + t_2) \phi\left(\frac{t_1}{t_1 + t_2} G_1 + \frac{t_2}{t_1 + t_2} G_2\right) \\ &= t_1 \phi(G_1) + t_2 \phi(G_2) \\ &= \widehat{\phi}(h_1) + \widehat{\phi}(h_2). \end{aligned}$$

The map is also positively homogeneous. Since $H(P)^+$ is generating, it extends uniquely by differences to a linear map $H(P) \rightarrow V_Z$. The same construction applied to ϕ^{-1} gives a positive linear inverse. Moreover,

$$u_Z \circ \widehat{\phi} = u_W,$$

so the extension is a unital order isomorphism. It is unique because P spans $H(P)$. Conversely, a unital order isomorphism carries the normalized base of $H(P)^+$ onto Ω_Z and restricts to an affine isomorphism. This proves (d). $\square$

Corollary 17.5 (Collapse of affine-base and generated-order tests). For finite-dimensional compact normalized bases with strictly positive deterministic effects, affine base representability is equivalent to unital order representability of the cones generated by those bases. Consequently, the affine-shape and ordinary generated-cone levels cannot be separated under these hypotheses. A further obstruction must involve an independently specified larger process cone, ancillary complete positivity, or categorical naturality.

Proof. Let P and Q be the two compact normalized bases and let $\phi : P \rightarrow Q$ be an affine isomorphism. Strict positivity of the deterministic effects implies that every nonzero element of the generated cone has a unique expression

$$h = t\omega, \quad t > 0, \quad \omega \in P,$$

where t is its deterministic-effect value. Define

$$\widehat{\phi}(0) = 0, \quad \widehat{\phi}(t\omega) = t\phi(\omega).$$

For positive elements $t_1\omega_1$ and $t_2\omega_2$, affinity gives

$$\begin{aligned}\widehat{\phi}(t_1\omega_1 + t_2\omega_2) &= (t_1 + t_2)\phi\left(\frac{t_1\omega_1 + t_2\omega_2}{t_1 + t_2}\right) \\ &= t_1\phi(\omega_1) + t_2\phi(\omega_2).\end{aligned}$$

Thus $\widehat{\phi}$ is additive and positively homogeneous on the generated cone and extends by differences to a positive linear map between the ordered spans. Applying the same construction to ϕ^{-1} gives its positive linear inverse. Preservation of the deterministic effect makes the extension unital.

Conversely, a unital order isomorphism maps the level set on which the deterministic effect equals one onto the corresponding level set in the target cone. These level sets are exactly P and Q , so restriction gives an affine base isomorphism. Hence affine-base and generated-order representability are equivalent. Any stronger obstruction must therefore use order data not generated by the normalized base, ancillary matrix cones, or categorical composition and naturality. $\square$

18 Complete Order and Operator-System Realization

Homogenization identifies the ordinary cone generated by a normalized process body. Positivity at arbitrary ancillary matrix levels is additional data and requires the separate criteria developed in this section.

18.1 Ancillary order and finite amplification

Proposition 18.1 (Strict ancillary boundary). Let $d > 1$ and let

$$\Theta : \text{Herm}(\mathbb{C}^d) \longrightarrow \text{Herm}(\mathbb{C}^d), \quad \Theta(A) = A^\top$$

be matrix transposition in a fixed basis. Then Θ is a trace-preserving order automorphism and restricts to an affine automorphism of the quantum state space, but $\Theta \otimes \text{id}_d$ is not positive. Hence affine-base and ordinary order compatibility do not imply ancillary complete-order compatibility.

Proof. Transposition preserves Hermiticity, positivity, and trace, and $\Theta^{-1} = \Theta$. It is therefore a trace-preserving order automorphism and maps the normalized positive base affinely onto itself.

Let $\{e_i\}_{i=1}^d$ be the defining basis and put

$$|\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^d e_i \otimes e_i, \quad \rho_\Omega = |\Omega\rangle\langle\Omega|.$$

Then $\rho_\Omega \geq 0$, while partial transposition gives

$$\begin{aligned}(\Theta \otimes \text{id}_d)(\rho_\Omega) &= \frac{1}{d} \sum_{i,j=1}^d |e_j\rangle\langle e_i| \otimes |e_i\rangle\langle e_j| \\ &= \frac{1}{d} F,\end{aligned}$$

where $F(x \otimes y) = y \otimes x$ is the swap operator. On the nonzero antisymmetric vector

$$e_1 \otimes e_2 - e_2 \otimes e_1,$$

the swap has eigenvalue -1 . Thus F/d is not positive, proving that $\Theta \otimes \text{id}_d$ is not positive. In particular, Θ is not completely positive and is not a complete-order isomorphism. $\square$

18.2 A finite matrix-level complete-order criterion

Let $\mathcal{S} \subseteq M_m(\mathbb{C})$ and $\mathcal{T} \subseteq M_n(\mathbb{C})$ be complex linear subspaces closed under adjoint. Equip them with the inherited matrix cones

$$M_k(\mathcal{S})^+ = M_k(\mathcal{S}) \cap \text{Pos}(\mathbb{C}^k \otimes \mathbb{C}^m),$$

and similarly for $\mathcal{T}$. For a $*$ -linear map $\phi : \mathcal{S} \rightarrow \mathcal{T}$, write

$$\phi^{(k)} = \text{id}_k \otimes \phi.$$

The map is completely positive if every $\phi^{(k)}$ is positive, and it is a complete-order isomorphism if both ϕ and ϕ^{-1} are completely positive.

Theorem 18.2 (Finite amplification criterion). Let $\phi : \mathcal{S} \rightarrow \mathcal{T}$ be a $*$ -linear bijection and put $N = \max\{m, n\}$. The following are equivalent:

- (i) ϕ is a complete-order isomorphism;
- (ii) the single level- N ampliation is a cone isomorphism,

$$\phi^{(N)}(M_N(\mathcal{S})^+) = M_N(\mathcal{T})^+;$$

- (iii) $\phi^{(n)}$ is positive and $(\phi^{-1})^{(m)}$ is positive.

Thus, for concrete finite matrix-ordered spaces, compatibility with every ancilla is decided by one finite amplification level.

Proof. The implication (i) $\Rightarrow$ (ii) is immediate. If (ii) holds, then $\phi^{(N)}$ and $(\phi^{-1})^{(N)}$ are positive. Positivity at every lower level follows by embedding a positive block matrix as a principal block at level N and then compressing. In particular, (iii) follows.

It remains to prove that (iii) implies complete positivity in both directions. Let $\mathcal{V}$ be any concrete matrix-ordered $*$ -space and let

$$\psi : \mathcal{V} \longrightarrow M_r(\mathbb{C})$$

be a $*$ -linear, r -positive map. Then ψ is completely positive. Let $k \geq 1$, $X \in M_k(\mathcal{V})^+$, and $\eta \in \mathbb{C}^k \otimes \mathbb{C}^r$. The Schmidt rank of η is some $s \leq r$. Hence there are an isometry $U : \mathbb{C}^s \rightarrow \mathbb{C}^k$ and a vector $\zeta \in \mathbb{C}^s \otimes \mathbb{C}^r$ such that

$$\eta = (U \otimes I_r)\zeta.$$

The scalar compression

$$X_U = (U^* \otimes I)X(U \otimes I)$$

belongs to $M_s(\mathcal{V})^+$. Since r -positivity implies s -positivity,

$$\psi^{(s)}(X_U) \geq 0.$$

Moreover, ampliation commutes with scalar compression, and therefore

$$\begin{aligned} \langle \eta, \psi^{(k)}(X)\eta \rangle &= \langle \zeta, (U^* \otimes I_r)\psi^{(k)}(X)(U \otimes I_r)\zeta \rangle \\ &= \langle \zeta, \psi^{(s)}(X_U)\zeta \rangle \geq 0. \end{aligned}$$

This holds for every η , so $\psi^{(k)}(X) \geq 0$. Since k was arbitrary, ψ is completely positive.

Under (iii), compose ϕ with the inclusion $\mathcal{T} \hookrightarrow M_n(\mathbb{C})$. The resulting map is n -positive and hence completely positive by the lemma. Because the matrix cones of $\mathcal{T}$ are inherited from M_n , this says exactly that ϕ is completely positive as a map into $\mathcal{T}$. Applying the same argument to

$$\phi^{-1} : \mathcal{T} \longrightarrow \mathcal{S} \subseteq M_m(\mathbb{C})$$

shows that ϕ^{-1} is completely positive. Thus (i) holds. $\square$

Corollary 18.3 (Finite ancillary test for a restricted process body). Let P be a compact normalized process body, let Z be a candidate representing object, and let

$$\widehat{\phi} : H(P) \longrightarrow V_Z$$

be the homogeneous order isomorphism induced by an affine isomorphism $P \cong \Omega_Z$. Suppose the complexifications of $H(P)$ and V_Z are realized as concrete $*$ -subspaces

$$\mathcal{S} \subseteq M_m(\mathbb{C}), \quad \mathcal{T} \subseteq M_n(\mathbb{C}),$$

with their physical matrix cones, and let $\widehat{\phi}_{\mathbb{C}} : \mathcal{S} \rightarrow \mathcal{T}$ be the complex-linear extension. Then the affine representation is compatible with all ancillary matrix levels if and only if, for $N = \max\{m, n\}$,

$$\widehat{\phi}_{\mathbb{C}}^{(N)}(M_N(\mathcal{S})^+) = M_N(\mathcal{T})^+.$$

Proof. Suppose first that the affine representation is compatible with every ancillary matrix level. Its homogeneous complex extension and its inverse are positive at every level. In particular, their level- N extensions are mutually inverse positive maps, and therefore

$$\widehat{\phi}_{\mathbb{C}}^{(N)}(M_N(\mathcal{S})^+) = M_N(\mathcal{T})^+.$$

Conversely, assume the displayed level- N equality. It implies that $\widehat{\phi}_{\mathbb{C}}$ and its inverse are both N -positive. Since $N \geq n$, the map into $\mathcal{T} \subseteq M_n(\mathbb{C})$ is n -positive. The Schmidt-support compression established in Theorem 18.2 then makes it completely positive. Likewise, $N \geq m$ makes the inverse map into $\mathcal{S} \subseteq M_m(\mathbb{C})$ completely positive. The two maps therefore preserve every ancillary cone and remain inverse at every matrix level. Their restrictions to the normalized bases give ancillary compatibility of the original affine representation. $\square$

18.3 Semialgebraic complete-order realization

Theorem 18.4 (Semialgebraic complete-order representer criterion). Let $\mathcal{S} \subseteq M_m(\mathbb{C})$ and $\mathcal{T} \subseteq M_n(\mathbb{C})$ be concrete $*$ -subspaces whose self-adjoint parts have the same real dimension d . Choose Hermitian bases

$$S_1, \dots, S_d, \quad T_1, \dots, T_d,$$

and let $\mathfrak{A} \subseteq \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d}$ be any finite semialgebraic set of prescribed side constraints. A real matrix C defines the $*$ -linear map

$$\phi_C(S_r) = \sum_{s=1}^d C_{sr} T_s,$$

and D analogously defines $\psi_D : \mathcal{T} \rightarrow \mathcal{S}$. Assume that the real and imaginary parts of the basis entries and the coefficients of the defining polynomials of $\mathfrak{A}$ are real algebraic. Then:

- (a) Existence of a pair $(C, D) \in \mathfrak{A}$ for which ϕ_C is a complete-order isomorphism with inverse ψ_D is decidable by a finite semialgebraic feasibility problem.
- (b) If $\mathcal{S}$ and $\mathcal{T}$ are operator systems, the existence problem is equivalent to the following explicit semialgebraic semidefinite feasibility system. There must exist $(C, D) \in \mathfrak{A}$ and Hermitian matrices

$$J \in M_m(\mathbb{C}) \otimes M_n(\mathbb{C}), \quad K \in M_n(\mathbb{C}) \otimes M_m(\mathbb{C})$$

such that

$$\begin{aligned}
CD &= DC = I_d, & J &\geq 0, & K &\geq 0, \\
\text{Tr}_m[(S_r^\top \otimes I_n)J] &= \sum_{s=1}^d C_{sr} T_s & (1 \leq r \leq d), \\
\text{Tr}_n[(T_s^\top \otimes I_m)K] &= \sum_{r=1}^d D_{rs} S_r & (1 \leq s \leq d).
\end{aligned}$$

When $\mathfrak{A}$ is affine-linear, this is a bilinear semidefinite feasibility problem. Unitality, deterministic-effect preservation, and finitely many prescribed interpolation conditions may be imposed by adding linear equations to $\mathfrak{A}$.

Proof. Put $N = \max\{m, n\}$. A Hermitian element of $M_N(\mathcal{S})$ has a unique expansion

$$X = \sum_{r=1}^d X_r \otimes S_r, \quad X_r \in \text{Herm}(\mathbb{C}^N),$$

and

$$\phi_C^{(N)}(X) = \sum_{r,s} C_{sr} X_r \otimes T_s.$$

Thus positivity of $\phi_C^{(N)}$ is the first-order real condition

$$\forall(X_1, \dots, X_d), \quad \sum_r X_r \otimes S_r \geq 0 \implies \sum_{r,s} C_{sr} X_r \otimes T_s \geq 0.$$

There is an analogous condition for $\psi_D^{(N)}$. Matrix positivity is a finite semialgebraic condition: after splitting complex entries into real and imaginary parts, it is equivalent to nonnegativity of all principal minors. The equations

$$CD = DC = I_d$$

are polynomial and state that the two maps are inverse. Hence the existence of $(C, D) \in \mathfrak{A}$ satisfying the two level- N positivity conditions is a finite first-order formula over the real closed field of real numbers. Real quantifier elimination makes its solution set semialgebraic and decides its nonemptiness for real-algebraic input data [27]. By Theorem 18.2, level- N positivity in both directions is equivalent to complete-order isomorphism. This proves (a).

Suppose now that $\mathcal{S}$ and $\mathcal{T}$ are operator systems. If the system in (b) is feasible, the positive matrix J is the Choi matrix of a completely positive map

$$\Phi : M_m(\mathbb{C}) \rightarrow M_n(\mathbb{C}).$$

The inverse Choi formula and the displayed restriction equations give

$$\Phi(S_r) = \phi_C(S_r)$$

for every basis element. Hence ϕ_C is completely positive. The matrix K similarly gives a completely positive extension of ψ_D . Since $CD = DC = I_d$, the two restrictions are inverse, so ϕ_C is a complete-order isomorphism satisfying the side constraints.

Conversely, suppose such a complete-order isomorphism exists. Regard ϕ_C as a completely positive map from the operator system $\mathcal{S}$ into $M_n(\mathbb{C})$. The finite-dimensional Arveson extension theorem [14] gives a completely positive extension $\Phi : M_m \rightarrow M_n$; let $J \geq 0$ be its Choi matrix. Apply the same theorem to $\phi_C^{-1} = \psi_D : \mathcal{T} \rightarrow M_m$ and let $K \geq 0$ be the Choi matrix of a completely positive extension. The inverse Choi formula gives exactly the displayed basis equations, and inverse coordinate matrices satisfy $CD = DC = I_d$. Thus the displayed semialgebraic semidefinite system is feasible.

Finally, the coefficient equations for $\phi_C(I_m) = I_n$, for preservation of a specified linear deterministic effect, and for the images of finitely many designated elements are linear in C (and similarly for D). They can therefore be included in $\mathfrak{A}$, proving the last assertion. $\square$

18.4 C^* -envelopes and finite certificates

An abstract operator system is a complex $*$ -vector space with an Archimedean matrix order unit. Its C^* -envelope $C_e^*(\mathcal{S})$ is the minimal C^* -cover generated by a unital complete-order copy of $\mathcal{S}$; equivalently, every other C^* -cover maps onto it by a surjective $*$ -homomorphism fixing $\mathcal{S}$ [15].

Theorem 18.5 (C^* -envelope criterion and minimal embedding size). Let $\mathcal{S}$ and $\mathcal{T}$ be finite-dimensional abstract operator systems, with canonical embeddings

$$\iota_{\mathcal{S}} : \mathcal{S} \rightarrow C_e^*(\mathcal{S}), \quad \iota_{\mathcal{T}} : \mathcal{T} \rightarrow C_e^*(\mathcal{T}).$$

Then:

- (a) $\mathcal{S}$ admits a unital complete-order embedding into some finite matrix algebra if and only if $C_e^*(\mathcal{S})$ is finite-dimensional.

- (b) If

$$C_e^*(\mathcal{S}) \cong \bigoplus_{i=1}^r M_{n_i}(\mathbb{C}),$$

then the least N for which $\mathcal{S}$ admits a unital complete-order embedding into $M_N(\mathbb{C})$ is

$$N_{\min}(\mathcal{S}) = \sum_{i=1}^r n_i.$$

- (c) A unital map $\phi : \mathcal{S} \rightarrow \mathcal{T}$ is a complete-order isomorphism onto $\mathcal{T}$ if and only if it extends uniquely to a $*$ -isomorphism

$$\tilde{\phi} : C_e^*(\mathcal{S}) \longrightarrow C_e^*(\mathcal{T})$$

satisfying $\tilde{\phi} \iota_{\mathcal{S}} = \iota_{\mathcal{T}} \phi$.

- (d) Suppose both envelopes are finite-dimensional. Write

$$C_e^*(\mathcal{S}) = \bigoplus_i M_{n_i}, \quad C_e^*(\mathcal{T}) = \bigoplus_j M_{m_j}.$$

A unital complete-order isomorphism exists exactly when the multisets $\{n_i\}$ and $\{m_j\}$ agree and, after a permutation of equal-sized blocks, there are unitaries $U_i \in M_{n_i}$ such that

$$\bigoplus_i \text{Ad}_{U_i}(\iota_{\mathcal{S}}(\mathcal{S})) = \iota_{\mathcal{T}}(\mathcal{T}).$$

For real-algebraic matrix coordinates, this is a finite semialgebraic feasibility problem.

Proof. Suppose first that $\phi : \mathcal{S} \rightarrow M_N(\mathbb{C})$ is a unital complete-order embedding and put

$$A = C^*(\phi(\mathcal{S})) \subseteq M_N(\mathbb{C}).$$

The universal property of the C^* -envelope gives a surjective $*$ -homomorphism

$$q : A \longrightarrow C_e^*(\mathcal{S})$$

whose restriction identifies $\phi(\mathcal{S})$ with $\iota_{\mathcal{S}}(\mathcal{S})$. Since A is finite-dimensional, so is its quotient $C_e^*(\mathcal{S})$. Conversely, if the envelope is finite-dimensional, its faithful block-diagonal representation on a finite Hilbert space, composed with $\iota_{\mathcal{S}}$, is a unital complete-order embedding. This proves (a).

For (b), the direct sum of the defining representations of the blocks is a faithful representation of the envelope on a Hilbert space of dimension $\sum_i n_i$, giving the upper bound. Conversely, retain

the cover $A = C^*(\phi(\mathcal{S})) \subseteq M_N$ from the preceding paragraph. Every finite-dimensional unital C^* -subalgebra of M_N has the form

$$A \cong \bigoplus_{j=1}^s M_{k_j}(\mathbb{C})$$

represented with positive multiplicities μ_j , so

$$N = \sum_j \mu_j k_j \geq \sum_j k_j.$$

Every ideal of A is a direct sum of some of its simple blocks. Therefore the quotient $C_e^*(\mathcal{S})$ consists of a subcollection of the blocks M_{k_j} , and

$$\sum_i n_i \leq \sum_j k_j \leq N.$$

This proves the lower bound and hence (b).

Suppose $\phi : \mathcal{S} \rightarrow \mathcal{T}$ is a unital complete-order isomorphism. The pair

$$(C_e^*(\mathcal{T}), \iota_{\mathcal{T}}\phi)$$

is a C^* -cover of $\mathcal{S}$, because $\iota_{\mathcal{T}}(\mathcal{T})$ generates its envelope. The envelope property gives a surjective $*$ -homomorphism

$$p : C_e^*(\mathcal{T}) \circ C_e^*(\mathcal{S})$$

with $p\iota_{\mathcal{T}}\phi = \iota_{\mathcal{S}}$. Applying the same argument to ϕ^{-1} gives a surjective $*$ -homomorphism

$$r : C_e^*(\mathcal{S}) \circ C_e^*(\mathcal{T})$$

with $r\iota_{\mathcal{S}} = \iota_{\mathcal{T}}\phi$. The composites pr and rp fix the generating copies of $\mathcal{S}$ and $\mathcal{T}$, respectively, and are therefore identities. Thus r is the required $*$ -isomorphism. It is unique because $\iota_{\mathcal{S}}(\mathcal{S})$ generates its envelope. Conversely, the restriction of any such $*$ -isomorphism is a unital complete-order isomorphism. This proves (c).

For (d), a $*$ -isomorphism between finite direct sums maps minimal central projections bijectively and therefore matches blocks of equal matrix size. Every complex $*$ -automorphism of M_n is inner, so after the resulting block permutation the isomorphism has the displayed form $\bigoplus_i \text{Ad}_{U_i}$. Part (c) shows that its restriction is a complete-order isomorphism exactly when it carries the canonical copy of $\mathcal{S}$ onto that of $\mathcal{T}$.

There are only finitely many admissible block permutations. For each one, the equations $U_i^* U_i = U_i U_i^* = I$ are polynomial in the real and imaginary parts of the entries. Equality of the two operator-system subspaces is expressed by introducing mutually inverse real coefficient matrices between chosen Hermitian bases and equating their block-matrix entries. With real-algebraic input coordinates, these are finitely many real polynomial equalities, so real quantifier elimination decides feasibility. This proves (d). $\square$

Theorem 18.6 (Finite-cover envelope algorithm and absence of a dimension bound). Let $\mathcal{S}$ be an operator system contained in a finite-dimensional C^* -cover

$$A = C^*(\mathcal{S}) \cong \bigoplus_{i=1}^r M_{n_i}(\mathbb{C}),$$

with real-algebraic matrix coordinates. Then:

- (a) The Shilov ideal and C^* -envelope of $\mathcal{S}$ are computable by at most 2^r finite semialgebraic complete-order tests. Explicitly, for each $I \subseteq \{1, \dots, r\}$ let

$$q_I : A \longrightarrow A_I := \bigoplus_{i \in I} M_{n_i}$$

be the block projection. The removed ideal $\bigoplus_{i \notin I} M_{n_i}$ is a boundary ideal exactly when $q_I|_{\mathcal{S}}$ is a unital complete-order embedding. The largest such ideal is the Shilov ideal, and the corresponding A_I is $C_e^*(\mathcal{S})$.

- (b) No bound on finite embedding size can depend only on $\dim_{\mathbb{C}} \mathcal{S}$. For every $n \geq 2$, let

$$J_n = \sum_{i=1}^{n-1} E_{i,i+1}, \quad \mathcal{S}_n = \text{span}_{\mathbb{C}}\{I_n, J_n, J_n^*\} \subseteq M_n.$$

Then

$$\dim_{\mathbb{C}} \mathcal{S}_n = 3, \quad C_e^*(\mathcal{S}_n) = M_n, \quad N_{\min}(\mathcal{S}_n) = n.$$

Proof. Every ideal of the finite-dimensional algebra A is the direct sum of a subset of its simple matrix blocks. By definition, an ideal $J \triangleleft A$ is a boundary ideal for $\mathcal{S}$ exactly when the quotient map

$$A \longrightarrow A/J$$

restricts to a unital complete-order embedding of $\mathcal{S}$. Consequently all boundary ideals occur among the finitely many kernels of the block projections q_I .

For each I , injectivity of $q_I|_{\mathcal{S}}$ is a finite linear rank test. The map itself is unital and completely positive. When it is injective, it is a complete-order embedding exactly when its inverse on $q_I(\mathcal{S})$ is completely positive. This is decided by Theorem 18.2, or equivalently by the Choi and semialgebraic feasibility conditions of Theorem 18.4. Thus every boundary ideal is decidable by a finite semialgebraic test. The Shilov ideal is the unique largest boundary ideal, and the defining universal property of the C^* -envelope identifies the quotient by that ideal with $C_e^*(\mathcal{S})$. Enumerating the 2^r block subsets therefore computes the envelope, proving (a).

For (b), the three matrices I_n, J_n, J_n^* are linearly independent when $n \geq 2$. Moreover,

$$E_{11} = I_n - J_n^* J_n,$$

and for all $1 \leq i, j \leq n$,

$$E_{ij} = (J_n^*)^{i-1} E_{11} J_n^{j-1}.$$

Hence J_n and J_n^* generate every matrix unit, so

$$C^*(\mathcal{S}_n) = M_n.$$

The universal property gives a quotient map

$$M_n \longrightarrow C_e^*(\mathcal{S}_n)$$

that is completely isometric on $\mathcal{S}_n$. Since M_n is simple, its kernel is either zero or all of M_n ; it cannot be all because the quotient is unital on $\mathcal{S}_n$. The kernel is therefore zero and $C_e^*(\mathcal{S}_n) = M_n$. Part (b) of Theorem 18.5 then gives $N_{\min}(\mathcal{S}_n) = n$. Since n is arbitrary while $\dim \mathcal{S}_n = 3$, no dimension-only upper bound exists. $\square$

Definition 18.1 (Finite matrix-state certificate). Let $\mathcal{S}$ be a finite-dimensional abstract operator system. A *finite matrix-state certificate* is a finite family of unital completely positive maps

$$\varphi_j : \mathcal{S} \rightarrow M_{n_j}(\mathbb{C}) \quad (1 \leq j \leq r)$$

such that, for every $k \geq 1$ and every $X \in M_k(\mathcal{S})_{\text{sa}}$,

$$X \in M_k(\mathcal{S})^+ \iff (\text{id}_k \otimes \varphi_j)(X) \geq 0 \quad \text{for all } j.$$

Its total matrix size is $N = \sum_j n_j$. Equivalently, the direct sum $\Phi = \bigoplus_j \varphi_j : \mathcal{S} \rightarrow \bigoplus_j M_{n_j}$ is a unital complete-order embedding.

Theorem 18.7 (Finite matrix-state certificates and constructive covers). Let $\mathcal{S}$ be a finite-dimensional abstract operator system.

- (a) A finite matrix-state certificate exists if and only if $C_e^*(\mathcal{S})$ is finite-dimensional.
- (b) Given such a certificate, the direct-sum image generates the explicit finite-dimensional C^* -cover

$$A = C^*(\Phi(\mathcal{S})) \subseteq \bigoplus_{j=1}^r M_{n_j}(\mathbb{C}) \subseteq M_N(\mathbb{C}).$$

The Shilov ideal and $C_e^*(\mathcal{S})$ are then computed by the finite-cover algorithm of Theorem 18.6.

- (c) If

$$C_e^*(\mathcal{S}) \cong \bigoplus_{i=1}^q M_{m_i}(\mathbb{C}),$$

then the least possible total size of a finite matrix-state certificate is

$$N_{\text{cert}}(\mathcal{S}) = \sum_{i=1}^q m_i = N_{\min}(\mathcal{S}).$$

- (d) Suppose a Hermitian basis of $\mathcal{S}$, the unit, and all certificate matrices have real-algebraic coordinates. Starting from $V_0 = \text{span}_{\mathbb{C}} \Phi(\mathcal{S})$, the iteration

$$V_{t+1} = \text{span}_{\mathbb{C}}(V_t \cup V_t^* \cup V_t V_t)$$

stabilizes at A after at most $N^2 - \dim_{\mathbb{C}} V_0$ strict dimension increases. Exact linear algebra computes its multiplication and adjoint structure constants and its center. If its computed Wedderburn decomposition has s simple blocks, then at most 2^s semialgebraic complete-order tests compute the envelope.

Thus a finite completely order-separating family of matrix states is necessary and sufficient finite input for constructive finite-envelope recovery; a bound on $\dim \mathcal{S}$ alone cannot replace this certificate.

Proof. Let $\{\varphi_j\}_{j=1}^r$ be a finite matrix-state certificate. The direct sum

$$\Phi = \bigoplus_j \varphi_j : \mathcal{S} \longrightarrow \bigoplus_j M_{n_j}$$

is unital and completely positive. The defining biconditional says that each amplification of Φ reflects positivity. Hence Φ is injective and its inverse on $\Phi(\mathcal{S})$ is completely positive. It is therefore a unital complete-order embedding into M_N . Part (a) of Theorem 18.5 makes $C_e^*(\mathcal{S})$ finite-dimensional.

Conversely, suppose

$$C_e^*(\mathcal{S}) = \bigoplus_{i=1}^q M_{m_i}(\mathbb{C})$$

and let $\iota : \mathcal{S} \rightarrow C_e^*(\mathcal{S})$ be the canonical complete-order embedding. Write π_i for the coordinate $*$ -representation onto the i th block and put

$$\varphi_i = \pi_i \iota : \mathcal{S} \rightarrow M_{m_i}.$$

Every φ_i is unital and completely positive. For $X \in M_k(\mathcal{S})_{\text{sa}}$, positivity of all $(\text{id}_k \otimes \varphi_i)(X)$ is exactly positivity of $(\text{id}_k \otimes \iota)(X)$ in the direct-sum algebra. Since ι is a complete-order embedding, this is equivalent to $X \in M_k(\mathcal{S})^+$. The coordinate maps therefore form a finite matrix-state certificate. This proves (a).

For (b), the certificate image is a concrete operator system in the finite algebra $\bigoplus_j M_{n_j}$. By definition, $A = C^*(\Phi(\mathcal{S}))$ is generated by that image, so (A, Φ) is a finite-dimensional C^* -cover. Every ideal of A is a sum of Wedderburn blocks. Theorem 18.6 tests each block quotient for complete-order preservation and selects the largest boundary ideal, whose quotient is the envelope.

Let a certificate have total size N . Its direct sum gives a unital complete-order embedding of $\mathcal{S}$ into M_N , so Theorem 18.5(b) gives

$$N \geq N_{\min}(\mathcal{S}) = \sum_i m_i.$$

The coordinate certificate constructed in the converse part of (a) has total size exactly $\sum_i m_i$. It attains this lower bound, proving (c).

For (d), every V_t is a finite-dimensional $*$ -stable subspace after the first iteration. If $V_{t+1} = V_t$, then V_t is closed under multiplication and adjoint and contains the unit; it is a C^* -subalgebra containing $\Phi(\mathcal{S})$, hence it equals A . Otherwise its complex dimension increases by at least one. Since $A \subseteq M_N$ has dimension at most N^2 , there are at most $N^2 - \dim V_0$ strict increases.

With real-algebraic entries, row reduction in the algebraic number field gives bases and exact multiplication and adjoint structure constants. Solving the linear commutator equations computes the center. Factoring the resulting finite-dimensional commutative algebra gives the minimal central projections, and the corresponding simple ideals give the Wedderburn blocks. The finite-cover theorem then enumerates their block ideals and uses the finite amplification and semialgebraic complete-order criteria for each quotient. This proves the stated termination and exact constructivity. $\square$

Corollary 18.8 (Quantum complete-order automorphisms). An affine automorphism α of the d -dimensional quantum state space has a complete-order-compatible homogeneous extension if and only if there is a unitary $U \in M_d(\mathbb{C})$ such that

$$\alpha(\rho) = U \rho U^\dagger \quad \text{for every density operator } \rho.$$

Proof. Unitary conjugation is completely positive, trace preserving, and has the completely positive inverse given by conjugation with $U^\dagger$.

Conversely, let $\Phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be the complex-linear homogeneous extension of α , and assume that Φ and $\Psi = \Phi^{-1}$ are completely positive. Because α maps the normalized state base onto itself, Φ is trace preserving. Put

$$B = \Phi(I_d).$$

The order isomorphism Φ maps the interior of the positive cone onto itself, so B is positive definite. Define

$$\begin{aligned} \tilde{\Phi}(A) &= B^{-1/2} \Phi(A) B^{-1/2}, \\ \tilde{\Psi}(X) &= \Psi(B^{1/2} X B^{1/2}). \end{aligned}$$

These maps are mutually inverse, unital, and completely positive.

A unital 2-positive map Λ satisfies the Kadison–Schwarz inequality

$$\Lambda(A^\dagger A) \geq \Lambda(A)^\dagger \Lambda(A).$$

Indeed, apply $\text{id}_2 \otimes \Lambda$ to the positive block matrix

$$\begin{pmatrix} A^\dagger A & A^\dagger \\ A & I \end{pmatrix} = \begin{pmatrix} A^\dagger \\ I \end{pmatrix} \begin{pmatrix} A & I \end{pmatrix}$$

and take the Schur complement of the lower-right identity block. Apply this inequality first to $\tilde{\Phi}$ and then to $\tilde{\Psi}$ at $\tilde{\Phi}(A)$. One obtains

$$\begin{aligned} A^\dagger A &\geq \tilde{\Psi}(\tilde{\Phi}(A)^\dagger \tilde{\Phi}(A)) \\ &\geq A^\dagger A. \end{aligned}$$

Both inequalities are equalities. Applying $\tilde{\Phi}$ gives

$$\tilde{\Phi}(A^\dagger A) = \tilde{\Phi}(A)^\dagger \tilde{\Phi}(A).$$

Polarizing this identity yields

$$\tilde{\Phi}(A^\dagger C) = \tilde{\Phi}(A)^\dagger \tilde{\Phi}(C) \quad (A, C \in M_d),$$

so $\tilde{\Phi}$ is a $*$ -automorphism. Every $*$ -automorphism of the full matrix algebra is inner: the images of the standard matrix units form a system of matrix units, and choosing unit vectors in the ranges of the images of the diagonal units, with phases adjusted by the off-diagonal relations, produces an implementing unitary. Hence

$$\tilde{\Phi}(A) = U A U^\dagger$$

for some unitary U . Therefore

$$\Phi(A) = S A S^\dagger, \quad S = B^{1/2} U.$$

Trace preservation gives

$$\text{Tr}(S^\dagger S A) = \text{Tr } A \quad (A \in M_d),$$

so $S^\dagger S = I_d$. Since S is square, it is unitary. Restriction to density operators proves the result. $\square$

Corollary 18.9 (Nonmaximal tensor-unit criterion). Let $P = \mathcal{C}(E, Y)$ be the actual compact convex process body of a finite-dimensional GPT. If $- \otimes E$ has a right adjoint and $Z = R(Y)$, then

$$(H(P), H(P)^+, u_W) \cong (V_Z, V_Z^+, u_Z)$$

as unital ordered vector spaces. In particular,

$$\dim V_Z = \dim_{\text{aff}} \mathcal{C}(E, Y) + 1 \leq \dim W(E, Y).$$

Equality holds if and only if $H(P) = W(E, Y)$, equivalently if and only if the affine hull of P is the full causal normalization fiber in $W(E, Y)$. In that case,

$$\dim V_Z = \dim V_E(\dim V_Y - 1) + 1.$$

Proof. Automatic affinity and the tensor-unit component of the adjunction give an affine isomorphism

$$P = \mathcal{C}(E, Y) \cong \mathcal{C}(I, Z) = \Omega_Z.$$

Theorem 17.4 gives the unital order isomorphism and the first dimension formula. Part (a) gives $H(P) \subseteq W(E, Y)$, and hence the dimension inequality. Equality of the finite dimensions is equivalent to $H(P) = W(E, Y)$. By part (c),

$$\text{aff}(P) = \{h \in H(P) : u_W(h) = 1\},$$

and each normalized hyperplane linearly spans its ambient space. Therefore $H(P) = W(E, Y)$ is equivalent to

$$\text{aff}(P) = \{h \in W(E, Y) : u_W(h) = 1\}.$$

The final formula follows from Theorem 5.2. □

18.5 Synthesis for GPT representability

These results yield the following collection of parallel representability conditions. The enumeration records consequences of a right adjoint rather than a sequence of implications in the displayed order.

Theorem 18.10 (GPT representability hierarchy). Let the GPT be finite-dimensional and locally tomographic, and assume that the allowed transformations $E \rightarrow Y$ have affine hull equal to the full causal normalization fiber

$$\{G \in W(E, Y) : u_W(G) = 1\}.$$

If $- \otimes E$ has a right adjoint represented by an object $Z = R(Y)$, then:

(i)

$$\dim V_Z = \dim W(E, Y) = \dim V_E(\dim V_Y - 1) + 1;$$

(ii) the physical process set $\mathcal{C}(E, Y)$ is affinely isomorphic to the normalized state space Ω_Z ;

(iii) with $P = \mathcal{C}(E, Y)$, the affine isomorphism extends uniquely to a unital order isomorphism

$$\widehat{\phi} : (H(P), H(P)^+, u_W) \longrightarrow (V_Z, V_Z^+, u_Z).$$

Under the full-span hypothesis $H(P) = W(E, Y)$; if the theory is maximal, this is the full positive cone on $W(E, Y)$;

(iv) suppose the complexifications are concretely realized as $\mathcal{S} \subseteq M_m$ and $\mathcal{T} \subseteq M_n$, and suppose that for every k the homogenized physical ancillary ordered spaces are identified with

$$(M_k(\mathcal{S}), M_k(\mathcal{S})^+), \quad (M_k(\mathcal{T}), M_k(\mathcal{T})^+),$$

in such a way that the homogeneous ancillary transposition is $\widehat{\phi}_{\mathbb{C}}^{(k)}$. Then $\widehat{\phi}_{\mathbb{C}}$ is a complete-order isomorphism. Equivalently, for $N = \max\{m, n\}$,

$$\widehat{\phi}_{\mathbb{C}}^{(N)}(M_N(\mathcal{S})^+) = M_N(\mathcal{T})^+.$$

If the representing map is not fixed, its existence under finite semialgebraic side constraints is decided by Theorem 18.4. For abstract operator systems, Theorem 18.5 decides whether a finite concrete realization exists and gives its least matrix size. From a finite cover, Theorem 18.6 computes the envelope and shows that no bound follows from operator-system dimension alone;

- (v) the representing isomorphisms are natural in the test and codomain objects and are compatible with composition.

Failure of any necessary condition obstructs a right adjoint.

Proof. Assume $- \otimes E \dashv R$ and put $Z = R(Y)$. Since tensoring and composition are affine on GPT hom-sets, Proposition 2.3 makes the adjunction transpositions affine. Setting the test object equal to the tensor unit gives the affine state-space isomorphism

$$\mathcal{C}(E, Y) \cong \mathcal{C}(I, Z) = \Omega_Z,$$

which proves (ii).

By hypothesis, $\mathcal{C}(E, Y)$ has affine hull equal to the hyperplane $u_W = 1$ in $W(E, Y)$. Consequently,

$$\dim_{\text{aff}} \mathcal{C}(E, Y) = \dim W(E, Y) - 1.$$

Lemma 17.1 gives

$$\dim_{\text{aff}} \Omega_Z = \dim V_Z - 1.$$

The affine isomorphism in (ii) preserves affine dimension, so

$$\dim W(E, Y) - 1 = \dim V_Z - 1.$$

Theorem 5.2 supplies $\dim W(E, Y) = \dim V_E(\dim V_Y - 1) + 1$, proving (i).

Part (iii) follows by applying Theorem 17.4 to the affine isomorphism in (ii). The full-span hypothesis gives $H(P) = W(E, Y)$ by Corollary 18.9; maximality further identifies $H(P)^+$ with the full positive cone chosen on $W(E, Y)$.

Under the ancillary identifications in (iv), the homogeneous transposition at level k is exactly $\widehat{\phi}_C^{(k)}$. The physical adjunction transposition and its inverse preserve the specified positive cones at that level. Therefore $\widehat{\phi}_C^{(k)}$ maps $M_k(\mathcal{S})^+$ bijectively onto $M_k(\mathcal{T})^+$ for every k . This is precisely the assertion that $\widehat{\phi}_C$ is a complete-order isomorphism. Theorem 18.2 makes this equivalent to the displayed cone equality at $N = \max\{m, n\}$. When the map is allowed to vary under fixed concrete embeddings, Theorem 18.4 gives the stated finite existence test. Theorem 18.5 identifies exactly when an abstract operator system admits such a finite realization, and Theorem 18.6 gives the finite-cover computation and dimension-bound obstruction. This proves (iv).

Finally, naturality in both variables is part of the definition of an adjunction. The transpose and inverse transpose are constructed from the unit, counit, functorial action, and composition; hence they are compatible with composition. This proves (v). $\square$

Remark 18.11 (Instantiation of the representability hierarchy). At the all-codomain level, deterministic quantum channels fail condition (i) of Theorem 18.10: for $B = E$, the causal internal-hom dimension is not the Hermitian-space dimension of any quantum object. Theorem 6.3 additionally shows that every nontrivial fixed-codomain channel body fails the affine state-space condition, even when its dimension is compatible. Deterministic classical channels satisfy the dimension condition but fail the same affine condition because the required process space is a product of simplices rather than a simplex.

The graded instrument and deterministic-superchannel theorems exhibit analogous failures outside the specific GPT hypotheses of Theorem 18.10. All instrument grades together fail through an affine-hyperplane relation, while every nontrivial fixed grade fails the affine state-space test by Theorems 9.8 and 9.10. Fixed-type superchannel decorations fail through input-coefficient comparison or, for pure output decoration, the discrete output-dimension spectrum; their diagonal tensor-unit components are affinely representable exactly when the target output is one-dimensional. In a symmetric compact-closed category, compact transposition supplies the representing object and the required naturality. Theorem 15.6 shows that this structure cannot coexist nontrivially with a terminal tensor unit.

Table 2: Constructive representability criteria and their finite certificates.

Goal	Hypotheses or data	Finite test or construction	Conclusion
Universal evaluation	Surjective programming and separating affine program statistics that factor through evaluated processes	$d = \dim_{\text{aff}} P$ independent scalar statistics	Evaluator bijectivity and an assembled right adjoint (Theorem 13.4).
Experimental tomography	Affine basis programs and a calibrated response matrix C	$\text{rank } C = d$, equivalently $C^\top W C > 0$	Explicit weighted decoder, stability constant, and unique evaluation (Theorem 13.5).
Symmetry-generated tomography	Accessible seed effects closed under a finite or compact symmetry action	$K_\lambda > 0$ on every multiplicity space; compact Haar frames use at most $d(d+1)/2$ weighted outcomes	A finite separating orbit tester with explicit frame bound (Theorems 13.7 and 13.9).
Ancillary complete order	Concrete realizations $\mathcal{S} \subseteq M_m, \mathcal{T} \subseteq M_n$	One cone equality at $N = \max\{m, n\}$	Complete-order compatibility at every ancillary level (Theorem 18.2).
Finite abstract realization	A finite matrix-state certificate reflecting every matrix cone	Algebra closure, center and block computation, followed by at most 2^s block-ideal tests	Constructive C^* -cover, Shilov ideal, envelope, and minimal matrix size (Theorem 18.7).
Positive globalization	A local normalized map and either polyhedral, Choi, or operator-system data	Linear programming, semidefinite feasibility, or Arveson extension	Exact extension decision and, when closed, dual nonextension certificates (Section 20).

Table 2 summarizes the principal positive and constructive criteria complementing the obstruction theorems.

19 Composites Without Local Tomography

Local tomography identifies $V_X \otimes V_E$ with the physical composite. Without that identification, product preparation is represented by a linear map whose kernel records local relations and whose cokernel records global degrees of freedom. Let product preparation induce

$$\tau = \tau_{X,E} : V_X \otimes V_E \longrightarrow V_{X \otimes E}$$

that is compatible with discarding:

$$u_{X \otimes E} \circ \tau = u_X \otimes u_E.$$

Neither injectivity nor surjectivity of τ is assumed.

Theorem 19.1 (Relation and global-excess decomposition). Put

$$K = \ker \tau, \quad S = \text{im } \tau, \quad Q = V_{X \otimes E} / S,$$

and write

$$x = \dim V_X, \quad e = \dim V_E, \quad v = \dim V_{X \otimes E}, \quad t = \text{rank } \tau, \quad y = \dim V_Y.$$

For a linear map $\tilde{F} : V_X \rightarrow W(E, Y)$, let $\tilde{F}^\flat : V_X \otimes V_E \rightarrow V_Y$ denote its ordinary uncurrying. Define

$$\begin{aligned}\mathcal{N}_{XE,Y} &= \{F : V_{X \otimes E} \rightarrow V_Y : u_Y \circ F = u_{X \otimes E}\}, \\ \mathcal{N}_\tau(X, E; Y) &= \{\tilde{F} : V_X \rightarrow W(E, Y) : u_W \circ \tilde{F} = u_X, \tilde{F}^\flat|_K = 0\}.\end{aligned}$$

Then restriction along τ followed by ordinary currying defines a surjective affine map

$$\text{Res}_\tau : \mathcal{N}_{XE,Y} \longrightarrow \mathcal{N}_\tau(X, E; Y).$$

The kernel of its linear map on difference spaces is canonically isomorphic to

$$\text{Hom}(Q, \ker u_Y).$$

Consequently,

$$\begin{aligned}\dim_{\text{aff}} \mathcal{N}_\tau(X, E; Y) &= t(y - 1), \\ \dim \text{Hom}(Q, \ker u_Y) &= (v - t)(y - 1), \\ \dim_{\text{aff}} \mathcal{N}_{XE,Y} &= v(y - 1).\end{aligned}$$

Equivalently, if

$$k = \dim K = xe - t, \quad q = \dim Q = v - t,$$

then local relations remove $k(y - 1)$ dimensions from unrestricted local currying, while global degrees of freedom add $q(y - 1)$ dimensions. Local tomography is the special case $k = q = 0$.

Proof. Let $F \in \mathcal{N}_{XE,Y}$. Compatibility of τ with discarding gives

$$u_Y \circ F \circ \tau = u_{X \otimes E} \circ \tau = u_X \otimes u_E.$$

Theorem 5.2 therefore curries $F \circ \tau$ to a map $\tilde{F} : V_X \rightarrow W(E, Y)$ satisfying $u_W \tilde{F} = u_X$. Since $F \circ \tau$ vanishes on $K = \ker \tau$, this map belongs to $\mathcal{N}_\tau(X, E; Y)$. This defines Res_τ , and it is affine because restriction and currying are linear.

To prove surjectivity, take $\tilde{F} \in \mathcal{N}_\tau(X, E; Y)$. Its uncurrying $f = \tilde{F}^\flat$ vanishes on K , so it factors uniquely through S as a linear map $f_S : S \rightarrow V_Y$ satisfying

$$f_S \circ \tau = f.$$

For every $z \in V_X \otimes V_E$,

$$\begin{aligned}u_Y(f_S(\tau z)) &= u_Y(f(z)) \\ &= (u_X \otimes u_E)(z) \\ &= u_{X \otimes E}(\tau z).\end{aligned}$$

Thus

$$u_Y \circ f_S = u_{X \otimes E}|_S.$$

Choose $y_0 \in V_Y$ with $u_Y(y_0) = 1$ and define

$$F_0(w) = u_{X \otimes E}(w)y_0 \quad (w \in V_{X \otimes E}).$$

The map $f_S - F_0|_S$ takes values in $\ker u_Y$. Choose a linear extension

$$D : V_{X \otimes E} \rightarrow \ker u_Y$$

of this map; such an extension exists after choosing a vector-space complement of S . Then $F = F_0 + D$ satisfies

$$u_Y \circ F = u_{X \otimes E}, \quad F|_S = f_S.$$

Hence $\text{Res}_\tau(F) = \tilde{F}$, proving surjectivity.

The difference space of $\mathcal{N}_{X \otimes E, Y}$ consists of maps $H : V_{X \otimes E} \rightarrow V_Y$ satisfying $u_Y H = 0$. Such a map lies in the kernel of the restriction map exactly when $H|_S = 0$. It then factors uniquely through $Q = V_{X \otimes E}/S$, and the equation $u_Y H = 0$ says exactly that the induced map has image in $\ker u_Y$. This proves the canonical kernel identification.

The affine space of maps $S \rightarrow V_Y$ normalized by $u_{X \otimes E}|_S$ has dimension $t(y-1)$. Indeed, the map

$$\text{Hom}(S, V_Y) \longrightarrow S^*, \quad g \longmapsto u_Y \circ g,$$

is surjective: for $\varphi \in S^*$, the map $g_\varphi(s) = \varphi(s)y_0$ is sent to φ . Its kernel therefore has dimension $ty - t$. Factoring through S and currying identify this affine space with $\mathcal{N}_\tau(X, E; Y)$. Since $\dim Q = v - t$ and $\dim \ker u_Y = y - 1$, the restriction fiber has dimension $(v - t)(y - 1)$. Adding base and fiber dimensions gives

$$t(y - 1) + (v - t)(y - 1) = v(y - 1).$$

Finally, rank-nullity for τ gives $k = xe - t$, while the quotient definition gives $q = v - t$. The unrestricted local normalization fiber has dimension $xe(y - 1)$; imposing annihilation of K leaves $t(y - 1) = xe(y - 1) - k(y - 1)$ dimensions, and the quotient fiber contributes $q(y - 1)$. If τ is an isomorphism, then $K = Q = 0$, recovering the exact normalization-fiber currying of Theorem 5.2. $\square$

Remark 19.2 (Physical transformation sets). Theorem 19.1 is an exact statement about linear normalization fibers. For maximal positive transformations, Proposition 17.2 applied to the composite system $X \otimes E$ shows that positivity has the full affine hull $\mathcal{N}_{X \otimes E, Y}$ and therefore gives affine dimension $v(y - 1)$. In a restricted theory, the actual process body is governed instead by Theorem 17.4 and its dimension profile. Surjectivity of the linear restriction map does not assert that every positive local map has a positive global extension.

20 Positive Extension Across the Product Image

The extension problem can be stated independently of a particular composite. Let $\mathcal{V}, \mathcal{W}$ be finite-dimensional ordered real vector spaces with cones $\mathcal{V}^+, \mathcal{W}^+$ and strictly positive deterministic effects $u \in \mathcal{V}^*$ and $v \in \mathcal{W}^*$. Let $S \subseteq \mathcal{V}$ be a linear subspace, equipped with the induced cone $S^+ = S \cap \mathcal{V}^+$, and let

$$f : S \longrightarrow \mathcal{W}$$

be positive and normalized:

$$v \circ f = u|_S.$$

For a composite without local tomography, one takes $\mathcal{V} = V_{X \otimes E}$ and $S = \text{im } \tau$.

20.1 Retract and polyhedral regimes

Proposition 20.1 (Positive-retract extension criterion). Suppose there is a positive normalized retraction

$$r : \mathcal{V} \longrightarrow S, \quad r|_S = \text{id}_S, \quad u|_S \circ r = u.$$

Then every positive normalized map $f : S \rightarrow \mathcal{W}$ has the positive normalized extension

$$F = f \circ r : \mathcal{V} \longrightarrow \mathcal{W}.$$

If the spaces carry matrix orderings and both r and f are completely positive, then F is completely positive.

Proof. The equality $r|_S = \text{id}_S$ gives $F|_S = f$. Positivity is preserved under composition, and

$$vF = vfr = u|_S r = u,$$

so F is normalized. The matrix-ordered statement follows because a composition of completely positive maps is completely positive at every amplification level. $\square$

Theorem 20.2 (Exact polyhedral extension criterion). Assume the cones are polyhedral and choose finite descriptions

$$\mathcal{V}^+ = \text{cone}\{x_1, \dots, x_R\}, \quad \mathcal{W}^+ = \{w : \lambda_j(w) \geq 0, j = 1, \dots, T\}.$$

Let $Q = \mathcal{V}/S$, let $\pi : \mathcal{V} \rightarrow Q$ be the quotient map, and choose any normalized linear extension F_0 of f . Then every normalized linear extension of f is uniquely of the form

$$F = F_0 + G \circ \pi, \quad G \in \text{Hom}(Q, \ker v).$$

Such an extension is positive if and only if the finite system of linear inequalities

$$\lambda_j(F_0(x_i) + G(\pi x_i)) \geq 0 \quad (1 \leq i \leq R, 1 \leq j \leq T) \quad (26)$$

has a solution G .

Moreover, failure has a finite Farkas certificate. Define linear functionals on $\text{Hom}(Q, \ker v)$ by

$$\mathfrak{L}_{ij}(G) = \lambda_j(G(\pi x_i))$$

and scalars $c_{ij} = \lambda_j(F_0(x_i))$. System (26) is infeasible if and only if there are coefficients $\mu_{ij} \geq 0$ such that

$$\sum_{i,j} \mu_{ij} \mathfrak{L}_{ij} = 0, \quad \sum_{i,j} \mu_{ij} c_{ij} < 0.$$

Proof. A normalized linear extension always exists. Choose $w_0 \in \mathcal{W}$ with $v(w_0) = 1$ and set $F_{\text{base}}(x) = u(x)w_0$. The map $f - F_{\text{base}}|_S$ takes values in $\ker v$ and therefore has a linear extension $D : \mathcal{V} \rightarrow \ker v$. Then $F_0 = F_{\text{base}} + D$ is normalized and extends f .

If F and F_0 are normalized extensions of f , their difference $H = F - F_0$ vanishes on S and satisfies $vH = 0$. Hence H factors uniquely through Q as $H = G\pi$ with image in $\ker v$. Conversely, every such G produces a normalized extension. This proves the affine parametrization.

A linear map F is positive exactly when $F(x_i) \in \mathcal{W}^+$ for every cone generator x_i . By the half-space description of $\mathcal{W}^+$, this is equivalent to all inequalities in (26). They are linear in the matrix entries of G , so they form a finite linear feasibility problem.

Write the system as

$$\mathfrak{L}_{ij}(G) \geq -c_{ij}.$$

The finite-dimensional Farkas alternative says that it is infeasible exactly when nonnegative multipliers μ_{ij} annihilate the left-hand linear map and have strictly positive pairing with the right-hand side. These conditions are precisely

$$\sum_{i,j} \mu_{ij} \mathfrak{L}_{ij} = 0, \quad -\sum_{i,j} \mu_{ij} c_{ij} > 0,$$

which proves the certificate statement. $\square$

20.2 General conic duality and closed-image criteria

Theorem 20.3 (General conic extension, duality, and stability). Retain the finite-dimensional ordered spaces, normalized map $f : S \rightarrow \mathcal{W}$, quotient $Q = \mathcal{V}/S$, and normalized linear extension F_0 above. Let

$$\mathfrak{E} = \text{Hom}(\mathcal{V}, \mathcal{W}), \quad \mathfrak{G} = \text{Hom}(Q, \ker v),$$

let

$$L : \mathfrak{G} \rightarrow \mathfrak{E}, \quad L(G) = G\pi,$$

and let $q : \mathfrak{E} \rightarrow \mathfrak{E}/\text{im } L$ be the quotient map. Define the closed positive-map cone

$$\mathfrak{P} = \{F \in \mathfrak{E} : F(\mathcal{V}^+) \subseteq \mathcal{W}^+\}.$$

Then:

- (a) f has a positive normalized extension if and only if

$$q(F_0) \in q(\mathfrak{P}).$$

- (b) Let

$$\mathfrak{P}^* = \{\Lambda \in \mathfrak{E}^* : \Lambda(P) \geq 0 \text{ for every } P \in \mathfrak{P}\}, \quad (\text{im } L)^\perp = \{\Lambda : \Lambda|_{\text{im } L} = 0\}.$$

Then

$$q(F_0) \in \overline{q(\mathfrak{P})} \iff \Lambda(F_0) \geq 0 \text{ for every } \Lambda \in \mathfrak{P}^* \cap (\text{im } L)^\perp. \quad (27)$$

If $q(\mathfrak{P})$ is closed, the closure may be removed. In that case, nonextendability is equivalent to a certificate

$$\Lambda \in \mathfrak{P}^* \cap (\text{im } L)^\perp, \quad \Lambda(F_0) < 0.$$

- (c) If some normalized extension belongs to $\text{int } \mathfrak{P}$, then $q(F_0) \in \text{int } q(\mathfrak{P})$. Consequently, all sufficiently small perturbations of the extension data in the quotient remain positively extendable.

If $\mathcal{V}^+$ has a compact normalized base B and $F(B) \subseteq \text{int } \mathcal{W}^+$, then $F \in \text{int } \mathfrak{P}$; hence this condition implies the stability in part (c).

Proof. Every normalized linear extension is $F_0 + L(G)$ by the affine-parametrization argument in Theorem 20.2, which does not use polyhedrality. Hence such an extension belongs to $\mathfrak{P}$ exactly when there is $P \in \mathfrak{P}$ with $P - F_0 \in \text{im } L$, equivalently $q(P) = q(F_0)$. Indeed, quotient equality supplies $G \in \mathfrak{G}$ with $L(G) = P - F_0$, and therefore

$$F_0 + L(G) = P \in \mathfrak{P}.$$

Conversely, every feasible correction has the same quotient class as F_0 . This proves (a).

Put $\mathfrak{K} = q(\mathfrak{P})$. A linear functional on the quotient pulls back along q to a functional on $\mathfrak{E}$ that vanishes on $\text{im } L$, and every functional annihilating $\text{im } L$ factors uniquely in this way. Under this identification, the dual cone of $\mathfrak{K}$ is precisely

$$\mathfrak{P}^* \cap (\text{im } L)^\perp.$$

The finite-dimensional bipolar theorem identifies the double dual cone with $\overline{\mathfrak{K}}$. This gives (27). If $q(\mathfrak{P})$ is closed, the ordinary finite-dimensional separation theorem gives a functional strictly negative on $q(F_0)$ and nonnegative on $q(\mathfrak{P})$ exactly when $q(F_0) \notin q(\mathfrak{P})$, proving the certificate claim.

For (c), let $P_0 = F_0 + L(G_0) \in \text{int } \mathfrak{P}$. A quotient map between finite-dimensional vector spaces is open, so it maps a neighborhood of P_0 contained in $\mathfrak{P}$ to a neighborhood of $q(P_0) = q(F_0)$ contained in $q(\mathfrak{P})$.

Finally, suppose B is compact and $F(B)$ lies in the interior of $\mathcal{W}^+$. After choosing norms, compactness gives a uniform positive distance from $F(B)$ to the complement of $\mathcal{W}^+$. Every sufficiently small operator-norm perturbation of F therefore remains positive on B . Since B generates $\mathcal{V}^+$ by nonnegative scaling, the perturbation is positive on the whole cone. Thus F is interior to $\mathfrak{P}$. $\square$

Proposition 20.4 (Transverse-kernel closed-image criterion). Let $\mathfrak{C}$ be a closed cone in a finite-dimensional real normed space E , and let $A : E \rightarrow F$ be linear. If

$$\mathfrak{C} \cap \ker A = \{0\},$$

then $A(\mathfrak{C})$ is closed. If $\mathfrak{C} \neq \{0\}$, there is moreover a constant $\kappa > 0$ such that

$$\|Ax\| \geq \kappa\|x\| \quad (x \in \mathfrak{C}).$$

Consequently, in Theorem 20.3, the sufficient condition

$$\mathfrak{P} \cap \text{im } L = \{0\}$$

implies that $q(\mathfrak{P})$ is closed, so the strict nonextension certificate in part (b) applies without a separate closed-image assumption.

Proof. If $\mathfrak{C} = \{0\}$, then $A(\mathfrak{C}) = \{0\}$ is closed. Suppose $\mathfrak{C} \neq \{0\}$ and set

$$K = \{x \in \mathfrak{C} : \|x\| = 1\}.$$

The set K is nonempty and compact. The kernel hypothesis gives $Ax \neq 0$ for every $x \in K$. Hence the continuous function $x \mapsto \|Ax\|$ has a strictly positive minimum

$$\kappa = \min_{x \in K} \|Ax\| > 0.$$

For nonzero $x \in \mathfrak{C}$, the cone property gives $x/\|x\| \in K$, and positive homogeneity yields

$$\|Ax\| = \|x\| \left\| A \left(\frac{x}{\|x\|} \right) \right\| \geq \kappa\|x\|.$$

The inequality is also immediate for $x = 0$.

Let Ax_n be a convergent sequence with $x_n \in \mathfrak{C}$. The coercive estimate makes (x_n) bounded. Finite dimensionality supplies a convergent subsequence $x_{n_j} \rightarrow x$. Since $\mathfrak{C}$ is closed, $x \in \mathfrak{C}$, and continuity gives $Ax_{n_j} \rightarrow Ax$. The original image sequence has the same limit, so that limit belongs to $A(\mathfrak{C})$. Thus $A(\mathfrak{C})$ is closed.

For the final assertion, take $\mathfrak{C} = \mathfrak{P}$ and $A = q$. The quotient map has kernel $\text{im } L$, so the displayed transversality condition is exactly $\mathfrak{P} \cap \ker q = \{0\}$. The first part therefore makes $q(\mathfrak{P})$ closed, and Theorem 20.3(b) applies. $\square$

A closed convex cone K is called *nice* when

$$K^* + F^\perp$$

is closed for every face F of K , where $F^\perp$ is the orthogonal complement of $\text{span } F$. Polyhedral, positive-semidefinite, second-order, and standard p -cones are nice [26].

Theorem 20.5 (Facial feasible-direction criterion). Let $\mathfrak{C}$ be a closed convex cone in a finite-dimensional Euclidean space, let $A : E \rightarrow F$ be linear, and put

$$K = \mathfrak{C}^*, \quad R = \text{range } A^*.$$

Choose

$$w \in \text{ri}(K \cap R)$$

and let F_w be the minimal face of K containing w . Define the feasible-direction cone

$$\text{dir}(w, K) = \{v : \exists \epsilon > 0, w + \epsilon v \in K\}.$$

Then

$$\text{dir}(w, K) = K + \text{span } F_w.$$

Moreover:

(a) if $A(\mathfrak{C})$ is closed, then

$$R \cap (\overline{K + \text{span } F_w} \setminus (K + \text{span } F_w)) = \emptyset;$$

(b) if $K = \mathfrak{C}^*$ is nice, the displayed condition is also sufficient for $A(\mathfrak{C})$ to be closed.

Thus, for nice dual cones, closedness is decided by one minimal face and one feasible-direction closure test.

Proof. Since $w \in \text{ri } F_w$, every $f \in \text{span } F_w$ satisfies $w + \delta f \in F_w$ for all sufficiently small $|\delta|$. Hence, if $v = k + f$ with $k \in K$ and $f \in \text{span } F_w$, then $w + \epsilon v \in K$ for sufficiently small $\epsilon > 0$. This proves

$$K + \text{span } F_w \subseteq \text{dir}(w, K).$$

Conversely, if $w + \epsilon v \in K$, then

$$v = \epsilon^{-1}(w + \epsilon v) - \epsilon^{-1}w \in K + \text{span } F_w,$$

so equality holds.

Apply the closed-linear-image theorem of Pataki [26] with cone K , linear map $M = A^*$, and point $w \in \text{ri}(K \cap \text{range } M)$. Since

$$M^* K^* = A\mathfrak{C},$$

that theorem makes absence of vectors in

$$\text{range } M \cap (\overline{\text{dir}(w, K)} \setminus \text{dir}(w, K))$$

necessary for every closed cone and sufficient when K is nice. Substituting $M = A^*$ and the direction-cone identity proves (a) and (b). $\square$

Corollary 20.6 (Positive-semidefinite block criterion). Let $A : \text{Herm}(\mathbb{C}^n) \rightarrow \mathbb{R}^q$ be linear. Choose

$$X \in \text{ri}(\text{Pos}(\mathbb{C}^n) \cap \text{range } A^*)$$

and a unitary basis in which

$$X = \begin{pmatrix} X_{11} & 0 \\ 0 & 0 \end{pmatrix}, \quad X_{11} > 0,$$

with $r = \text{rank } X$. Write each $V \in \text{range } A^*$ conformally as

$$V = \begin{pmatrix} V_{11} & V_{12} \\ V_{12}^* & V_{22} \end{pmatrix}.$$

Then $A(\text{Pos}(\mathbb{C}^n))$ is closed if and only if there is no $V \in \text{range } A^*$ such that

$$V_{22} \geq 0, \quad \ker V_{22} \not\subseteq \ker V_{12}.$$

Proof. The minimal face generated by X consists of the positive-semidefinite matrices supported on the first r coordinates. Its linear span is the full Hermitian upper-left block. A standard positive block-matrix completion gives

$$\text{Pos}(\mathbb{C}^n) + \text{span } F_X = \left\{ \begin{pmatrix} U & B \\ B^* & C \end{pmatrix} : U = U^*, C \geq 0, \ker C \subseteq \ker B \right\}.$$

Indeed, the kernel condition is necessary for positivity of a completion, and it is sufficient by taking an upper-left block larger than $BC^\dagger B^*$, where $C^\dagger$ is the Moore–Penrose inverse. Its closure is

$$\left\{ \begin{pmatrix} U & B \\ B^* & C \end{pmatrix} : U = U^*, C \geq 0 \right\},$$

because replacing C by $C + \epsilon I$ makes the kernel condition automatic. The positive-semidefinite cone is self-dual and nice. Theorem 20.5 therefore says that the image is closed exactly when the range of A^* contains no element of the difference between these two sets, which is the stated block condition. $\square$

Theorem 20.7 (Dual complementary-face criterion). Retain the notation of Theorem 20.5 and define the complementary face

$$F_w^\Delta = \mathfrak{C} \cap F_w^\perp.$$

Then closedness of $A(\mathfrak{C})$ implies

$$A(F_w^\Delta) = A(F_w^\perp).$$

If the dual cone $\mathfrak{C}^*$ is nice, the converse also holds. Hence for nice dual cones,

$$A(\mathfrak{C}) \text{ is closed} \iff A(\mathfrak{C} \cap F_w^\perp) = A(F_w^\perp).$$

Proof. Apply the complementary-face form of the closed-linear-image theorem of [26] to $K = \mathfrak{C}^*$, $M = A^*$, and the face F_w selected by a relative-interior point of $K \cap \text{range } M$. In that theorem the complementary face is

$$K^* \cap F_w^\perp = \mathfrak{C} \cap F_w^\perp = F_w^\Delta,$$

and $M^* = A$. The image equality is necessary for arbitrary closed K and sufficient when K is nice, giving precisely the two assertions above. $\square$

Theorem 20.8 (Semialgebraic closed-image decision and conditioning). Let $\mathfrak{C} \subseteq \mathbb{R}^p$ be a nonempty closed semialgebraic cone and let $A : \mathbb{R}^p \rightarrow \mathbb{R}^q$ be linear. Assume that the defining polynomials of $\mathfrak{C}$ and the entries of A are real algebraic, and put

$$D = A(\mathfrak{C}).$$

Then:

- (a) The sets D and $\overline{D}$ are semialgebraic, and whether $D = \overline{D}$ is decidable by real quantifier elimination. This remains valid when $\mathfrak{C} \cap \ker A$ contains nonzero recession directions.
- (b) If D is not closed, a real-algebraic point $y_0 \in \overline{D} \setminus D$ and continuous semialgebraic curves

$$y : (0, \varepsilon) \rightarrow D, \quad x : (0, \varepsilon) \rightarrow \mathfrak{C}$$

may be chosen so that

$$Ax(t) = y(t), \quad y(t) \longrightarrow y_0, \quad \|x(t)\| \longrightarrow \infty.$$

(c) For $y \in D$, define the minimal lifting norm

$$\mu(y) = \min\{\|x\| : x \in \mathfrak{C}, Ax = y\}.$$

The minimum is attained, μ is semialgebraic and positively homogeneous, and D has a finite semialgebraic stratification on which μ is continuous. In part (b), the lift may be chosen with $\|x(t)\| = \mu(y(t))$. After restricting ε if necessary, there are $c_1, c_2 > 0$ and a rational exponent $\alpha > 0$ such that

$$c_1 t^{-\alpha} \leq \mu(y(t)) \leq c_2 t^{-\alpha} \quad (0 < t < \varepsilon).$$

Consequently, for semialgebraic GPT cones or spectrahedral quantum cones, the closed-image question in Theorem 20.3 is decidable even when $\mathfrak{P} \cap \text{im } L \neq \{0\}$. If the quotient image is not closed, part (b) supplies an exact weak-feasibility curve and part (c) quantifies the divergence of minimal positive lifts.

Proof. Membership in D is the first-order condition

$$y \in D \iff \exists x (x \in \mathfrak{C} \wedge Ax = y).$$

The Tarski–Seidenberg theorem therefore makes D semialgebraic. Membership in its closure is expressed by

$$y \in \overline{D} \iff \forall \epsilon > 0 \exists x \in \mathfrak{C} : \|Ax - y\|^2 < \epsilon^2.$$

Thus $\overline{D}$ is semialgebraic, and closedness is the first-order sentence

$$\forall y (y \in \overline{D} \implies y \in D).$$

Real quantifier elimination decides this sentence for real-algebraic input data [27]. No kernel-transversality assumption occurs, proving (a).

If D is not closed, the nonempty semialgebraic set $\overline{D} \setminus D$ has a real-algebraic sample point y_0 . The semialgebraic curve-selection lemma gives a continuous semialgebraic curve $y(t) \in D$ converging to y_0 . The relation

$$\{(t, x) : 0 < t < \varepsilon, x \in \mathfrak{C}, Ax = y(t)\}$$

is semialgebraic and has nonempty fibers. Semialgebraic choice supplies a semialgebraic lift $x(t)$; after a finite cell decomposition, the interval may be shortened so that the lift is continuous. If $\|x(t)\|$ did not tend to infinity, there would be a sequence $t_j \downarrow 0$ for which $x(t_j)$ is bounded. A convergent subsequence would have a limit $x_0 \in \mathfrak{C}$ with $Ax_0 = y_0$, contradicting $y_0 \notin D$. This proves (b).

For fixed $y \in D$, the feasible fiber $\mathfrak{C} \cap A^{-1}(y)$ is nonempty and closed. The Euclidean norm is coercive, so it attains its minimum on that fiber. The graph of μ is defined by the first-order formula

$$(y, r) \in \text{graph } \mu \iff r \geq 0 \wedge \exists x \in \mathfrak{C} : Ax = y, \|x\|^2 = r^2, \\ \forall z \in \mathfrak{C} : Az = y \implies \|z\|^2 \geq r^2.$$

Hence μ is semialgebraic. Since $\mathfrak{C}$ is a cone and A is linear,

$$\mu(\lambda y) = \lambda \mu(y) \quad (\lambda \geq 0).$$

Semialgebraic cell decomposition gives a finite partition of D into cells on which μ is continuous.

Apply semialgebraic choice to the nonempty argmin fibers to choose $x(t)$ with $\|x(t)\| = \mu(y(t))$. The argument in part (b) shows that this quantity tends to infinity. The one-variable semialgebraic

growth theorem then gives a rational $\alpha > 0$ and positive constants c_1, c_2 with the displayed power bounds after shortening the interval [27]. This proves (c).

Finally, if the source and target GPT cones are semialgebraic, positivity of a linear map is itself a first-order semialgebraic condition; for spectrahedral cones it is given directly by positive-semidefinite constraints. Thus the positive-map cone $\mathfrak{P}$ and the quotient map in Theorem 20.3 satisfy the hypotheses above. Parts (a)–(c) give the asserted decision and conditioning statements. $\square$

Example 20.9 (The quotient positive cone need not be closed). Let Pos_2 be the cone of real positive-semidefinite 2×2 matrices and consider

$$\Pi : \text{Sym}_2(\mathbb{R}) \rightarrow \mathbb{R}^2, \quad \Pi \begin{pmatrix} a & b \\ b & c \end{pmatrix} = (a, b).$$

Then

$$\Pi(\text{Pos}_2) = \{(a, b) : a > 0\} \cup \{(0, 0)\}.$$

Indeed, positivity forces $a \geq 0$ and $ac \geq b^2$; if $a = 0$, then $b = 0$, while for $a > 0$ one may choose $c = b^2/a$. The sequence $(1/n, 1)$ belongs to the image and converges to $(0, 1)$, which does not. For the Frobenius norm, the minimal lift of (a, b) with $a > 0$ has $c = b^2/a$ and therefore

$$\mu(a, b)^2 = a^2 + 2b^2 + \frac{b^4}{a^2}.$$

Along $y(t) = (t, 1)$,

$$\mu(y(t)) = \sqrt{t^2 + 2 + t^{-2}} \sim t^{-1},$$

giving the exponent $\alpha = 1$ in Theorem 20.8. Thus the closed-image hypothesis in the exact dual certificate cannot be omitted: without it, dual inequalities characterize only approximate extendability, namely membership in the closure.

20.3 Quantum and operator-system regimes

Theorem 20.10 (Positive normalized extension can fail). Let

$$\mathcal{V} = \mathbb{R}^3, \quad \mathcal{V}^+ = \mathbb{R}_+^3, \quad u(x) = x_1 + x_2 + x_3,$$

and let $\mathcal{W} = \mathbb{R}^2$ have the analogous positive cone and deterministic effect. Put

$$p = \frac{1}{2}(1, 1, 0), \quad q = \frac{1}{2}(1, 0, 1), \quad S = \text{span}\{p, q\}.$$

The map

$$f(p) = (1, 0), \quad f(q) = (0, 1)$$

is positive and normalized on S , but it has no positive normalized extension $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$.

Under the diagonal embeddings into $M_3(\mathbb{C})$ and $M_2(\mathbb{C})$, the complexification of f is a complete-order isomorphism onto the diagonal subspace spanned by E_{11}, E_{22} , but no CPTP map $M_3(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ extends it.

Proof. Every element of S has the form

$$\alpha p + \beta q = \frac{1}{2}(\alpha + \beta, \alpha, \beta).$$

It belongs to $\mathbb{R}_+^3$ exactly when $\alpha, \beta \geq 0$. Thus the induced cone on S is generated by p, q , and f maps those generators to the two generators of $\mathbb{R}_+^2$. Moreover,

$$u(p) = u(q) = 1, \quad v(f(p)) = v(f(q)) = 1,$$

so f is positive and normalized.

Suppose a positive normalized extension F existed, and let $y_i = F(e_i) \in \mathbb{R}_+^2$. Normalization gives $v(y_i) = 1$, so each y_i is a probability vector. The prescribed values of f imply

$$\frac{y_1 + y_2}{2} = (1, 0), \quad \frac{y_1 + y_3}{2} = (0, 1).$$

The two points on the right are extreme points of the probability segment. The first equality forces $y_1 = y_2 = (1, 0)$, whereas the second forces $y_1 = y_3 = (0, 1)$, a contradiction.

For the matrix-ordered statement, identify

$$p = \text{diag}(1/2, 1/2, 0), \quad q = \text{diag}(1/2, 0, 1/2).$$

At level k , an element of $M_k(\text{span}_{\mathbb{C}}\{p, q\})$ has the unique form $A \otimes p + B \otimes q$ and is block diagonal with diagonal blocks

$$\frac{A+B}{2}, \quad \frac{A}{2}, \quad \frac{B}{2}.$$

It is positive if and only if $A, B \geq 0$. Its image under $\text{id}_k \otimes f$ is the block-diagonal operator

$$A \otimes E_{11} + B \otimes E_{22},$$

which is positive under the same condition. Thus f and its inverse on the range are completely positive.

If a CPTP extension $\Phi : M_3 \rightarrow M_2$ existed, the states $\rho_i = \Phi(E_{ii})$ would satisfy

$$\frac{\rho_1 + \rho_2}{2} = E_{11}, \quad \frac{\rho_1 + \rho_3}{2} = E_{22}.$$

Since rank-one density operators are extreme, the first equality gives $\rho_1 = \rho_2 = E_{11}$ and the second gives $\rho_1 = \rho_3 = E_{22}$, again a contradiction. $\square$

Proposition 20.11 (Finite Choi feasibility criterion). Let $\mathcal{R} \subseteq M_a(\mathbb{C})$ be a $*$ -closed linear subspace and let $\varphi : \mathcal{R} \rightarrow M_b(\mathbb{C})$ be complex linear. There exists a CPTP extension $\Phi : M_a(\mathbb{C}) \rightarrow M_b(\mathbb{C})$ of φ if and only if there is an operator $J \in M_a(\mathbb{C}) \otimes M_b(\mathbb{C})$ satisfying

$$\begin{aligned} J &\geq 0, \\ \text{Tr}_b J &= I_a, \\ \text{Tr}_a \left[(A^\top \otimes I_b) J \right] &= \varphi(A) \quad (A \in \mathcal{R}). \end{aligned}$$

It is enough to impose the last family on a basis of $\mathcal{R}$; hence CPTP extendability is an exact finite semidefinite feasibility problem.

Proof. For a linear map $\Phi : M_a \rightarrow M_b$, use the Choi convention

$$J_\Phi = \sum_{i,j=1}^a E_{ij} \otimes \Phi(E_{ij}).$$

The Choi theorem gives

$$\Phi \text{ is completely positive} \iff J_\Phi \geq 0,$$

and trace preservation is equivalent to $\text{Tr}_b J_\Phi = I_a$ [13, 38]. The inverse Choi formula is

$$\Phi(A) = \text{Tr}_a \left[(A^\top \otimes I_b) J_\Phi \right].$$

Therefore agreement with φ on $\mathcal{R}$ is exactly the displayed family of linear constraints. Conversely, any J satisfying those constraints defines by the inverse formula a CPTP extension of φ . Finite dimensionality reduces agreement to any chosen basis. $\square$

Proposition 20.12 (Operator-system extension regime). Let $\mathcal{R} \subseteq M_a(\mathbb{C})$ be an operator system, and let $\varphi : \mathcal{R} \rightarrow M_b(\mathbb{C})$ be completely positive. Then φ has a completely positive extension to $M_a(\mathbb{C})$. If φ is unital, the extension is unital.

Proof. This is the finite-dimensional case of the Arveson extension theorem [14]. The extension agrees with φ on the identity, so it is unital whenever φ is unital. $\square$

Remark 20.13 (Schrödinger and Heisenberg normalization). Proposition 20.12 is a unital, Heisenberg-picture extension statement. Theorem 20.10 concerns trace-preserving Schrödinger-picture extension from a subspace that does not contain the identity. Thus the failure theorem does not contradict Arveson extension. For finite quantum systems, Proposition 20.11 gives the exact trace-preserving criterion in either case.

21 Ambient Codimension and Structural Origin

The causal internal-hom theorem gives the primary structural explanation. The following result isolates the additional ambient hypothesis under which normalization codimensions must agree.

Lemma 21.1 (Dimension of a normalization fiber). Let $L : V \rightarrow W$ be a surjective linear map between finite-dimensional vector spaces. Every nonempty fiber $L^{-1}(w)$ is an affine translate of $\ker L$ and has dimension

$$\dim V - \dim W.$$

Proof. If $v_0 \in L^{-1}(w)$, then $L^{-1}(w) = v_0 + \ker L$. Surjectivity gives $\text{rank } L = \dim W$, and rank-nullity gives $\dim \ker L = \dim V - \dim W$. $\square$

Theorem 21.2 (Transport of normalization codimension). Let V_i, W_i be finite-dimensional vector spaces, let $L_i : V_i \rightarrow W_i$ be surjective, and let $N_i = L_i^{-1}(w_i)$ be nonempty. Suppose a linear isomorphism $C : V_1 \rightarrow V_2$ satisfies $C(N_1) = N_2$. Then

$$C(\ker L_1) = \ker L_2, \quad \dim W_1 = \dim W_2.$$

Proof. For every nonempty affine fiber,

$$N_1 - N_1 = \ker L_1.$$

Since C is linear and maps N_1 bijectively onto N_2 , it maps their difference spaces bijectively. Thus the kernels have equal dimension. The spaces V_1, V_2 have equal dimension, so rank-nullity gives $\text{rank } L_1 = \text{rank } L_2$. Surjectivity identifies these ranks with $\dim W_1$ and $\dim W_2$. $\square$

Remark 21.3 (Necessity of the ambient hypothesis). An affine bijection of normalized fibers alone does not force equal ambient codimensions. The line

$$\{(1, t) : t \in \mathbb{R}\} \subset \mathbb{R}^2$$

is a rank-1 fiber, while

$$\{(1, 1, t) : t \in \mathbb{R}\} \subset \mathbb{R}^3$$

is a rank-2 fiber, yet $(1, t) \mapsto (1, 1, t)$ is an affine bijection. Thus an affine adjunction on normalized hom-sets does not by itself transport ambient codimension; that conclusion requires an ambient linear extension.

For quantum processes, the normalization target at input X has real dimension d_X^2 , and environmental extension scales it by d_E^2 . For classical processes it has dimension $|X|$ and is scaled by $|E|$. The profile and grade arguments show when this mismatch survives every possible choice of representing object. Theorem 5.2 identifies the corresponding ambient causal internal-hom space.

22 Scope of the Results

The results primarily concern representability of hom-functors and existence of right adjoints; Theorem 10.4 also records the dimension-regular left-adjoint consequence of the same explicit profile law. Failure of a right adjoint does not imply that every individual process lacks a left inverse.

Dimension-profile invariance is necessary but not sufficient for representability. The classical stochastic vertex obstruction and the quantum extreme-boundary obstruction demonstrate the need for finer affine invariants when the dimension condition is satisfied.

The autonomous controller theorem is a finite classical-comb construction. Its program values are perfectly distinguishable classical states, and a desired control is selected by preparation of the initial program state. It does not assert coherent quantum control, indefinite causal order, or representable descent. Likewise, a finite matrix-state certificate includes the assertion that its maps reflect positivity at every matrix level. The certificate theorem constructs a cover from such certified data; it does not derive the certificate from operator-system dimension or from an uncertified finite truncation of the matrix order.

The profile and affine-rank defects are geometric quantities. Without a specified operational norm and continuity assumptions, they do not yield metric error bounds; Proposition 16.6 gives an explicit finite-dimensional example. The stability estimate in Theorem 13.5 has different hypotheses: it is relative to a calibrated tester, a positive weight matrix, and the resulting smallest singular value. Informational completeness on evaluated processes also cannot eliminate a nontrivial evaluator fiber. The finite response-rank condition is the exact experimental separation boundary. The symmetry-orbit criteria assume an exact finite or compact group action and closure of the accessible outcomes under that action. The perturbation corollary permits approximate implementation only when the actual affine outcomes still factor through the evaluator and the explicit frame-error threshold holds. The metric transposition theorem separately controls inverse, naturality, and triangle defects on specified hom-set metrics. It does not supply tensorial error accumulation, asymptotic regularization, or a noncompact-group theorem without further Lipschitz and analytic hypotheses.

Proposition 22.1 (Recovery, Bayesian inversion, and entropy change). The following constructions are typed independently of environmental right-adjoint representability.

- (a) Let $\mathcal{N} : A \rightarrow B$ be a quantum channel, let $\sigma > 0$ be a faithful density operator, and assume $\mathcal{N}(\sigma) > 0$. The Petz map

$$\mathcal{R}_{\sigma, \mathcal{N}}(X) = \sigma^{1/2} \mathcal{N}^\dagger \left(\mathcal{N}(\sigma)^{-1/2} X \mathcal{N}(\sigma)^{-1/2} \right) \sigma^{1/2}$$

is a channel $B \rightarrow A$ satisfying $\mathcal{R}_{\sigma, \mathcal{N}}(\mathcal{N}(\sigma)) = \sigma$. It depends on the pair $(\sigma, \mathcal{N})$.

- (b) For a finite stochastic map $N(y | x)$ and a prior $p(x)$, put $q(y) = \sum_x N(y | x)p(x)$. On the support of q , the Bayesian inverse is

$$N_p^\dagger(x | y) = \frac{N(y | x)p(x)}{q(y)}.$$

Columns with $q(y) = 0$ may be chosen arbitrarily to obtain a stochastic map. For the unique map from a nontrivial finite set X to the singleton, this inverse is exactly preparation of p and therefore varies with the prior.

- (c) In **FinStoch** on a set X with $|X| > 1$, channels exist whose Shannon entropy change is zero, negative, or positive: respectively the identity, a pure-state replacement applied to a nonpure state, and uniform replacement applied to a pure state.

Consequently, failure of an environmental right adjoint neither excludes state-dependent recovery or Bayesian inversion nor determines a sign of entropy change.

Proof. For (a), conjugation by a fixed operator is completely positive, and $\mathcal{N}^\dagger$ is completely positive because $\mathcal{N}$ is. Hence $\mathcal{R}_{\sigma, \mathcal{N}}$ is completely positive. For every X , cyclicity of trace and the defining adjoint relation give

$$\begin{aligned}\mathrm{Tr} \mathcal{R}_{\sigma, \mathcal{N}}(X) &= \mathrm{Tr} \left[\mathcal{N}(\sigma) \mathcal{N}(\sigma)^{-1/2} X \mathcal{N}(\sigma)^{-1/2} \right] \\ &= \mathrm{Tr} X.\end{aligned}$$

Thus the map is trace preserving. Since $\mathcal{N}^\dagger(I_B) = I_A$, substitution of $X = \mathcal{N}(\sigma)$ gives recovery of σ . The stronger characterization of equality in relative-entropy data processing by this recovery map is the Petz sufficiency theorem [34, 35]; it is a channel-and-reference-state statement, not a hom-functor representation.

For (b), summing the displayed expression over x gives one, and

$$q(y) N_p^\dagger(x | y) = p(x) N(y | x),$$

which is Bayes' identity. If the codomain is a singleton, then $q(*) = 1$ and $N_p^\dagger(x | *) = p(x)$, proving explicit prior dependence.

For (c), write

$$H(p) = - \sum_x p(x) \log p(x).$$

The identity leaves H unchanged. A replacement channel with pure output has output entropy zero, so it strictly decreases H on every nonpure input. Uniform replacement has output entropy $\log |X|$, so it strictly increases H on a pure input. These three dynamics occur in the same category and for the same environmental functor, while Theorem 7.1 excludes its right adjoint for every nontrivial environment. Corollary 6.6 gives the analogous quantum-channel no-go while part (a) remains available for each admissible pair $(\sigma, \mathcal{N})$. Parts (a) and (b) therefore exhibit recovery data that depend on a channel and a state or prior rather than on universal natural hom-set bijections. This proves the conclusion. $\square$

23 Conclusion

Environmental currying requires more than a formal reassignment of an environmental wire. It requires the physical category to represent the causal internal-hom process space. The first necessary test is invariance of the realizable dimension-profile locus. Quantum channels fail the all-codomain requirement because, at $B = E$, the required vector-space dimension is not a square. At individual dimension-compatible codomains they still fail the affine state-space test because their extreme-point set is not closed. Classical channels fail the same test by product-of-simplices vertex counts. Instruments fail globally by a nonzero affine-hyperplane intercept; grade by grade, every nontrivial quantum and classical process body also fails an affine state-space invariant. For deterministic superchannels, tensoring by a fixed channel type has a right adjoint exactly when that type is the tensor unit. Input decoration fails by all-source coefficient comparison, whereas pure output decoration fails by the arithmetic output-dimension spectrum. At the tensor unit, polygonal sections show that a diagonal environmental component is an affine state space exactly when the target output is one-dimensional; the weaker dimension equation has a complete factor-pair classification. Transfer to larger categories is equivalent to relative representability on the objects $R^{\mathcal{H}} IY$, a condition strictly weaker than global coreflectivity. Set-valued and convex-causal presheaf completions always supply ambient right adjoints; the latter retains positive normalized process bodies and affine mixing pointwise. Nontrivial classical environments and nonunit superchannel decorations nevertheless force some ambient values outside the representable

physical image. For a finite base category, every ambient finite presheaf compiles into a fixed autonomous classical two-stage comb. A finite program monoid summarizes every control word, its action quotient is the exact minimal classical program register, and one time-homogeneous run-halt channel executes the program. Representable descent remains strictly stronger. Even identical complete-ordered hom-bodies need not assemble naturally. Universal evaluators force lax module and causal coherence. Operationally, surjective programming plus a full-column-rank finite tester response is sufficient. The calibrated pseudoinverse recovers all affine program statistics, exactly the affine dimension many independent probabilities suffice, and its smallest singular value gives the corresponding noise bound. Finite and compact symmetry closure gives an exact isotypic criterion for forcing this rank from a restricted seed family, with a tight one-seed frame in the irreducible case. Haar frames have exact finite weighted orbit quadratures, and a spectral perturbation bound preserves completeness under approximate implementation. More generally, metric defects of inverse transposition and naturality control both triangle identities, while hom-set inradii turn affine-dimension mismatch into sharp reconstruction lower bounds. A maximally entangled probe followed by spanning binary POVMs provides the finite quantum tester, while a nontrivial evaluator fiber remains invisible to every evaluated-process experiment. Recursive comb-state dimensions and superchannel hom-set dimensions are treated under their respective causal and categorical conventions. For any separately defined arbitrary-order category, the explicit two-variable profile hypotheses of Theorem 10.4 give the exact conditional right- and dimension-regular left-adjoint conclusions.

The causal internal-hom space

$$W(E, Y) = \{G : V_E \rightarrow V_Y : u_Y G \in \text{span}(u_E)\}$$

exhibits the common linear normalization mechanism. Monoidality of discarding and vector-space currying identify the corresponding normalization fibers. Environmental currying further requires the resulting vector space, cone, base, and functorial structure to be represented among the physical objects. For a restricted GPT process body P , the exact tensor-unit ordered space is its homogenization $H(P)$ rather than the maximal cone. Affine-base and generated-order representability coincide, whereas matrix transposition shows that ancillary complete order is a genuinely stronger property. For concrete matrix realizations, that entire ancillary tower is decided at the finite level $\max\{m, n\}$; quantifying over the representing map is a finite semialgebraic problem with Choi witnesses. At the abstract level, the C^* -envelope gives the exact finite-realizability criterion, minimal matrix size, and block-unitary classification. A finite matrix-state certificate constructs a cover and has minimum total size equal to the minimal embedding size; the cover then computes the Shilov ideal by finitely many complete-order tests. Three-dimensional examples exclude every dimension-only embedding bound. For a composite without local tomography, local relations and global quotient degrees modify causal currying by explicit rank-nullity terms. Positive globalization is governed in general by the quotient image of the positive-map cone, with exact dual certificates when that image is closed, a transverse sufficient condition, and robustness under strict feasibility. For nice dual cones, one minimal face yields equivalent feasible-direction and complementary-face tests. For general semialgebraic cones, closedness is decidable; nonclosed images admit weak-feasibility curves whose minimal lift norms diverge at a rational power rate. Extension is a finite LP in polyhedral theories and a finite Choi SDP in quantum theory, and can fail even for a completely positive local identification. At the tensor-unit level, an affine storage retract is weaker than a state-space isomorphism, and a state-space isomorphism is weaker than an all-test-object adjunction. Once an evaluator has unique factorization for every test object, however, those objectwise universal data assemble uniquely into the adjunction. The grade-by-grade and superchannel tensor-unit results make these distinctions explicit in the concrete categories.

The representability hierarchy consists of dimension profiles; affine shape and extreme-boundary topology; ancillary complete-order structure; and categorical naturality. Ordinary

generated-cone representability coincides with affine-base representability under the compact-base hypotheses above. The channel, instrument, superchannel, and compact-closure results instantiate these distinct necessary tests. The Bell completely positive adjunction shows in particular that ambient compact currying may exist while deterministic restriction fails simultaneously at its trace-scaling unit and rank-deficient counit; the reciprocal-weight calculation also gives the sharp $1 - \delta^{-2}$ triangle defect under trace-nonincreasing weighted caps. Conversely, Petz maps, Bayesian inverses, and entropy-increasing or entropy-decreasing dynamics can exist in categories whose environmental functor has no right adjoint. Dimension is therefore the first representability invariant, not the last.

AI Declaration

Qwen (Alibaba), ChatGPT (OpenAI), and DeepSeek AI were used to assist with drafting, editing, and stress-testing proofs. The author reviewed and edited the resulting content and takes full responsibility for the accuracy of the work.

Supplementary Material

The computational-verification code, theorem-to-evidence matrix, exact witness ledger, and selective Lean source are deposited in the article’s Zenodo repository [5].

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